LOGARITHMIC LAPLACIAN, HADAMARD FORMULA AND BOUNDARY BALANCE FOR THE INFIMUM OF THE TRUNCATED WEIL FORM

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Abstract. Let $\lambda(c)$ be the infimum of the truncated Weil form in the even sector over functions supported in $[-a,a]$, $a=(\ln c)/2$. Every bounded weak solution of the Dirichlet problem for the logarithmic Laplacian on an interval with bounded right-hand side has, at distance δ from the boundary, an expansion $u(a-\delta)=\beta(\ln(1/\delta))^{-1/2}+O((\ln(1/\delta))^{-1})$ with a boundary coefficient β. The expansion is derived from a first-order linear equation for the averages of the solution against bump functions shrinking to the boundary. The Pohozaev identity for the problem $(L_\Delta-\mu_0)z=f$, valid for every μ_0, yields $\beta^2=1/(2a)$ for every real normalized eigenfunction of this operator on $(-a,a)$, and with it the simplicity of its spectrum. The function λ is locally Lipschitz in a and differentiable outside an at most countable set, and at points of differentiability $\mathrm{d}\lambda/\mathrm{d}a=-2\beta^2$, where β is the boundary coefficient of any minimizer. The terms with powers p^m of primes do not enter this formula, and the derivative of λ exists at every point where $\lambda(c)$ is a simple eigenvalue, including $c=p^m$. The boundary balance r_c is the ratio of the logarithmic decay rates of λ and of the explicit scale Θ_c of order $c^5e^{-4\pi c}$. If $r_c\le\rho$ with a constant $\rho\ge 0$ on the intervals where $\lambda>0$, then $\lambda(c)\ge\lambda(5)(\Theta_c/\Theta_5)^\rho>0$ for all $c\ge 5$. Boundedness of r_c does not follow from the proved properties of λ and remains an open question. Rigorous bounds for λ give the bound 1.0022 for the weighted mean of r_c over $[5,71]$, but not the values of r_c at individual points.

1. Introduction

How the least eigenvalue changes as the domain varies is described by the classical Hadamard formula: for the Laplacian with the Dirichlet condition, the decay rate of the eigenvalue when the boundary is moved outward equals the square of the normal derivative of the eigenfunction on it [14, (1.1)]. In the present work the same question is posed for the truncated Weil form. The role of the domain in it is played by the interval on which the test functions are concentrated. The principal part of the form is the logarithmic Laplacian, and its eigenfunctions have no normal derivative at the boundary: at distance δ from the endpoint they decay as $(\ln(1/\delta))^{-1/2}$. What should replace it is the first question of this work.

Let us recall the setup. For $c>1$ set $a=(\ln c)/2$ and $I_c=[-a,a]$. The explicit formula [21] defines a Hermitian form on test functions, and restricting their supports to an interval I_c yields the *truncated Weil form* Q_c (in Bombieri's notation [3, Theorem 2, p. 193]), in which, for a fixed c, only terms with powers of primes $p^m<c$ appear. Its nonnegativity for all $c>1$ is equivalent to the Riemann hypothesis by Weil's criterion [3, 21]. To verify nonnegativity it suffices to consider the even sector (see [22, Proposition 1(2), p. 285] and [5, Proposition 2.5]) and smooth functions supported inside I_c [5, Lemma 2.2]. Denote by $\lambda(c)$ the infimum of Q_c in this sector.

Where the infimum has been computed, it is exponentially small and agrees up to a factor of order one with the *scale* $\Theta_c=(2\pi\Phi(a)/\kappa)^4$, in which Φ is the Riemann kernel (2.2) and $\kappa=2\pi c-9/2$. More precisely, it is proved that $\lambda(c)\ge 0.97\,\Theta_c$ at integer points $c=3,\dots,31$ and for $c=45$, 47 [5, Theorem 1.1], where for $3\le c\le 31$ the ratio $\lambda(c)/\Theta_c$ is not less than 1.04 [5, Table A.1], while $\lambda(71)>0$ [5, Proposition 1.3]. In the same place the bound $\lambda(c)\le 3.0005\,\Theta_c$ is rigorously established for $c=5,\dots,8$ [5, Proposition 7.1, Table A.3(a)]. The scale was introduced in [5, Definition 2.7], and its form, including the constant $9/2$, was chosen according to rigorous computations of λ, whereas the asymptotics $\ln\Theta_c=-4\pi c+5\ln c+O(1)$ is proved (Lemma 5.3). The meaning of this constant is explained in Section 5.2.

There are three main results of this work, and two of them are unconditional. The first describes the solutions of the Dirichlet problem for the logarithmic Laplacian L_Δ, that is, the operator with symbol $2\ln|r|$ [6], near the endpoint of an interval.

Theorem 1.1 (boundary expansion)**.** *Let u be a bounded weak solution of the Dirichlet problem $L_\Delta u=f$ on an interval $(-a,a)$ with bounded right-hand side f. Then there exists a* boundary coefficient $\beta=$

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$\lim_{\delta\to 0+} u(a-\delta)(\ln(1/\delta))^{1/2}$, *and*

$$u(a-\delta) = \beta\left(\ln\frac{1}{\delta}\right)^{-1/2} + O\left(\left(\ln\frac{1}{\delta}\right)^{-1}\right), \quad \delta\to 0+.$$

Moreover, for every real normalized eigenfunction φ of the operator L_Δ itself on $(-a,a)$ the boundary coefficients $\beta_\pm(\varphi)$ at the endpoints $\pm a$ satisfy the equality $\beta_\pm(\varphi)^2 = 1/(2a)$.

The first part of Theorem 1.1 is proved as Theorem 3.8, the second as Corollary 4.9. No sign condition on the solution is assumed, and the case $\beta = 0$ is not excluded. The boundary coefficient plays the same role here as the normal derivative in the classical Hadamard formula. This is seen from the second result, where β is the boundary coefficient of a minimizer, that is, of a normalized eigenfunction corresponding to the infimum of Q_c in the even sector.

Theorem 1.2 (Hadamard formula for the infimum)**.** *The function $a\mapsto\lambda(e^{2a})$ is locally Lipschitz on $(0,\infty)$ and differentiable everywhere except on an at most countable set of points at which $\lambda(e^{2a})$ is a multiple eigenvalue. At every point of differentiability,*

$$\frac{\mathrm{d}\lambda}{\mathrm{d}a} = -2\beta(u)^2$$

for every normalized minimizer u with $c = e^{2a}$. In particular, λ is differentiable at a point a if $\lambda(e^{2a})$ is a simple eigenvalue, including when $e^{2a} = p^m$.

Lipschitz continuity is proved in Proposition 4.2, the formula for a simple eigenvalue $\lambda(c)$ in Theorem 4.5(i), and in the general case the statement is deduced in Corollary 8.2 from the virial identity (Corollary 7.5) via one-sided derivatives (Theorem 8.1). Note that the terms with powers of primes do not enter the formula: the derivative of the infimum is determined solely by the behavior of the minimizer near the endpoints of the interval, even at points $c = p^m$ at which a new term is added to the form.

The third result reduces the positivity of λ to an estimate of its logarithmic derivative. Set $a_0 = (\ln 5)/2$ and $\kappa_\Theta = -(1/4)\,\mathrm{d}\ln\Theta_c/\,\mathrm{d}a$, that is, one quarter of the logarithmic rate of decay of the scale.

Theorem 1.3 (reduction to the boundary balance)**.** *Suppose that there exists a constant $\rho\ge 0$ with the following property: on any interval $[a_0,b]$ on which $\lambda(e^{2a})>0$, for almost all a one has*

$$-\frac{\mathrm{d}\lambda}{\mathrm{d}a} \le 4\rho\,\kappa_\Theta\,\lambda(e^{2a}). \tag{1.1}$$

Then, since $\lambda(5)>0$ [5, Theorem 1.1], for all $c\ge 5$

$$\lambda(c) \ge \lambda(5)\left(\frac{\Theta_c}{\Theta_5}\right)^{\rho} > 0.$$

We call the *boundary balance* the ratio

$$r_c = \frac{\beta(u_a)^2}{2\kappa_\Theta\lambda}$$

of the logarithmic rates of decay of the infimum and of the scale, where u_a is any normalized minimizer at $c = e^{2a}$. By Theorem 1.2, at the points of differentiability of λ the ratio r_c does not depend on the choice of u_a, and the inequality (1.1) means that $r_c\le\rho$. Theorem 1.3 with $\rho = 1$ yields the *scale law* $\lambda(c)\ge(\lambda(5)/\Theta_5)\Theta_c$. The exact form of Θ_c is needed only for this law, whereas for the positivity of λ the quantity κ_Θ in the hypothesis can be replaced by $2\pi c$ (Section 6).

The hypothesis is local in c and is needed only where λ is already positive. When proving (1.1) on an interval, one may use the fact that $\lambda>0$ on it (Lemma 5.1). The mean of r_c over an interval is determined by the values of λ/Θ_c at its endpoints (Corollary 5.7). If this ratio is bounded above and away from zero, then the means of r_c over long intervals are close to one, but they say nothing about the values of r_c at individual points. Only the Galerkin values of r_c away from prime powers, found numerically (Appendix A), are consistent with the possibility of taking ρ near 1.

The following is known rigorously about the hypothesis. The scale law $\lambda(c)\ge\vartheta_*\Theta_c$ with constant $\vartheta_*>0$ is equivalent to the boundedness above of the integrals $\int_{a_0}^{a} 4\kappa_\Theta(r_c-1)\,\mathrm{d}a'$ over a (Proposition 5.6), and the weighted average $\langle r\rangle_{[5,71]}$ of the boundary balance over the interval $[5,71]$ by rigorous bounds [5] does not exceed 1.0022 (Corollary 5.7). The boundedness of r_c does not follow from the positivity of λ and the properties of λ proved in this paper (Proposition 8.16).

Method. The principal part of the form Q_c is the logarithmic Laplacian, while the terms of the explicit formula with prime powers (atoms at the points $\pm\ln n$) and the rank-one term arising from the pole of

the zeta function (the pole term) enter it as bounded perturbations, which allows one to start from the behavior of the eigenfunctions at the boundary. Their boundedness is proved by the method of Feulefack, Jarohs and Weth [10], in which the kernel of the Archimedean term is split into parts with $|z| < \delta$ and $|z| \geq \delta$. The boundary expansion is obtained because on bump functions, which shrink to the boundary with their profile preserved, the weak form of the problem for L_Δ turns into a first-order linear equation $2\xi U'(\xi) - U(\xi) = O(1)$ for $U(\xi) = \int_{e^{-\xi}}^{2a} u(a-t)\,\mathrm{d}t/t$, $\xi = \ln(1/\delta)$, whose solutions grow like $2\beta\xi^{1/2}$. We then compare the minimizers at two close values of c using the Hadamard identity in finite differences. The atoms and the pole term contribute to it an amount small compared with the increment of a even at $c = p^m$, when a new atom is added, and only $-2\beta^2$ remains. Finally, Grönwall's inequality for the Lipschitz function $a \mapsto \lambda(e^{2a})$ together with a continuity argument leads to Theorem 1.3.

What is known and what is new. The monotonicity of λ in c is known (see [3, Theorem 5] and [8, (3.27)]). In the same work [3] Bombieri claims the continuity of the infima of the form in each sector, but, as Suzuki [19, §4.5] notes, the details of the proof are not given there in full, while the continuity of the smallest eigenvalue of the form in a (without separation into sectors) was established in [19, Theorem 1.3]. A Hadamard formula for the eigenvalues of nonlocal operators has been obtained for the fractional Laplacian [9] and for kernels without a singularity [2], and its classical analogue for prolate spheroidal functions expresses the derivative with respect to the bandwidth through their value at the endpoint of the interval [11] (see also [4, (39)]). For the logarithmic Laplacian, upper bounds near the boundary of order $(\ln(1/\delta))^{-\tau}$ with $\tau < 1/2$ [6, Theorem 1.11] and with $\tau = 1/2$ [12, Theorem 1.1] have been proved, as well as Hopf-type lemmas [12,17]. Two-sided estimates of the Green function for the operator $(I-\Delta)^{\log}$, corresponding to the geometric 2-stable process, in bounded sets of class $C^{1,1}$ for $N \geq 3$ were obtained by Kim and Mimica [13, Theorem 1.2]. In [1] the existence of the limit $u(a-\delta)(\ln(1/\delta))^{1/2}$ is assumed, and the Pohozaev identity is stated, but the preprint was withdrawn by the authors (version v2) due to an error in its proof. To the best of our knowledge, the following results are new:

(i) the existence of a boundary coefficient for every bounded weak solution of the Dirichlet problem for the logarithmic Laplacian on an interval with bounded right-hand side, with a rate of convergence and an integral formula (3.5) (Theorem 1.1);
(ii) the Pohozaev identity for the Dirichlet problem $(L_\Delta - \mu_0)z = f$ on an interval for every $\mu_0 \in \mathbb{R}$ (Theorem 4.8), the value $\beta^2 = 1/(2a)$ for all eigenfunctions of the operator L_Δ on $(-a, a)$, and the simplicity of its spectrum on this interval (Corollary 4.10);
(iii) the Lipschitz continuity of λ, its differentiability outside an at most countable set and the Hadamard formula $\mathrm{d}\lambda/\,\mathrm{d}a = -2\beta^2$ for the form with atoms and a pole term, independent of the choice of minimizer and valid also at points $c = p^m$, provided $\lambda(c)$ is a simple eigenvalue there (Theorem 1.2), as well as the one-sided derivatives of λ at every point, including points where $\lambda(c)$ is a multiple eigenvalue (Theorem 8.1);
(iv) *logarithmic cusp* of the derivative of λ to the right of a point $c_* = p^m$: when the right-hand derivative is negative, the function λ does not lie there in $C^{1,\alpha}$ for any $\alpha > 0$ (Corollary 8.6), and if $\lambda(c_*)$ is a simple eigenvalue, the correction to the derivative at distance δ has order $1/\ln(1/\delta)$ with an explicit constant (Theorem 8.14);
(v) the reduction of the positivity of λ for all c to the boundedness of the boundary balance (Theorem 1.3). The reduction itself to a bound for the logarithmic derivative is elementary for a Lipschitz λ, and what is new in it is writing this derivative in terms of the boundary coefficient via the Hadamard formula.

What is proved along the way. The proof of Theorem 3.1 carries over to forms with shift weights of any sign (Proposition 3.2). The eigenfunctions of the form lie in $W^{1,1}$ (Theorem 3.11), and for each of them a virial identity holds (Corollary 7.5), which for a minimizer coincides with the dilation identity (Proposition 7.2). The derivative of λ has modulus of continuity $O((\ln(1/h))^{-5/4})$ on compact sets without points $c = p^m$ at every point of which $\lambda(c)$ is a simple eigenvalue, and at a point $c = p^m$ with a simple eigenvalue $\lambda(c)$, from the left of it (Corollary 8.13). For the scale, the asymptotic expansion (5.2) and the exact representation (5.1) have been found, from which it is seen that Θ_c differs from the Fuchs benchmark [11] by a factor of order $\sqrt{c}$. Over an interval of length 1 the scale decreases by no more than a factor of $1/C_\Theta < 6.7 \cdot 10^5$ (Lemma 5.5). Therefore the limit superior of the mean $\langle r \rangle_{[5,c]}$ of the boundary balance as $c \to \infty$ does not exceed 1 already when the lower bounds $\lambda \geq Cc^{-A_0}\Theta_c$ are known on a sequence with gaps $o(c)$ (Corollary 5.8).

Status of the statements. All numbered statements of Sections 2–8 are proved. The inequality $\lambda(5) > 0$ from [5, Theorem 1.1] is used by Lemma 5.1, Propositions 5.2 and 5.6(b), Corollary 5.7(c) and

Theorem 1.3, and the same corollary and Proposition 8.16(d) also rely on other rigorous bounds from that paper. Statements that require the simplicity of the eigenvalue $\lambda(c)$, that is, uniqueness of the minimizer up to sign, contain it among their hypotheses. Open questions are listed at the end of Section 9.

Structure of the paper. In Section 2 the notation is collected, the form is defined and the monotonicity of the infimum is proved. Then in Section 3 the behavior of the eigenfunctions at the boundary is studied and their derivatives are shown to be integrable, while in Section 4 the Lipschitz continuity of λ, the Hadamard formula, the equality $\beta^2 = 1/(2a)$ for the logarithmic Laplacian, and the Pohozaev identity on an interval are established. The scale, the boundary balance and its means are studied in Section 5, while Section 6 contains the proof of Theorem 1.3. In Section 7 the hypothesis is written in terms of the function $\Psi(x) - x$ and a virial identity for the eigenfunctions of the form is derived. From it, in Section 8, the one-sided derivatives of the infimum at every point and its behavior near prime powers are obtained. There, in Section 8.6, it is also shown that the boundedness of r_c does not follow from the proved properties of λ. The results are summarized in Section 9, after which it is indicated where the computational results are placed, while the numerical data on the boundary balance are collected in Appendix A.

Statements, definitions and remarks are numbered jointly within each section. Numerical results in the text are marked with the word "numerically", Appendix A is entirely numerical, and they do not enter the proofs.

2. Setup

2.1. **Notation.** Below is the notation used in several sections, with the location of their definition indicated in parentheses. All other notation is introduced where it appears. An integral without limits is taken over $\mathbb{R}$.

c, a, a_0, I_c	truncation parameter $c > 1$, $a = (\ln c)/2$, $a_0 = (\ln 5)/2$, $I_c = [-a, a]$
κ, $\Lambda(n)$, σ_c, p^m	$\kappa = 2\pi c - 9/2$; the von Mangoldt function; $\sigma_c = \sum_{n<c} \Lambda(n) n^{-1/2}$; a power of a prime
$\hat f$, $\tilde f$, $f \star \tilde h$	$\hat f(r) = \int f(x) e^{-irx}\,\mathrm{d}x$, $\tilde f(x) = \overline{f(-x)}$, the correlation
$\mathbb{1}_X$, γ_{E}	indicator of the set X, Euler's constant
Φ, Θ_c, κ_Θ	Riemann kernel (2.2), the scale $\Theta_c = (2\pi\Phi(a)/\kappa)^4$, $\kappa_\Theta = -(1/4)\,\mathrm{d}\ln\Theta_c/\,\mathrm{d}a$ (Section 1)
Q_c, $D(Q_c)$, $\lambda(c)$	the truncated form (2.1), its domain, the infimum of Q_c in the even sector, in [5, Definition 2.1] denoted $\lambda_{\min}(Q_c)$ of that sector (Section 2.2)
h_+, ϖ, $\mathcal{A}$, $P(f)$	symbol, kernel, and Archimedean term, the pole functional (Section 2.2)
$\mathcal{B}$, A_c, Π, $\mathcal{V}$	energy, the operator of atoms, the pole operator, the bounded part of the equation (Section 3.1, Lemma 3.3)
u, v, u_a	eigenfunctions of Q_c; minimizer for $c = e^{2a}$ (Proposition 2.1, Theorem 4.3)
$u^{(t)}$	the dilation $u^{(t)}(x) = t^{-1/2}u(x/t)$, $t > 0$, which preserves the norm (Section 7.1)
L_Δ, E_{L}, $H(I)$	logarithmic Laplacian, its form, and space (Section 3.2)
ξ, $\ell(r)$	$\xi = \ln(1/\delta)$, $\ell(r) = 1/\lvert\ln\min(r, 0.1)\rvert$ (Section 3.2)
β, $\mathfrak{J}$	boundary coefficient (Theorem 3.8), boundary flux of the infimum (Theorem 4.3)
$\mathcal{P}$	the set of a at which $\lambda(e^{2a})$ is a simple eigenvalue (Section 4.3)
c_*, a_*, w_*	a power of a prime $c_* = p^m$, $a_* = (\ln c_*)/2$, $w_* = \Lambda(c_*)c_*^{-1/2}$ (Section 8.2)
r_c, ϑ	boundary balance (Sections 1, 5.3), level $\vartheta = \lambda/\Theta_c$ (Proposition 5.6)

2.2. **The form.** For $f \in L^2(I_c)$, extended by zero, and $g = f \star \tilde f$ the truncated form is

$$Q_c(f) = \mathcal{A}(g) - \sum_{n<c} \frac{\Lambda(n)}{\sqrt{n}} (g(\ln n) + g(-\ln n)) + 2\,\mathrm{Re}[\hat f(i/2)\overline{\hat f(-i/2)}], \tag{2.1}$$

where $\mathcal{A}$ is the Archimedean term in Weil's regularization [5, §2.2]:

$$\mathcal{A}(g) = C_0 g(0) + \int_0^\infty \left[\frac{2e^{-2y}}{1-e^{-2y}} g(0) - \varpi(y)(g(y) + g(-y))\right] \mathrm{d}y,$$
$$\varpi(y) = \frac{e^{|y|/2}}{2\sinh|y|}, \quad C_0 = -\gamma_{\mathrm{E}} - \ln\pi.$$

The terms with $\Lambda(n)$ we call *atoms* (these are the point masses of the measure of the explicit formula at the points $\pm\ln n$), and the last one the *pole term*.

A second formula for the Archimedean term. In the definition of $\mathcal{A}$ the integrals of the two summands individually diverge at zero. For the estimates in Section 3.1 a convenient formula is one in which g enters

only through the difference $g(0)-g(y)$. It follows from the identity [5, Lemma A(a)] for $r=0$: subtracting it from the definition of $\mathcal{A}(g)$, we obtain for an even Lipschitz g with integrable function $g(y)e^{|y|/2}$

$$\mathcal{A}(g) = h_+(0)g(0) + \int_0^\infty 2\varpi(y)(g(0)-g(y))\,\mathrm{d}y, \quad h_+(0) = -\gamma_{\mathrm{E}} - \pi/2 - 3\ln 2 - \ln\pi.$$

The domain and the sectors. The form Q_c is closed and semibounded below on the domain $D(Q_c) = \{f \in L^2(I_c) : \int \ln(2+|r|)|\hat{f}(r)|^2\,\mathrm{d}r < \infty\}$. The symbol of the Archimedean term $h_+(r) = \operatorname{Re}\psi(1/4+ir/2) - \ln\pi$, where $\psi = \Gamma'/\Gamma$, differs from $\ln(2+|r|)$ by a bounded quantity, and smooth functions supported inside I_c form a form core [5, Lemma A(c), Lemma 2.2]. Since the even and odd sectors are orthogonal with respect to Q_c [5, Lemma 2.3], below, unless otherwise stated, the even sector is considered, and $\lambda(c)$ is the infimum of Q_c on it. The pole term is expressed through the functional $P(f) = \int f(x)\cosh(x/2)\,\mathrm{d}x$ for functions on $\mathbb{R}$: for an even real f supported in I_c it equals $2P(f)^2$ by the same lemma.

The scale Θ_c is constructed from the Riemann kernel [5, Definition 2.7]

$$\Phi(x) = \sum_{n\ge 1} 2(2\pi^2 n^4 e^{9x/2} - 3\pi n^2 e^{5x/2})\exp(-\pi n^2 e^{2x}). \tag{2.2}$$

It is even, and by formula [20, (2.16.1), p. 36] together with the inversion formula its Fourier transform is the function Ξ on the critical line.

2.3. **Monotonicity and the minimizer.**

Proposition 2.1 (monotonicity and existence of a minimizer)**.** *The function $c \mapsto \lambda(c)$ is nonincreasing. For every c the infimum is attained: there exists an even real-valued $u \in D(Q_c)$, $\|u\| = 1$, with $Q_c(u) = \lambda(c)$.*

Proof. Monotonicity is known (Bombieri [3, Theorem 5, p. 199], and also [8, (3.27)])[1]. We give a short argument. Extension by zero embeds $L^2(I_c)$ into $L^2(I_{c'})$ for $c \le c'$, and $Q_{c'} = Q_c$ on functions supported in I_c [5, proof of Proposition 2.6], which proves monotonicity. The same argument shows that the infimum of Q_c in the odd sector is also non-increasing. The form Q_c is closed, semibounded and has discrete spectrum [8, Proposition 3.3, Theorem 3.6], and the infimum is the smallest eigenvalue and is attained. From the realness of the form it follows that the real and imaginary parts of an eigenfunction correspond to the same number, and the minimizer can be chosen real-valued. □

We call the normalized real-valued function u of Proposition 2.1 a *minimizer*. If $\lambda(c)$ is a simple eigenvalue of Q_c in the even sector, that is, the minimizer is unique up to sign, we say briefly that $\lambda(c)$ is *simple*, and otherwise that it is *multiple*.

3. Behavior of eigenfunctions near the boundary

To find the derivative of the infimum with respect to a, one needs to know how the eigenfunctions of Q_c behave at the endpoints of the interval. To this end, in Section 3.1 we prove their boundedness and vanishing at the endpoints, and in Section 3.2 we find the boundary expansion for any bounded weak solution of the Dirichlet problem for the logarithmic Laplacian with bounded right-hand side. In Section 3.3 it is carried over to the eigenfunctions of Q_c with constants uniform in c that are needed for Hadamard's formula (Section 4), and in Section 3.4 it is proved that the derivatives of these functions are integrable.

3.1. **Boundedness.**

We write the Archimedean term in terms of the *energy* $\mathcal{B}$, that is, the quadratic form with kernel ϖ. By Lemma A(a) of [5] and the proof of its part (b) for $f \in L^2(I_c)$, extended by zero,

$$\mathcal{A}(f\star\tilde{f}) = h_+(0)\|f\|^2 + \mathcal{B}(f), \quad \mathcal{B}(f) = \frac{1}{2}\iint_{\mathbb{R}\times\mathbb{R}} (f(x)-f(y))^2\varpi(x-y)\,\mathrm{d}x\,\mathrm{d}y.$$

The kernel ϖ is even, $\varpi(z) = 1/(2|z|) + O(1)$ at zero and $\varpi(z) = O(e^{-|z|/2})$ at infinity. Set $(A_c u)(x) = \sum_{n<c}\Lambda(n)n^{-1/2}[u(x+\ln n) + u(x-\ln n)]$ (the function u is extended by zero) and $\Pi u = 2\langle u, \cosh(\cdot/2)\rangle\cosh(\cdot/2)$. The operator A_c is positive, that is, maps nonnegative functions to nonnegative ones. Its norms in L^2 and L^∞ do not exceed $2\sigma_c$ (notation from Section 2.1).

[1] In both papers the interval is written multiplicatively, with endpoints $c^{\mp 1/2}$.

Theorem 3.1 (boundedness of eigenfunctions)**.** *Let $u \in L^2(I_c)$, $\|u\| = 1$, be an eigenfunction of the form Q_c (in any sector) with eigenvalue λ, and $\mu = \lambda - h_+(0)$. For $\delta > 0$ set $\varpi_\delta = \varpi \mathbb{1}_{\{|z| \ge \delta\}}$ and $m_\delta = \int \varpi_\delta$. Then $u \in L^\infty(\mathbb{R})$, namely for every δ with $m_\delta > \mu + 2\sigma_c$*

$$\|u\|_\infty \le \frac{\|\varpi_\delta\|_2 + C_\Pi}{m_\delta - \mu - 2\sigma_c}, \quad C_\Pi = 2\|\cosh(\cdot/2)\|_{L^2(I_c)} \cosh(a/2). \tag{3.1}$$

Proof. We apply the truncation method with the functions $(u-t)_+$. The singular part of the kernel, $|z| < \delta$, is kept on the left-hand side, where it has the required sign, while its integrable part, the atoms, and the pole term are estimated by $\int (u-t)_+$. For t equal to the right-hand side of (3.1), the truncation vanishes. Assume first that u is real-valued, which is needed for the truncation. By orthogonality of the sectors $Q_c(u,v) = \lambda\langle u, v\rangle$ for all $v \in D(Q_c)$, that is,

$$\mathcal{B}(u,v) = \langle \mu u + A_c u - \Pi u, v\rangle, \quad v \in D(Q_c). \tag{3.2}$$

Admissible δ exist: for the kernel ϖ, nonintegrable near zero, $m_\delta \to \infty$ as $\delta \to 0$. Decompose $\mathcal{B}(u,v) = \mathcal{B}^\delta(u,v) + m_\delta\langle u, v\rangle - \langle \varpi_\delta \star u, v\rangle$, where $\mathcal{B}^\delta$ is the same bilinear form with kernel $\varpi \mathbb{1}_{\{|z|<\delta\}}$. Let $t \ge 0$ and $q = (u-t)_+$. Here $q \in D(Q_c)$, the support of q lies in I_c, and pointwise $(u(x) - u(y))(q(x) - q(y)) \ge (q(x) - q(y))^2$. We have $0 \le \mathcal{B}^\delta(q,q) \le \mathcal{B}^\delta(u,q)$ and

$$0 \le \langle (\mu - m_\delta)u + A_c u + \varpi_\delta \star u - \Pi u, q\rangle.$$

On the support of q the function u equals $t + q$, while everywhere on $\mathbb{R}$ we have $u \le t\mathbb{1}_{I_c} + q$. Therefore $\langle u, q\rangle = t\int q + \|q\|^2$. By the positivity of A_c we have $\langle A_c u, q\rangle \le t\langle A_c \mathbb{1}_{I_c}, q\rangle + \langle A_c q, q\rangle \le 2\sigma_c t \int q + 2\sigma_c \|q\|^2$. By Young's inequality, $\langle \varpi_\delta \star u, q\rangle \le \|\varpi_\delta\|_2 \int q$. Finally, $-\langle \Pi u, q\rangle \le C_\Pi \int q$. Altogether,

$$0 \le -(m_\delta - \mu - 2\sigma_c)\|q\|^2 + [\|\varpi_\delta\|_2 + C_\Pi - (m_\delta - \mu - 2\sigma_c)t] \int q.$$

For t equal to the right-hand side of (3.1), the nonpositivity of the bracket implies $q = 0$, that is, $u \le t$ almost everywhere. The same argument for $-u$ yields $u \ge -t$. In the odd sector the pole term $-2\langle u, \sinh(\cdot/2)\rangle \sinh(\cdot/2)$ is estimated in the same way.

It remains to treat the complex case. The form Q_c is real: the real and imaginary parts of a complex eigenfunction correspond to the same eigenvalue and belong to the same sector. Applying the obtained inequality to the normalized $\operatorname{Re} u$ and $\operatorname{Im} u$, provided they are nonzero, we obtain for the complex u the bound $|u|^2 \le T^2(\|\operatorname{Re} u\|^2 + \|\operatorname{Im} u\|^2) = T^2$, where T is the right-hand side of (3.1). □

The proof transfers the method of Feulefack, Jarohs and Weth [10, Theorem 3.1] to the form with shifts. The shifts do not need to be mollified in this argument, and only the constant of the estimate depends on the positivity of A_c (Proposition 3.2).

Proposition 3.2 (perturbations of arbitrary sign)**.** *Suppose that in the form Q_c the atom operator is replaced by an operator $(Au)(x) = \sum_n b_n[u(x+\tau_n) + u(x-\tau_n)]$ with real weights b_n of arbitrary sign, shifts $\tau_n \ge 0$ and $\sigma_A = 2\sum_n |b_n| < \infty$, and that u is a normalized real-valued eigenfunction of the resulting form with eigenvalue λ, $\mu = \lambda - h_+(0)$. For $m_\delta > \mu + 2\sigma_A$ and even or odd u one has $\|u\|_\infty \le (\|\varpi_\delta\|_2 + C_\Pi)/(m_\delta - \mu - \sigma_A)$, and in the general case the same bound holds with the factor $\sqrt{2}$.*

Proof. The operator A commutes with the reflection $x \mapsto -x$: the shifts enter it in pairs. The sectors remain orthogonal. Equation (3.2) persists with A instead of A_c. Set $q = (u-t)_+$ and $\tilde{q} = (-u-t)_+$. The bound $|u| \le t\mathbb{1}_{I_c} + q + \tilde{q}$ implies $\langle Au, q\rangle \le \sigma_A t \int q + \sigma_A\|q\|^2 + \sigma_A\|\tilde{q}\|\|q\|$, and likewise for $\tilde{q}$. Add the inequalities from the proof of Theorem 3.1 for q and $\tilde{q}$ and set $X = \|q\|^2 + \|\tilde{q}\|^2$, $Y = \int(q + \tilde{q})$. Taking $2\sigma_A\|q\|\|\tilde{q}\| \le \sigma_A X$ into account, we obtain

$$(m_\delta - \mu - 2\sigma_A)X + (m_\delta - \mu - \sigma_A)tY \le (\|\varpi_\delta\|_2 + C_\Pi)Y.$$

For $t = (\|\varpi_\delta\|_2 + C_\Pi)/(m_\delta - \mu - \sigma_A)$ this gives $X = 0$. The boundedness of u is not assumed here: for u from one sector the estimate $-\langle \Pi u, q\rangle \le C_\Pi \int q$ relies only on $\|u\| = 1$. In the general case $u = u_+ + u_-$, where even u_+ and odd u_- correspond to the same eigenvalue, since A commutes with the reflection, and $\|u_+\|^2 + \|u_-\|^2 = 1$. The bound is homogeneous, so $\|u\|_\infty \le (\|u_+\| + \|u_-\|)t \le \sqrt{2}\,t$. Otherwise the multiplier cannot be removed: the pole term of the form on functions without parity has the kernel $\cosh((x-y)/2)$, and the largest value of $-\langle \Pi u, q\rangle / \int q$ exceeds C_Π (for $c = 5$ by a factor of 1.0038). □

For $\sigma_A = 2\sigma_c$ (atoms) the bound coincides with (3.1), but under the stronger condition $m_\delta > \mu + 4\sigma_c$.

For the boundary behavior it is convenient to write the equation as a Dirichlet problem for the logarithmic Laplacian, moving everything else to the right-hand side. The form of this operator on functions u, φ from the space $H(I)$ of the paper [6, p. 3], where I is an interval, is equal to

$$E_{\mathrm{L}}(u,\varphi) = \frac{1}{2}\iint_{|x-y|<1} \frac{(u(x)-u(y))(\varphi(x)-\varphi(y))}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y - \iint_{|x-y|\ge 1} \frac{u(x)\varphi(y)}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y - 2\gamma_{\mathrm{E}}\int u\varphi\,\mathrm{d}x.$$

In dimension one, the normalization constants [6, (1.5)] are equal to 1 and $-2\gamma_{\mathrm{E}}$.

Lemma 3.3 (eigenfunction as a solution of the Dirichlet problem)**.** *The eigenfunction u of Theorem 3.1 is a weak solution of the Dirichlet problem for L_Δ on I_c in the sense of [6, p. 5] with right-hand side $f = 2(\mu u + A_c u - \Pi u) + \mathcal{V}u$ (in the odd sector Π is replaced by the pole term with* sinh*), where the operator $\mathcal{V}$ is bounded on L^∞ with norm uniformly bounded in c on any interval $[c_0, c_1]$. The space $H(I_c)$ of [6] coincides here with $D(Q_c)$.*

Proof. Let us single out in ϖ the kernel of the logarithmic Laplacian: $\varpi(z) = 1/(2|z|) + \varpi_r(z)$ for $|z| < 1$ with bounded ϖ_r. The local part of the form E_{L} coincides with the contribution of the term $1/(2|z|)$ to $2\mathcal{B}$. For the remainder $k = \varpi_r \mathbb{1}_{\{|z|<1\}} + \varpi \mathbb{1}_{\{|z|\ge 1\}} \in L^1$

$$\iint (u(x)-u(y))(v(x)-v(y))k(x-y)\,\mathrm{d}x\,\mathrm{d}y = 2\Big(\int k\Big)\langle u, v\rangle - 2\langle k \star u, v\rangle,$$

hence $2\mathcal{B}(u,v) = E_{\mathrm{L}}(u,v) - \langle \mathcal{V}u, v\rangle$, where

$$\mathcal{V}u = 2k \star u - (\mathbb{1}_{\{|z|\ge 1\}}/|z|) \star u - \Big(2\int k + 2\gamma_{\mathrm{E}}\Big)u,$$

and equation (3.2), multiplied by 2, takes the stated form. The three terms of $\mathcal{V}$ are bounded in L^∞ with norms depending on c only through a. The spaces coincide because $2\varpi - \mathbb{1}_{\{|z|<1\}}/|z| \in L^1$. □

Corollary 3.4 (boundary value)**.** *The eigenfunction u of Theorem 3.1 is continuous on $\mathbb{R}$ and $|u(x)| \le C(\ln(1/(a-|x|)))^{-1/2}$ near the boundary. In particular, $u(\pm a \mp 0) = 0$.*

Proof. By Lemma 3.3, u is a weak solution of the Dirichlet problem for L_Δ on I_c with right-hand side f, and by Theorem 3.1 $u \in L^\infty$, and with it also $f \in L^\infty$. The interval satisfies the uniform exterior sphere condition, and for such domains [6, Theorem 1.11] gives continuity and the bound $|u| = O((\ln(1/\delta))^{-\tau})$ for any $\tau < 1/2$, while [12, Theorem 1.1] yields the bound with $\tau = 1/2$. □

3.2. **Boundary expansion for the logarithmic Laplacian.** We show that every bounded solution of the Dirichlet problem for this operator with bounded right-hand side has a boundary coefficient

$$\beta = \lim_{\delta\to 0} u(a-\delta)(\ln(1/\delta))^{1/2}.$$

On bump functions that contract towards the boundary while preserving their profile, the weak form of the problem turns into a first-order linear equation whose right-hand side is also bounded. Lemma 3.6 derives this equation for the averages of u over bump functions and finds their limit, while Lemma 3.7 passes from the averages to the values of u. The arguments of this section concern the operator L_Δ on an interval (symbol $2\ln|r|$, kernel $1/|z|$) and do not use the sign of the solution. Let $I = (-a, a)$, and let $H(I)$ and E_{L} be the space and the form of the operator L_Δ from Section 3.1. A function $u \in H(I)$ is called a weak solution of the problem $L_\Delta u = f$ in I, $u = 0$ outside I, if $E_{\mathrm{L}}(u,\varphi) = \int f\varphi$ for all $\varphi \in H(I)$ [6, p. 5], but below only smooth φ compactly supported in I are used. For $\delta \in (0,1)$ set $\xi = \ln(1/\delta)$ and $\ell(r) = 1/|\ln\min(r, 0.1)|$, so that $\ell(\delta) = 1/\xi$ for $\delta < 0.1$.

Lemma 3.5 (weak form with kernel cut-off)**.** *Let $u \in H(I) \cap L^\infty(\mathbb{R})$, and let φ be a Lipschitz function with compact support in I. Then for every $r \in (0,1]$*

$$E_{\mathrm{L}}(u,\varphi) = \frac{1}{2}\iint_{|x-y|<r} \frac{(u(x)-u(y))(\varphi(x)-\varphi(y))}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y + \Big(2\ln\frac{1}{r} - 2\gamma_{\mathrm{E}}\Big)\int u\varphi\,\mathrm{d}x - \iint_{|x-y|\ge r} \frac{u(y)\varphi(x)}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y,$$

and all three integrals converge absolutely.

Proof. In the layer $r \le |x-y| < 1$ expand the product $(u(x)-u(y))(\varphi(x)-\varphi(y))$. The terms $u(x)\varphi(x)$ and $u(y)\varphi(y)$ sum to $\int u\varphi \cdot \int_{r\le|z|<1} \mathrm{d}z/|z| = 2\ln(1/r)\int u\varphi$. The cross terms reduce by symmetry to $-\iint_{r\le|x-y|<1} u(y)\varphi(x)|x-y|^{-1}\,\mathrm{d}x\,\mathrm{d}y$, and the term with $|x-y|\ge 1$ for the same reason equals $-\iint_{|x-y|\ge 1} u(y)\varphi(x)|x-y|^{-1}\,\mathrm{d}x\,\mathrm{d}y$. The integrand of the local term does not exceed $2\|u\|_\infty \operatorname{Lip}(\varphi)$ and is concentrated on a set of finite measure. The remaining integrals converge absolutely for bounded u and φ with compact support. □

Lemma 3.6 (equation for the averages near the boundary)**.** *Let $u \in H(I)\cap L^\infty(\mathbb{R})$ be a weak solution of the problem $L_\Delta u = f$ with $f \in L^\infty(I)$, and let χ be a Lipschitz function with constant L_χ, $\chi \ge 0$, $\operatorname{supp}\chi \subset [1/2,2]$, $\int\chi = 1$, and*

$$\varphi_\delta(x) = \delta^{-1}\chi((a-x)/\delta), \quad \overline{u}(\delta) = \langle u, \varphi_\delta\rangle, \quad U(\xi) = \int_{e^{-\xi}}^{2a} u(a-t)\frac{\mathrm{d}t}{t}.$$

(a) *For $0 < \delta < \min(1,a)/4$*

$$2\xi\,\overline{u}(\delta) - U(\xi - \ln 4) = e(\delta), \quad |e(\delta)| \le C_\chi(\|u\|_\infty + \|f\|_\infty), \tag{3.3}$$

where C_χ depends only on χ.

(b) *There exists a limit $B = \lim_{\xi\to\infty}\xi^{-1/2}U(\xi)$ independent of χ, and for $\xi > \xi_0$, where $\xi_0 = \ln(4/\min(1,a))$,*

$$|\xi^{1/2}\overline{u}(e^{-\xi}) - B/2| \le C'_\chi(\|u\|_\infty + \|f\|_\infty)\xi^{-1/2},$$
$$|U(\xi) - B\xi^{1/2}| \le C'_\chi(\|u\|_\infty + \|f\|_\infty).$$

Proof. We apply Lemma 3.5 to the bump functions φ_δ and estimate separately the local term, the far term and the right-hand side. The main contribution turns out to be the part of the far term, which equals $U(\xi - \ln 4)$.

(a) The support of φ_δ consists of points x with $t = a - x \in [\delta/2, 2\delta]$ and lies in I. We apply Lemma 3.5 with $\varphi = \varphi_\delta$ and $r = \delta/4$. Here $2\ln(1/r) = 2\xi + 2\ln 4$, $\int u\varphi_\delta = \overline{u}(\delta)$ and $|\overline{u}(\delta)| \le \|u\|_\infty$ for $\chi \ge 0$ and $\int\chi = 1$.

Local term. We have $|\varphi_\delta(x) - \varphi_\delta(y)| \le L_\chi|x-y|/\delta^2$, while the pairs with $|x-y| < \delta/4$ for which at least one point lies in the support of φ_δ form a set of measure at most $2\delta^2$. The contribution is at most $2L_\chi\|u\|_\infty$.

Far term. Let $t' = a - y$. Outside I the function u vanishes. For $t' \ge 4\delta$ one has $t' - t \ge t'/2$ and $1/(t'-t) = 1/t' + t/(t'(t'-t))$. The correction is at most $4\delta/t'^2$, and its integral over $t' \ge 4\delta$ does not exceed 1. The main part equals $\int\varphi_\delta \cdot \int_{4\delta}^{2a} u(a-t')\,\mathrm{d}t'/t' = U(\xi - \ln 4)$. The remainder $t' < 4\delta$, $|t'-t| \ge \delta/4$ contributes at most $2\ln 16 \cdot \|u\|_\infty$.

Right-hand side. We have $|\langle f, \varphi_\delta\rangle| \le \|f\|_\infty$. Substituting these estimates into $E_{\mathrm{L}}(u,\varphi_\delta) = \langle f,\varphi_\delta\rangle$, we obtain (3.3).

(b) We have $U'(\xi) = u(a - e^{-\xi})$ and $\overline{u}(e^{-\xi}) = (\zeta \star U')(\xi)$, where $\zeta(\sigma) = \chi(e^\sigma)e^\sigma \ge 0$, $\operatorname{supp}\zeta \subset [-\ln 2, \ln 2]$, $\int\zeta = 1$. The function $U_\zeta = \zeta\star U$ satisfies $U'_\zeta = \overline{u}$ and $|U_\zeta(\xi) - U(\xi - \ln 4)| \le 3\ln 2 \cdot \|u\|_\infty$. By (a)

$$2\xi U'_\zeta - U_\zeta = \tilde{e}, \quad |\tilde{e}| \le C_1 = (C_\chi + 3\ln 2)(\|u\|_\infty + \|f\|_\infty),$$

and $(\xi^{-1/2}U_\zeta)' = \tilde{e}/(2\xi^{3/2})$. The integrability of the right-hand side on $[\xi_0,\infty)$ implies the existence of the limit $B = \lim\xi^{-1/2}U_\zeta$ and the estimate $|\xi^{-1/2}U_\zeta - B| \le C_1\xi^{-1/2}$. Next

$$\xi^{1/2}\overline{u} = \xi^{1/2}U'_\zeta = (\xi^{-1/2}U_\zeta + \xi^{-1/2}\tilde{e})/2 = B/2 + O(C_1\xi^{-1/2}).$$

Finally, from $|U - U_\zeta| \le \ln 2 \cdot \|u\|_\infty$ there follow $B = \lim\xi^{-1/2}U(\xi)$ and, by the triangle inequality, the second bound (b), $|U - B\xi^{1/2}| \le \ln 2 \cdot \|u\|_\infty + C_1$. In particular, B does not depend on χ. □

Lemma 3.7 (deviation from the mean)**.** *Suppose the hypotheses of Lemma 3.6 hold and $x = a - \delta$, where $0 < \delta \le \min(1,a)/16$, while χ_0 is an even Lipschitz function, $\chi_0 \ge 0$, $\operatorname{supp}\chi_0 \subset \{1/2 \le |t| \le 1\}$, $\int\chi_0 = 1$, and $\varphi^x(y) = (4/\delta)\chi_0(4(y-x)/\delta)$. Then*

$$|u(x) - \langle u, \varphi^x\rangle| \le C_{\chi_0}\frac{\|u\|_\infty + \|f\|_\infty}{\ln(1/\delta)},$$

where C_{χ_0} depends only on χ_0.

Proof. We repeat the derivation of Lemma 3.6, contracting the bump function not to the boundary but to the point x. For averages over bump functions of width $(\delta/4)e^{-s}$ one obtains an equation of the same form. For $s \to \infty$ its solution tends to $u(x)$, for $s = 0$ it equals the average $\langle u, \varphi^x \rangle$, and an estimate uniform in s compares these values.

The function u is continuous in I [6, Theorem 1.11]. For $s \geq 0$ set

$$d_s = (\delta/4)e^{-s}, \quad \varphi_s(y) = d_s^{-1}\chi_0((y-x)/d_s), \quad \alpha(s) = \langle u, \varphi_s \rangle,$$

and also $m(t) = (u(x+t) + u(x-t))/2$ and $\overline{m}(\sigma) = m((\delta/4)e^{-\sigma})$. By the evenness of χ_0

$$\alpha(s) = \int_0^{\ln 2} \overline{m}(s+\theta)\zeta_0(\theta)\,\mathrm{d}\theta, \quad \zeta_0(\theta) = 2\chi_0(e^{-\theta})e^{-\theta} \geq 0, \quad \int \zeta_0 = 1,$$

and $\alpha(0) = \langle u, \varphi^x \rangle$, while $\alpha(s) \to u(x)$ for $s \to \infty$. We apply Lemma 3.5 with $\varphi = \varphi_s$ and $r = d_s/4$. Now $2\ln(1/r) = 2(\xi + s) + 2\ln 16$. The local term, as in Lemma 3.6, equals $O(\|u\|_\infty)$. The far term, containing $u(z)$, we split according to the magnitude of $|z - x|$.

- For $|z - x| \geq \delta/2$ we have $|y - x| \leq d_s \leq |z - x|/2$, so replacing $1/|y - z|$ by $1/|x - z|$ introduces an error of at most $\int 2d_s|x - z|^{-2}\,\mathrm{d}z \leq 2$ in units of $\|u\|_\infty$. There remains a quantity $V = \int_{|z-x|\geq\delta/2} u(z)|x - z|^{-1}\,\mathrm{d}z$ independent of s.
- For $2d_s \leq |z - x| < \delta/2$ the same substitution introduces an error of at most $2\|u\|_\infty$, while the main part equals $\int_{2d_s}^{\delta/2} 2m(t)\,\mathrm{d}t/t = 2\int_{-\ln 2}^{s-\ln 2} \overline{m} = 2\int_0^s \overline{m} + O(\|u\|_\infty)$.
- For $|z - x| < 2d_s$ and $|y - z| \geq r$ the contribution is at most $2\ln 12 \cdot \|u\|_\infty$.

Set $\mathcal{M}(s) = \int_0^s \alpha$. The inequality $\zeta_0 \geq 0$ implies $|\mathcal{M}(s) - \int_0^s \overline{m}| \leq 2\ln 2 \cdot \|u\|_\infty$. Substituting the three bounds into the equality of the lemma on the weak form with the truncated kernel for $E_{\mathrm{L}}(u, \varphi_s) = \langle f, \varphi_s \rangle$ and dividing it by 2, we arrive at the equation

$$(\xi + s)\mathcal{M}'(s) - \mathcal{M}(s) = V/2 + e(s), \quad |e(s)| \leq C_1 = C(\|u\|_\infty + \|f\|_\infty), \quad \mathcal{M}(0) = 0.$$

This means that $(\mathcal{M}/(\xi + s))' = (V/2 + e)/(\xi + s)^2$, and

$$\frac{\mathcal{M}(s)}{\xi + s} = \frac{V}{2}\left(\frac{1}{\xi} - \frac{1}{\xi + s}\right) + e_1(s), \quad e_1(s) = \int_0^s \frac{e(\sigma)}{(\xi + \sigma)^2}\,\mathrm{d}\sigma, \quad |e_1(s)| \leq \frac{C_1}{\xi}.$$

Differentiating, we obtain

$$\alpha(s) = \mathcal{M}'(s) = V/(2\xi) + e_1(s) + e(s)/(\xi + s), \quad |\alpha(s) - V/(2\xi)| \leq 2C_1/\xi, \quad s \geq 0.$$

Comparing the values at $s \to \infty$ and at $s = 0$, we find $|u(x) - \alpha(0)| \leq 4C_1/\xi$. □

The quantity $1/\ln(1/\delta)$ in Lemma 3.7 is small compared with the boundary factor $(\ln(1/\delta))^{-1/2}$, which makes it possible to pass from averages near the boundary to values of the function.

Theorem 3.8 (boundary expansion)**.** *Let $u \in H(I) \cap L^\infty(\mathbb{R})$ be a weak solution of the Dirichlet problem $L_\Delta u = f$ in $I = (-a, a)$, with $u = 0$ outside I and $f \in L^\infty(I)$. Then the limits $\beta_\pm = \lim_{\delta\to 0+} u(\pm(a - \delta))(\ln(1/\delta))^{1/2}$ exist, and for $0 < \delta \leq \min(1, a)/16$*

$$|u(a - \delta)(\ln(1/\delta))^{1/2} - \beta_+| \leq C(\|u\|_\infty + \|f\|_\infty)(\ln(1/\delta))^{-1/2} \tag{3.4}$$

with an absolute constant C. The same holds for β_- and the point $-a + \delta$. In other words, $u(a - \delta) = \beta_+\ell(\delta)^{1/2} + O(\ell(\delta))$. Moreover,

$$2\beta_+ = \lim_{\xi\to\infty} \xi^{-1/2} \int_{e^{-\xi}}^1 u(a - t)\frac{\mathrm{d}t}{t}. \tag{3.5}$$

Proof. Fix χ_0 as in Lemma 3.7 and put $\chi(\tau) = 4\chi_0(4(1 - \tau))$. For $x = a - \delta$ the bump function φ^x in the variable $t = a - y$ equals $\delta^{-1}\chi(t/\delta)$. Here $\chi \geq 0$, $\operatorname{supp}\chi \subset [3/4, 5/4] \subset [1/2, 2]$ and $\int \chi = 1$. Moreover, $\langle u, \varphi^x \rangle = \overline{u}(\delta)$ in the sense of Lemma 3.6. Its part (b) gives $\xi^{1/2}\overline{u}(\delta) = B/2 + O((\|u\|_\infty + \|f\|_\infty)\xi^{-1/2})$, and the lemma on the deviation from the mean gives the bound $\xi^{1/2}|u(x) - \overline{u}(\delta)| \leq C_{\chi_0}(\|u\|_\infty + \|f\|_\infty)\xi^{-1/2}$. The two bounds together yield $\beta_+ = B/2$ and (3.4). Replacing the upper limit $2a$ in U by 1 changes the integral by at most $\|u\|_\infty|\ln 2a|$, and formula (3.5) is proved. The constants depend only on the chosen χ_0. The left boundary is treated in the same way, with the reflection $x \mapsto -x$. □

Let us compare Theorem 3.8 with known results. In [12, Theorem 1.1] the upper bound $|u| \leq C\ell^{1/2}$ was proved for bounded open sets in $\mathbb{R}^N$ satisfying a uniform exterior sphere condition, and in dimension one, on an interval, it also follows from the boundary expansion. From lemmas of Hopf type (see [12, Theorem 1.4] and [17, Lemmas 3.4, 3.11]) one obtains $\liminf u/\ell^{1/2} > 0$ for nonnegative or antisymmetric solutions

under the assumptions of those lemmas (the inequality $L_\Delta u \ge Vu$ with small V), but not the existence of the limit, which in [1, Theorem 2.3] is assumed and in [1, Corollary 3.8] is used without proof. In the boundary expansion the sign of u is not used, and the case $\beta_+ = 0$ is not excluded: the proof relies not on the maximum principle but on the exact form of the kernel $1/|z|$. On the half-line, the same order $\ell^{1/2}$ near the boundary is exhibited by the generalized eigenfunctions of the operator with symbol $\ln(1+r^2)$ [15, Lemma 4.27, Example 6.4].

3.3. **Boundary coefficient of the eigenfunctions of the form.** We now carry the boundary expansion over to eigenfunctions of the form Q_c with constants uniform in c.

Corollary 3.9 (boundary coefficient of the eigenfunctions of the form)**.** *Let $1 < c_0 < c_1$ and $\overline{\lambda} \in \mathbb{R}$. The eigenfunctions of Q_c with $c \in [c_0, c_1]$ (in either sector) and eigenvalue at most $\overline{\lambda}$ have boundary coefficients $\beta_\pm$, and the bounds of Theorem 3.8 hold for them with a common constant. More precisely, there exist C_* and K, depending only on c_0, c_1, and $\overline{\lambda}$, such that for any such normalized function u, for $0 < \delta \le \delta_0$, $\delta_0 = \min(1, (\ln c_0)/2)/16$, and $\xi = \ln(1/\delta)$,*

$$|u(a-\delta)\xi^{1/2} - \beta_+(u)| \le C_*\xi^{-1/2}, \quad |U_u(\xi) - 2\beta_+(u)\xi^{1/2}| \le C_*, \quad |\beta_+(u)| \le C_*,$$

where $U_u(\xi) = \int_{e^{-\xi}}^1 u(a-t)\,\mathrm{d}t/t$. In particular, $|u(x)| \le K\ell(a-|x|)^{1/2}$. For an even function $\beta_- = \beta_+$, for an odd one $\beta_- = -\beta_+$.

Proof. By Theorem 3.1, $\|u\|_\infty$ is uniformly bounded: the bound (3.1) depends on c through a and $\sigma_c \le \sigma_{c_1}$, and on λ through $\mu \le \overline{\lambda} - h_+(0)$. Lemma 3.3 represents u as a weak solution of the Dirichlet problem for L_Δ on I_c with the right-hand side $f = 2(\mu u + A_c u - \Pi u) + \mathcal{V}u$. The operators A_c, Π and $\mathcal{V}$ are bounded in L^∞ uniformly in $c \in [c_0, c_1]$, and $|\mu| \le C$, since λ lies between the infimum of Q_{c_1} over both sectors (proof of Proposition 2.1) and $\overline{\lambda}$. For the right-hand side this gives $\|f\|_\infty \le K_1\|u\|_\infty$, where K_1 depends on the same quantities. The first inequality is (3.4): the threshold $\min(1,a)/16$ is not less than δ_0 for $a \ge (\ln c_0)/2$. The second follows from Lemma 3.6(b) with $B = 2\beta_+$, since replacing the upper limit $2a$ by 1 changes the integral by at most $\|u\|_\infty|\ln 2a|$. The third follows from the second for $\xi = \ln(1/\delta_0)$ and $|U_u(\xi)| \le \|u\|_\infty\xi$. The pointwise bound for $a - |x| \le \delta_0$ is obtained from the first inequality, and for $a - |x| > \delta_0$ from the boundedness of u: there $\ell(a-|x|) \ge 1/\ln(1/\delta_0)$. The relations between β_- and β_+ follow from the evenness or oddness of u. □

Below we write $\beta(u) = \beta_+(u)$. Minimizers for $c \in [c_0, c_1]$ have eigenvalue at most $\lambda(c_0)$, and to them, as to all eigenfunctions with eigenvalue not exceeding $\lambda(c_0)+1$, Corollary 3.9 applies with $\overline{\lambda} = \lambda(c_0)+1$, and the bound $|u(x)| \le K\ell(a-|x|)^{1/2}$ is uniform in $c \in [c_0, c_1]$. It is used in the comparison of minimizers for close c and in the Lipschitz continuity of λ in Section 4.

Theorem 3.10 (continuity of the boundary coefficient)**.** *Let $c_k \to c$, let $c_k, c \in [c_0, c_1]$, let u_k be normalized eigenfunctions of Q_{c_k} with eigenvalues $\lambda_k \le \overline{\lambda}$, and let $u_k \to u$ in $L^2(\mathbb{R})$. Then u is an eigenfunction of Q_c, $\beta(u_k) \to \beta(u)$, and λ_k converge to the eigenvalue of u.*

Proof. First we show that the limit u is an eigenfunction and that the eigenvalues converge, and then we derive the convergence of the boundary coefficients from the uniform estimate of Corollary 3.9. The numbers λ_k are bounded below by the infimum of Q_{c_1} over both sectors (proof of Proposition 2.1). Let λ_* be the limit of some subsequence. The support of u lies in I_c. The energies $\mathcal{B}(u_k)$ are bounded, since they differ from $\lambda_k - h_+(0)$ by the contribution of the atoms and of the pole term, which are uniformly bounded in L^2 with respect to k. By Fatou's lemma along a subsequence converging almost everywhere, $\mathcal{B}(u) \le \liminf \mathcal{B}(u_k) < \infty$ and $u \in D(Q_c)$. Let φ be a smooth function supported inside I_c. For large k its support lies also inside I_{c_k}. We have $\mathcal{B}(u_k, \varphi) = \langle u_k, \mathcal{L}\varphi\rangle$, where the Fourier transform of the function $\mathcal{L}\varphi$ equals $(h_+ - h_+(0))\hat{\varphi}$. The symbol h_+ grows logarithmically, and $\mathcal{L}\varphi \in L^2$. The atoms contribute $\langle u_k, A_{c_k}\varphi\rangle$. If $c = p^m$ and $c_k > c$, the atom $n = c$ shifts the support of φ by $\pm 2a$ out of I_{c_k} for large k, and therefore $\langle u_k, A_{c_k}\varphi\rangle \to \langle u, A_c\varphi\rangle$. In the remaining cases, for large k the atoms n of Q_{c_k} with $\Lambda(n) \ne 0$ are the same as those of Q_c. The pole term is continuous in L^2. Passing to the limit in $Q_{c_k}(u_k, \varphi) = \lambda_k\langle u_k, \varphi\rangle$, we obtain $Q_c(u, \varphi) = \lambda_*\langle u, \varphi\rangle$. Such φ form a core of the form [5, Lemma 2.2], and consequently u is an eigenfunction with eigenvalue λ_*, and $\|u\| = 1$. The number λ_* is determined uniquely by the function u, and hence the entire sequence λ_k converges.

Let $a_k = (\ln c_k)/2$. For fixed ξ

$$|U_{u_k}(\xi) - U_u(\xi)| \le e^{\xi}(\|u_k - u\|_{L^1} + \|u(\cdot + a_k - a) - u\|_{L^1}) \to 0,$$

since the supports lie in a common interval and the shift is continuous in L^1. For $v = u_k$ and $v = u$, by Corollary 3.9 we have

$$|\xi^{-1/2} U_v(\xi) - 2\beta(v)| \le C_* \xi^{-1/2}.$$

Therefore, $\limsup |\beta(u_k) - \beta(u)| \le C_* \xi^{-1/2}$ for every $\xi \ge \ln(1/\delta_0)$, that is, $\beta(u_k) \to \beta(u)$. □

3.4. **Integrability of the derivative.** The dilation identity in Section 7 requires the differentiability of the autocorrelation of the minimizer at the points $\ln n$, to which the shifts carry the boundary singularity. This differentiability follows from the integrability of the derivative of the eigenfunctions.

Theorem 3.11 (integrability of the derivative of the eigenfunctions)**.** *Every eigenfunction u of the form Q_c (in either sector), extended by zero, lies in $W^{1,1}(\mathbb{R})$: it is absolutely continuous, and $u' \in L^1(\mathbb{R})$. In particular, the autocorrelation $g = u \star \tilde{u}$ is continuously differentiable on $\mathbb{R}$, $g'(s) = \int u(x)u'(x+s)\,\mathrm{d}x$, and $g'(\pm 2a) = 0$.*

Proof. We shall write the equation for u on the whole line in the form $(L_{\mathcal{J}} + 1)u = \mathcal{U}$, estimate the kernel $\mathcal{G} = (L_{\mathcal{J}} + 1)^{-1}$ near zero, and derive the boundedness of the variation of u from an estimate of differences, and absolute continuity will follow from the convolution representation.

Expansion of the kernel. Put $\mathcal{J}(z) = e^{-|z|}/|z|$ and $k_1 = 2\varpi - \mathcal{J}$. The function k_1 is even, bounded ($k_1(0 \pm 0) = 3/2$), integrable, and of bounded variation: on $(0, \infty)$ it is smooth, its derivative is finite at zero and decreases exponentially. The form $\mathcal{B}_{\mathcal{J}}(f,h) = (1/2)\iint (f(x) - f(y))(h(x) - h(y))\mathcal{J}(x-y)\,\mathrm{d}x\,\mathrm{d}y$ has symbol $\int (1 - \cos rz)\mathcal{J}(z)\,\mathrm{d}z = \ln(1 + r^2)$, since the derivative of the integral with respect to r equals $2\int_0^\infty \sin(rz)e^{-z}\,\mathrm{d}z = 2r/(1+r^2)$. Since $k_1 \in L^1$, for the form $\mathcal{B}$ from Section 3.1 one has $2\mathcal{B}(f,h) = \mathcal{B}_{\mathcal{J}}(f,h) + (\int k_1)\langle f, h\rangle - \langle k_1 \star f, h\rangle$.

The equation on the line. By (3.2), $\mathcal{B}(u,v) = \langle \mu u + A_c u - \Pi u, v\rangle$ for all $v \in D(Q_c)$ (in the odd sector Π is replaced by the pole term with sinh). Put

$$f = \left(1 + 2\mu - \int k_1\right) u + 2A_c u + k_1 \star u - 2\Pi u, \quad \mathcal{U} = f\mathbb{1}_{I_c} - (\mathcal{J} \star u)\mathbb{1}_{\mathbb{R}\setminus\overline{I_c}},$$

where u is extended by zero, and let $L_{\mathcal{J}}$ be the operator of the form $\mathcal{B}_{\mathcal{J}}$. On test functions supported in I_c the equality $(L_{\mathcal{J}} + 1)u = \mathcal{U}$ is the weak equation. On test functions φ supported outside $\overline{I_c}$ we have $u\varphi = 0$ and $\mathcal{B}_{\mathcal{J}}(u,\varphi) = -\langle \mathcal{J} \star u, \varphi\rangle$. The distribution $T = (L_{\mathcal{J}} + 1)u - \mathcal{U}$ is concentrated at the points $\pm a$. Moreover, $\mathcal{U} \in L^2$. Indeed, f is bounded, since $u \in L^\infty$, and by Corollary 3.9 $|u(x)| \le K\ell(a - |x|)^{1/2}$. Consequently, $|(\mathcal{J} \star u)(\pm(a+\varepsilon))| \le C(1 + \ln^+(1/\varepsilon))^{1/2}e^{-\varepsilon}$ for all $\varepsilon > 0$ (the integral $\int_0^{2a} \ell(s)^{1/2}(s+\varepsilon)^{-1}\,\mathrm{d}s$ is split at the point $s = \varepsilon$). From $\int \ln(2 + |r|)|\hat{u}|^2\,\mathrm{d}r < \infty$ (Section 2.2) it follows that $\int |\hat{T}(r)|^2 \ln(2+|r|)^{-1}\,\mathrm{d}r < \infty$. The Fourier transform of a nonzero distribution concentrated at the points $\pm a$ is $Z_+(r)e^{-iar} + Z_-(r)e^{iar}$ with polynomials $Z_\pm$ not vanishing simultaneously. The mean of its squared modulus over a period π/a is bounded away from zero for large $|r|$, so the integral diverges, and $T = 0$, while $u = \mathcal{G} \star \mathcal{U}$, where $\hat{\mathcal{G}}(r) = (1 + \ln(1+r^2))^{-1}$.

The kernel $\mathcal{G}$. From $(1+r^2)^{-t} = \Gamma(t)^{-1}\int_0^\infty s^{t-1}e^{-s(1+r^2)}\,\mathrm{d}s$ it follows that

$$\mathcal{G} = \int_0^\infty e^{-t}\mathcal{H}_t\,\mathrm{d}t, \quad \mathcal{H}_t(x) = \frac{1}{\Gamma(t)}\int_0^\infty s^{t-1}e^{-s}\frac{e^{-x^2/(4s)}}{\sqrt{4\pi s}}\,\mathrm{d}s.$$

Each $\mathcal{H}_t$ is a mixture of Gaussian densities: it is positive, even, decreasing in $|x|$, and $\int \mathcal{H}_t = 1$. Integration over t with the weight e^{-t} carries these properties over to $\mathcal{G}$, which is moreover smooth away from zero, so that $\mathcal{G}$ decreases on $(0,\infty)$ and $\int \mathcal{G} = 1$. We estimate $\mathcal{H}_t$ for $0 < r \le 1$, using the equality $\int_0^\infty s^{-\alpha-1}e^{-b/s}\,\mathrm{d}s = b^{-\alpha}\Gamma(\alpha)$. If $t \le 1/4$, then, discarding e^{-s}, we obtain $\mathcal{H}_t(r) \le \Gamma(1/2 - t)2^{-2t}r^{2t-1}/(\sqrt{\pi}\Gamma(t)) \le 2.31\,tr^{2t-1}$, since $\Gamma(1/2 - t) \le \Gamma(1/4)$ and $1/\Gamma(t) = t/\Gamma(1+t) \le t/0.8856$. For $1/4 \le t \le 3/4$ splitting the integral over s at the point 1 leads to the estimate $\mathcal{H}_t(r) \le 1.64r^{-1/2} + 0.12$. Finally, for $t \ge 3/4$, without the Gaussian factor, we have $\mathcal{H}_t \le \Gamma(t - 1/2)/(2\sqrt{\pi}\Gamma(t)) \le \Gamma(1/4)/(2\sqrt{\pi}\Gamma(3/4)) < 0.835$. Integrating these estimates with the weight e^{-t} and using the equality $\int_0^\infty te^{-t(1+2\ln(1/r))}\,\mathrm{d}t = (1 + 2\ln(1/r))^{-2}$, we obtain

$$\mathcal{G}(r) \le \frac{2.31}{r(1 + 2\ln(1/r))^2} + 0.82r^{-1/2} + 0.9, \quad 0 < r \le 1. \tag{3.6}$$

Further, only the constant in the main term, $C_{\mathcal{G}} = 2.31$, is needed, while the other numerical values in (3.6) are used nowhere.

Estimate of differences. For $v \in L^1(\mathbb{R})$ and $h > 0$ put $\mathcal{T}_h(v) = \int |v(x+h) - v(x)|\,\mathrm{d}x$. The function v has bounded variation if and only if $\mathcal{T}_h(v) \le Ch$ for all $h > 0$. We write $u = u_1 + u_2$, $u_1 = \mathcal{G} \star (f\mathbb{1}_{I_c})$, $u_2 = \mathcal{G} \star \mathcal{U}_o$, $\mathcal{U}_o = -(\mathcal{J} \star u)\mathbb{1}_{\mathbb{R}\setminus\overline{I_c}}$. The contribution of f to the differences of u passes through the kernel $\mathcal{G}$, whose mass in a neighborhood of the origin is arbitrarily small, and the estimate closes.

Let $0 < r_0 < 1$ and $m_{\mathcal{G}}(r_0) = \int_{|x|<r_0} \mathcal{G}$. The variation of the kernel $\mathcal{G}\mathbb{1}_{\{|x|\ge r_0\}}$ equals $4\mathcal{G}(r_0)$, which gives $\mathcal{T}_h(u_1) \le m_{\mathcal{G}}(r_0)\mathcal{T}_h(f\mathbb{1}_{I_c}) + 4h\mathcal{G}(r_0)\|f\mathbb{1}_{I_c}\|_1$.

The operator A_c is a sum of shifts with weights of total mass $2\sigma_c$, and $\mathcal{T}_h$ does not change under a shift. Furthermore, $\mathcal{T}_h(k_1 \star u) \le h\,\mathrm{Var}(k_1)\|u\|_1$, the function Πu is smooth on $\overline{I_c}$, and multiplication by $\mathbb{1}_{I_c}$ adds at most $2h\|f\|_{L^\infty(I_c)}$. Altogether $\mathcal{T}_h(f\mathbb{1}_{I_c}) \le C_1\mathcal{T}_h(u) + C_2h$, $C_1 = C_\mu + 4\sigma_c$, $C_\mu = |1 + 2\mu - \int k_1|$.

On I_c the function u_2 is smooth. By the monotonicity of $\mathcal{G}$ on $(0,\infty)$ for $y \notin \overline{I_c}$ we have $\int_{I_c} |\mathcal{G}'(x-y)|\,\mathrm{d}x \le \mathcal{G}(\mathrm{dist}(y, I_c))$, and

$$\int_{I_c} |u_2'| \le \int_{\mathbb{R}\setminus\overline{I_c}} |\mathcal{U}_o(y)|\mathcal{G}(\mathrm{dist}(y, I_c))\,\mathrm{d}y < \infty.$$

Indeed, near the boundary

$$|\mathcal{U}_o| \le C(1 + \ln(1/\varepsilon))^{1/2}, \quad \mathcal{G}(\varepsilon) = O(1/(\varepsilon \ln^2(1/\varepsilon)))$$

by (3.6), while $\int_0 \mathrm{d}\varepsilon/(\varepsilon \ln^{3/2}(1/\varepsilon)) < \infty$. Outside $\overline{I_c}$, where the function u vanishes, $u_2 = -u_1$. The functions u and u_1 are continuous ($\mathcal{G} \in L^1$, f is bounded), and with them so is u_2. Splitting the pairs x, $x + h$ into those lying in I_c, outside $\overline{I_c}$ and on opposite sides of the boundary, we obtain

$$\mathcal{T}_h(u_2) \le h \int_{I_c} |u_2'| + \mathcal{T}_h(u_1) + 4h(\|u\|_\infty + \|u_1\|_\infty).$$

Altogether

$$\mathcal{T}_h(u) \le 2\mathcal{T}_h(u_1) + C_3h \le 2m_{\mathcal{G}}(r_0)C_1\mathcal{T}_h(u) + C_4(r_0)h.$$

Here $m_{\mathcal{G}}(r_0) \to 0$ for $r_0 \to 0$, while C_1 does not depend on r_0. Choose r_0 with $2m_{\mathcal{G}}(r_0)C_1 \le 1/2$. The quantity $\mathcal{T}_h(u) \le 2\|u\|_1$ is finite, which allows one to move the term containing it to the left and obtain $\mathcal{T}_h(u) \le 2C_4(r_0)h$ for all $h > 0$, that is, the functions u and $f\mathbb{1}_{I_c}$ have bounded variation.

Absolute continuity. The derivative of u_1 in the sense of distributions is the convolution $\mathcal{G} \star (f\mathbb{1}_{I_c})'$ of the kernel from L^1 with a finite measure $(f\mathbb{1}_{I_c})'$ and lies in L^1. The function u_2 is continuous, on I_c it has an integrable derivative, and outside $\overline{I_c}$ it coincides with $-u_1$. Thus, $u \in W^{1,1}(\mathbb{R})$. Finally, $g(s+h) - g(s) = \int u(x) \int_0^h u'(x+s+\tau)\,\mathrm{d}\tau\,\mathrm{d}x$, whence $g'(s) = \int u(x)u'(x+s)\,\mathrm{d}x$. Continuity in s follows from the continuity of the shift in L^1. For $s = 2a$ the factor $u'(x+2a)$ is nonzero only for $x < -a$, where $u(x) = 0$. The case $s = -2a$ is treated in the same way. □

The proof uses only the boundedness of u, the boundary estimate $|u| \le K\ell^{1/2}$, and the structure of the perturbation, that is, shifts, convolution with a kernel of bounded variation, and a finite-rank term, while the sign of u and the simplicity of the eigenvalue are not needed. The squared logarithm in (3.6) is essential, since for $\mathcal{G}(\varepsilon) = O(1/(\varepsilon \ln(1/\varepsilon)))$ the integral in the bound on the differences would diverge.

The proofs of Lemma 3.3, Corollaries 3.4, 3.9 and Theorems 3.10, 3.11 use the results of [6] for real functions. The complex case reduces to the real one just as in the proof of Theorem 3.1: $\mathrm{Re}\,u$ and $\mathrm{Im}\,u$ are eigenfunctions of the same eigenvalue and sector with norms at most one, and the estimates are homogeneous. The function $u \star \tilde{u}$ is equal to the sum of the autocorrelations of $\mathrm{Re}\,u$ and $\mathrm{Im}\,u$: for the same parity their cross-correlations cancel out.

4. The Hadamard formula for the infimum

We now turn to the derivative of the infimum. The minimizer vanishes at the endpoints of the interval (Corollary 3.4). However, this by itself does not yet allow one to find the derivative of λ, especially at the points $c = p^m$, where a new atom enters the form. We compare the minimizers at two close values of c and show that in the Hadamard identity in finite differences the atoms and the pole term contribute only a quantity small compared with the increment of a (Proposition 4.1). It implies the Lipschitz continuity of λ (Proposition 4.2) and a formula for the *boundary flux* $\mathfrak{J} = -\,\mathrm{d}\lambda/\,\mathrm{d}a$ (Theorem 4.3), and via the boundary coefficient from Section 3.3 this flux is computed (Theorem 4.5). At the end of the section the same argument is applied to the logarithmic Laplacian itself (Theorem 4.6), and then the Pohozaev identity on an interval is derived (Section 4.5).

4.1. **The finite-difference Hadamard identity.**

Proposition 4.1 (the finite-difference Hadamard identity)**.** *Let $a < a'$, $c = e^{2a}$, $c' = e^{2a'}$, $\Omega_{a,a'} = \{a < |x| < a'\}$, and $A_{c'}$ be the operator of the atoms $n < c'$.*

(a) *If u and v are normalized real eigenfunctions of Q_c and $Q_{c'}$ of the even sector with eigenvalues λ_u and λ_v, then*

$$\begin{aligned}(\lambda_v - \lambda_u)\langle u, v\rangle = &-\int_{\Omega_{a,a'}} v(x)(\varpi \star u)(x)\,\mathrm{d}x - \int_{\Omega_{a,a'}} v(x)(A_{c'}u)(x)\,\mathrm{d}x \\ &+ 2\langle u, \cosh(\cdot/2)\rangle \int_{\Omega_{a,a'}} v(x)\cosh(x/2)\,\mathrm{d}x,\end{aligned} \tag{4.1}$$

and the contribution of the atoms and the pole term to the right-hand side is at most

$$2(a'-a)\sup_{\Omega_{a,a'}} |v| \cdot (2\sigma_{c'}\|u\|_\infty + 2\|\cosh(\cdot/2)\|_{L^2(I_c)}\cosh(a'/2)).$$

(b) *Let u and v be normalized minimizers at c and c', which we shall call* inner *and* outer*. If the outer minimizer is fixed while $a \to a'$, or the inner one is fixed while $a' \to a$, then the contribution of the atoms and the pole term to the right-hand side of* (4.1) *is $o(a'-a)$, including for $c' = p^m$.*

Proof. (a) We have $Q_{c'}(v,u) = \lambda_v\langle v,u\rangle$ and $Q_{c'}(u,v) = Q_{c'}(u, v\mathbb{1}_{I_c}) + Q_{c'}(u, v\mathbb{1}_{\Omega_{a,a'}})$. The function $v\mathbb{1}_{I_c}$ lies in $D(Q_c)$. The norm $\int \ln(2+|r|)|\hat f|^2\,\mathrm{d}r$ is equivalent to $\|f\|^2 + \mathcal{B}(f)$, and $\mathcal{B}(v\mathbb{1}_{I_c})$ exceeds the contribution of the pairs $x, y \in I_c$, which is at most $\mathcal{B}(v)$, only by the integral of $v(x)^2\int_{|z|\ge d}\varpi = v(x)^2(\ln(1/d) + O(1))$ over $x \in I_c$, where $d = \operatorname{dist}(x, \partial I_c)$. For bounded v (Theorem 3.1) this integral is finite. The first term (the atoms $c \le n < c'$ shift the support of $v\mathbb{1}_{I_c}$ out of I_c and do not enter it) equals $Q_c(u, v\mathbb{1}_{I_c}) = \lambda_u\langle u, v\rangle$. The integrals converge absolutely: in the second one the supports of u and $v\mathbb{1}_{\Omega_{a,a'}}$ do not intersect. The contribution of the energy equals $-\langle \varpi \star u, v\mathbb{1}_{\Omega_{a,a'}}\rangle$, the term with $h_+(0)$ equals zero, the atoms of the form $Q_{c'}$ contribute $-\langle A_{c'}u, v\mathbb{1}_{\Omega_{a,a'}}\rangle$, and the pole term contributes $+2\langle u, \cosh\rangle\langle \cosh, v\mathbb{1}_{\Omega_{a,a'}}\rangle$. Thus (4.1) and the estimate of the contribution of the atoms and the pole term ($\|A_{c'}\|_{\infty\to\infty} \le 2\sigma_{c'}$) are proved.

(b) If the outer minimizer is fixed and $a \to a'$, then $\sup_{\Omega_{a,a'}}|v| \to 0$ by Corollary 3.4, and the norms $\|u\|_\infty$ are uniformly bounded (Theorem 3.1). If the inner one is fixed and $a' \to a$, then the norms $\|v\|_\infty$ are uniformly bounded, since the right-hand side of (3.1) is locally bounded in c and λ, although σ_c has jumps at the points $c = p^m$. Moreover, $\sup_{\Omega_{a,a'}}|v| \to 0$ uniformly in a' by Corollary 3.9 with $\overline{\lambda} = \lambda(c) + 1$. In both cases the estimate of (a) is $o(a'-a)$. □

Proposition 4.2 (Lipschitz continuity of the infimum)**.** *The function $a \mapsto \lambda(e^{2a})$ is Lipschitz on every interval $[\alpha_0, \alpha_1]$, $\alpha_0 > 0$. In particular, it is absolutely continuous, differentiable almost everywhere, and $\lambda(e^{2\alpha_1}) - \lambda(e^{2\alpha_0}) = \int_{\alpha_0}^{\alpha_1}(\mathrm{d}\lambda/\,\mathrm{d}a)\,\mathrm{d}a$.*

Proof. We truncate the minimizer at c' to the interval I_c and compare the values of the forms. The change is concentrated in a boundary layer of width $a'-a$. The boundary estimate for the eigenfunction makes this change a quantity of order $a'-a$. Let $a < a'$, $\varepsilon = a'-a$, v be the normalized minimizer at c' (Proposition 2.1), $\Omega = \Omega_{a,a'}$, $v_I = v\mathbb{1}_{I_c}$, $v_\Omega = v\mathbb{1}_\Omega$. By the monotonicity of λ it suffices to estimate $\lambda(c) - \lambda(c')$ from above. Since the function v_I lies in $D(Q_c)$, we have

$$\lambda(c)(1 - \|v_\Omega\|^2) \le Q_c(v_I) = Q_{c'}(v - v_\Omega) = \lambda(c') - 2\lambda(c')\|v_\Omega\|^2 + Q_{c'}(v_\Omega),$$

so that

$$(\lambda(c) - \lambda(c'))(1 - \|v_\Omega\|^2) \le Q_{c'}(v_\Omega) - \lambda(c')\|v_\Omega\|^2.$$

By Corollary 3.9 with $\overline{\lambda} = \lambda(e^{2\alpha_0}) + 1$ we have $\sup_\Omega|v| \le K(\ln(1/\varepsilon))^{-1/2}$ and $\|v_\Omega\|^2 = O(\varepsilon/\ln(1/\varepsilon))$. We now estimate $Q_{c'}(v_\Omega)$. The terms with $h_+(0)$, the atoms and the pole are of order $O(\|v_\Omega\|_1\|v\|_\infty)$, that is $O(\varepsilon)$, and the energy is written as $\mathcal{B}(v_\Omega) = \mathcal{B}(v, v_\Omega) - \mathcal{B}(v_I, v_\Omega)$. By the equation for v the first term

equals

$$\langle \mu' v + A_{c'} v - \Pi v, v_\Omega \rangle = O(\varepsilon).$$

In the second, the supports do not intersect, and $-\mathcal{B}(v_I, v_\Omega) = \int_\Omega v(\varpi \star v_I)\,\mathrm{d}x$. For $y \in \Omega$ and $s = y - x$ the same estimate gives $|v(x)| \le K\ell(\varepsilon + s)^{1/2}$, and, if the integral is split at the point $s = \varepsilon$,

$$\begin{aligned}(\varpi \star |v_I|)(y) &\le K \int_{y-a}^{1} \ell(\varepsilon + s)^{1/2} (2s)^{-1}\,\mathrm{d}s + K' \\ &= O((\ln(1/\varepsilon))^{1/2} + \ln(\varepsilon/(y-a))(\ln(1/\varepsilon))^{-1/2}).\end{aligned}$$

Multiply this by $|v(y)| \le K(\ln(1/\varepsilon))^{-1/2}$ and integrate over $y \in \Omega$. The integral $\int_a^{a+\varepsilon} \ln(\varepsilon/(y-a))\,\mathrm{d}y$ equals ε, which gives $O(\varepsilon)$. Summing up, $\lambda(c) - \lambda(c') = O(\varepsilon)$ with a constant uniform on $[\alpha_0, \alpha_1]$. □

4.2. **Boundary flux.** Set $\mathfrak{J}(a) = -\,\mathrm{d}\lambda/\,\mathrm{d}a$ wherever the derivative exists, that is, almost everywhere (Proposition 4.2). By u_b we denote the normalized minimizer at $c = e^{2b}$.

Theorem 4.3 (boundary flux of the infimum)**.** *Suppose λ is differentiable at the point a, $c = e^{2a}$. Then there exist $\varepsilon_k \to 0+$ and a minimizer $u_a = \lim_{k\to\infty} u_{a-\varepsilon_k}$ (limit in L^2) at c, for which*

$$\mathfrak{J}(a) = \lim_{k\to\infty} \frac{1}{\varepsilon_k} \int_{a-\varepsilon_k<|x|<a} u_a(x)(\varpi \star u_{a-\varepsilon_k})(x)\,\mathrm{d}x \ge 0. \tag{4.2}$$

The minimizer u_a is called limiting. *If $\lambda(c)$ is simple, the limit is taken over all $\varepsilon \to 0+$, with the sign of $u_{a-\varepsilon}$ chosen so that $\langle u_{a-\varepsilon}, u_a\rangle \ge 0$.*

Proof. We extract from the minimizers $u_{a-\varepsilon}$ a convergent subsequence and pass to the limit in the identity (4.1) for the pair $(a-\varepsilon, a)$. With the outer minimizer fixed, the contribution of the atoms and the pole term is $o(\varepsilon)$ by Proposition 4.1(b). The minimizers $u_{a-\varepsilon}$, extended by zero, lie in $D(Q_c)$ and are bounded in the form norm of Q_c: $Q_c(u_{a-\varepsilon}) = \lambda(e^{2(a-\varepsilon)})$ [5, proof of Proposition 2.6]. This norm, as shown in [5, proof of Lemma 2.2], is equivalent to the quantity $(\int \ln(2+|r|)|\hat f(r)|^2\,\mathrm{d}r)^{1/2}$, and $D(Q_c)$ coincides with the space $H(I_c)$ from [6] (Lemma 3.3). The embedding of $H(I_c)$ into $L^2(I_c)$ is compact [6, (3.3)], cf. also [8, Theorem 3.6], and some sequence $u_{a-\varepsilon_k}$ converges in L^2 and weakly in the norm of the form to a function u_a with $\|u_a\| = 1$. By the lower semicontinuity of the closed form $Q_c(u_a) \le \liminf Q_{c_k}(u_{a-\varepsilon_k}) = \liminf \lambda(c_k) = \lambda(c)$ (the continuity of λ is given by Proposition 4.2), that is, u_a is a minimizer at c, and $\langle u_{a-\varepsilon_k}, u_a\rangle \to 1$. We apply (4.1) to the pair $(a-\varepsilon_k, a)$ with the outer minimizer u_a and divide by ε_k. Since $\langle u_{a-\varepsilon_k}, u_a\rangle \to 1$, the left-hand side differs from the difference quotients of λ itself only by the factor $1 + o(1)$. This yields (4.2), and nonnegativity follows from Proposition 2.1. For simple $\lambda(c)$ every subsequence converges to $\pm u_a$. □

Theorem 4.3 is a rigorous version of Hadamard's formula for the infimum itself. The boundary flux is the interaction of the minimizer u_a with the minimizers at smaller c, written with the kernel of the Archimedean term and containing no atoms. The atoms, both at the points $c = p^m$ and between them, affect λ only through the minimizer itself. We express this flux through β, for which we need the continuity of the minimizer in a wherever $\lambda(c)$ is simple. Starting from Section 8.1, where Hadamard's formula is proved for every minimizer (Corollary 8.2), passage to the limit is no longer needed.

4.3. **Hadamard's formula.** *The simplicity set* $\mathcal{P}$ is the collection of those $a > 0$ for which $\lambda(e^{2a})$ is a simple eigenvalue of Q_c in the even sector, and its elements are *simplicity points*. Since at $a \in \mathcal{P}$ the minimizer is unique up to sign, the limiting minimizer u_a coincides with it.

Lemma 4.4 (continuity on the simplicity set)**.** *The set $\mathcal{P}$ is open. If $a \in \mathcal{P}$, $a_k \to a$ and $\langle u_{a_k}, u_a\rangle \ge 0$, then $u_{a_k} \to u_a$ in L^2. The function $\beta(u_a)^2$, uniquely defined on $\mathcal{P}$, is continuous on $\mathcal{P}$.*

Proof. Openness of $\mathcal{P}$ follows from the continuity of the second eigenvalue, which we verify separately from the right and from the left. Convergence of the minimizers then follows from compactness of the embedding and simplicity.

Continuity of λ_2. We denote by $\lambda_2(c)$ the second eigenvalue of Q_c in the even sector, so that $\mathcal{P} = \{a : \lambda(e^{2a}) < \lambda_2(e^{2a})\}$. Like λ, it is nonincreasing in c: extension by zero embeds the form domains, and the minimax preserves the ordering. Let $a_k \downarrow a$, and e_1^k, e_2^k be the orthonormal eigenfunctions of the first two eigenvalues of the even sector at $c_k = e^{2a_k}$. They are bounded in the form norm. By compactness of the embedding (as in the proof of Theorem 4.3) some subsequence converges in L^2 to an orthonormal pair e_1, e_2 with supports in I_c. For $v = \alpha_1 e_1 + \alpha_2 e_2$ and $v_k = \alpha_1 e_1^k + \alpha_2 e_2^k$ the energy is lower semicontinuous, while the terms with $h_+(0)$, the pole, and the atoms $n < c$ are continuous. The atom $n = c$ at $c = p^m$ enters

through the value of the autocorrelation $(v_k \star \tilde{v}_k)(2a) \to (v \star \tilde{v})(2a) = 0$. Hence $Q_c(v) \le \lim \lambda_2(c_k)\|v\|^2$, and $\lambda_2(c) \le \lim \lambda_2(c_k) \le \lambda_2(c)$. Now let $a_k \uparrow a$, $\varepsilon_k = a - a_k$, and $\mathcal{N}_2$ be the span of the first two eigenfunctions at c. We truncate $v \in \mathcal{N}_2$ to I_{c_k}. We repeat for v the proof of Proposition 4.2, replacing in it c' by c, and c by c_k. Let $v = \alpha_1 u^{(1)} + \alpha_2 u^{(2)}$, where $u^{(1)}$, $u^{(2)}$ are orthonormal eigenfunctions of the eigenvalues $\lambda(c)$ and $\lambda_2(c)$. Then $|\alpha_i| \le \|v\|$ and $Q_c(v) \le \lambda_2(c)\|v\|^2$. The equation enters that proof only through the terms $Q_c(v, v_\Omega)$ and $\mathcal{B}(v, v_\Omega)$, which are linear in v, and for v they equal the sums of the same quantities for $u^{(1)}$ and $u^{(2)}$ with coefficients α_1, α_2. The boundary estimate of Corollary 3.9 at $\overline{\lambda} = \lambda_2(c)$ holds for both functions with a common constant K, and for v with $2K\|v\|$. Therefore in the equality $Q_{c_k}(v\mathbb{1}_{I_{c_k}}) = Q_c(v) - 2Q_c(v, v_\Omega) + Q_c(v_\Omega)$, where $v_\Omega = v\mathbb{1}_{I_c \setminus I_{c_k}}$, the first term is at most $\lambda_2(c)\|v\|^2$, the second equals $-2\alpha_1\lambda(c)\langle u^{(1)}, v_\Omega\rangle - 2\alpha_2\lambda_2(c)\langle u^{(2)}, v_\Omega\rangle$, and the third is estimated like $Q_{c'}(v_\Omega)$ in that proof. The last two are both $O(\varepsilon_k)\|v\|^2$, so $Q_{c_k}(v\mathbb{1}_{I_{c_k}}) \le (\lambda_2(c) + O(\varepsilon_k))\|v\|^2$ and $\|v\mathbb{1}_{I_{c_k}}\|^2 \ge (1 - O(\varepsilon_k))\|v\|^2$. For small ε_k the cutoff is therefore injective on $\mathcal{N}_2$, its image is two-dimensional, and by the minimax $\lambda_2(c_k) \le \lambda_2(c) + O(\varepsilon_k)$, while monotonicity implies the reverse inequality. The function λ_2 is continuous together with λ, and the set $\mathcal{P} = \{\lambda < \lambda_2\}$ is open.

Convergence of the minimizers. Let $a \in \mathcal{P}$. For $a_k \uparrow a$, every subsequence u_{a_k} has a subsequence converging in L^2 to a minimizer at c (proof of Theorem 4.3). For $a_k \downarrow a$, the same is true by the argument of the preceding paragraph with one function instead of two. By simplicity, the limit equals $\pm u_a$, and the condition $\langle u_{a_k}, u_a\rangle \ge 0$ selects u_a. Continuity of $\beta(u_a)^2$ follows from Theorem 3.10. □

Theorem 4.5 (Hadamard formula for the infimum)**.**

(i) *If $\lambda(c)$, $c = e^{2a}$, is a simple eigenvalue of the even sector, then the function $a \mapsto \lambda(e^{2a})$ is differentiable at a, including at $c = p^m$, and*

$$\frac{\mathrm{d}\lambda}{\mathrm{d}a} = -2\beta(u_a)^2, \quad \frac{\mathrm{d}\ln\lambda}{\mathrm{d}a} = -\frac{2\beta(u_a)^2}{\lambda},$$

where the second formula is written for $\lambda > 0$. On the open set $\mathcal{P}$, the function λ is continuously differentiable.

(ii) *At every differentiability point of λ (without the simplicity assumption), $\mathfrak{J}(a) = 2\beta(u_a)^2$ with a limiting minimizer u_a of Theorem 4.3.*

Proof. Substitute the boundary expansions of the two minimizers into (4.1). The convolution of the principal part of the kernel $1/(2|z|)$ with the inner minimizer behaves near the boundary like $\beta(\ln(1/r))^{1/2}$, the outer minimizer equals $\beta(\ln(1/t))^{-1/2}$, and the integral of their product over the boundary layer of width ε is $\varepsilon\beta^2(1 + o(1))$.

(i) *Left derivative.* Let $\varepsilon > 0$ be small and u_ε a normalized minimizer at $e^{2(a-\varepsilon)}$ with sign $\langle u_\varepsilon, u_a\rangle \ge 0$. Apply (4.1) to the pair $(a - \varepsilon, a)$ with the inner minimizer u_ε and the outer u_a. The contribution of the atoms and of the pole term is $o(\varepsilon)$ (Proposition 4.1(b)). By evenness,

$$(\lambda(e^{2a}) - \lambda(e^{2(a-\varepsilon)}))\langle u_\varepsilon, u_a\rangle = -2\int_0^\varepsilon u_a(a-t)(\varpi \star u_\varepsilon)(a-t)\,\mathrm{d}t + o(\varepsilon).$$

The point $a - t$ is at distance $r = \varepsilon - t$ from the edge of the support of u_ε. Write the kernel for $|z| < 1$ in the form $\varpi(z) = 1/(2|z|) + \varpi_r(z)$. The contribution of ϖ_r and $\varpi\mathbb{1}_{\{|z|\ge 1\}}$ does not exceed $C\|u_\varepsilon\|_\infty$. In the remaining integral, make the substitution $y = a - \varepsilon - t'$, under which $|a - t - y| = r + t'$. The interval $t' < r$ is discarded: its contribution does not exceed $\ln 2 \cdot \|u_\varepsilon\|_\infty$. For $t' \ge r$, replace the kernel $1/(r + t')$ by $1/t'$. Since $|1/(r + t') - 1/t'| \le r/t'^2$, this introduces an error not exceeding $\|u_\varepsilon\|_\infty$. From

Corollary 3.9, we obtain

$$(\varpi \star u_\varepsilon)(a-t) = \frac{1}{2}\int_r^1 \frac{u_\varepsilon(a-\varepsilon-t')}{t'}\,\mathrm{d}t' + O(1)$$
$$= \frac{1}{2}U_{u_\varepsilon}\left(\ln\frac{1}{r}\right) + O(1) = \beta(u_\varepsilon)\left(\ln\frac{1}{r}\right)^{1/2} + O(1)$$

uniformly in ε and t. Moreover, $u_a(a-t) = \beta(u_a)(\ln(1/t))^{-1/2} + O((\ln(1/t))^{-1})$ by Theorem 3.8. Set $\xi_\varepsilon = \ln(1/\varepsilon)$ and $t = \varepsilon s$. In these variables,

$$\frac{1}{\varepsilon}\int_0^\varepsilon \left(\frac{\ln(1/(\varepsilon-t))}{\ln(1/t)}\right)^{1/2}\mathrm{d}t = \int_0^1 \left(\frac{\xi_\varepsilon + \ln(1/(1-s))}{\xi_\varepsilon + \ln(1/s)}\right)^{1/2}\mathrm{d}s \to 1$$

by the Lebesgue dominated convergence theorem with the majorant $(1+\ln(1/(1-s)))^{1/2}$ for $\xi_\varepsilon \geq 1$. The products of the principal terms with the remainders contribute $O(\varepsilon\xi_\varepsilon^{-1/2})$. Thus,

$$(\lambda(e^{2a}) - \lambda(e^{2(a-\varepsilon)}))\langle u_\varepsilon, u_a\rangle = -2\varepsilon\beta(u_a)\beta(u_\varepsilon) + o(\varepsilon).$$

By Lemma 4.4, $u_\varepsilon \to u_a$, and therefore $\langle u_\varepsilon, u_a\rangle \to 1$ and $\beta(u_\varepsilon) \to \beta(u_a)$ by Theorem 3.10. The left derivative equals $-2\beta(u_a)^2$.

Right derivative. Take a pair $(a, a+\varepsilon)$ with the inner minimizer u_a and the outer $v_\varepsilon = u_{a+\varepsilon}$, $\langle u_a, v_\varepsilon\rangle \geq 0$. The contribution of the atoms and of the pole term, including the atom $n = c$ at $c = p^m$, is $o(\varepsilon)$ by Proposition 4.1(b). The principal term equals $-2\int_0^\varepsilon v_\varepsilon(a+s)(\varpi \star u_a)(a+s)\,\mathrm{d}s$, where $(\varpi \star u_a)(a+s) = \beta(u_a)(\ln(1/s))^{1/2} + O(1)$ and $v_\varepsilon(a+s) = \beta(v_\varepsilon)(\ln(1/(\varepsilon-s)))^{-1/2} + O((\ln(1/(\varepsilon-s)))^{-1})$ uniformly in ε (Corollary 3.9). This is the same integral with t replaced by $\varepsilon - t$. Hence $(\lambda(e^{2(a+\varepsilon)}) - \lambda(e^{2a}))\langle u_a, v_\varepsilon\rangle = -2\varepsilon\beta(u_a)\beta(v_\varepsilon) + o(\varepsilon)$. Lemma 4.4 and Theorem 3.10 show that the right derivative also equals $-2\beta(u_a)^2$. From the continuity of $\beta(u_a)^2$ on $\mathcal{P}$, continuous differentiability follows.

(ii) At a point where the derivative exists, by Theorem 4.3,

$$\mathfrak{J}(a) = \lim_{k\to\infty} \varepsilon_k^{-1}\int_{a-\varepsilon_k<|x|<a} u_a(\varpi \star u_{a-\varepsilon_k})\,\mathrm{d}x$$

with $u_{a-\varepsilon_k} \to u_a$ in L^2. The computation in (i) shows that this expression equals $2\beta(u_a)\beta(u_{a-\varepsilon_k}) + o(1) \to 2\beta(u_a)^2$ by Theorem 3.10. □

4.4. **The logarithmic Laplacian.** The same reasoning applies to the operator L_Δ itself, which has no atoms or pole term, while a dilation of the interval changes the eigenvalues explicitly. Denote by $\lambda_1^{\mathrm{L}}(a)$ the smallest eigenvalue of the Dirichlet problem for L_Δ on $(-a,a)$, that is, the minimum of $E_{\mathrm{L}}(u,u)/\|u\|^2$ over $u \in H((-a,a))$.

Theorem 4.6 (boundary coefficient of the first eigenfunction of the logarithmic Laplacian)**.** *Let φ_a be the normalized nonnegative eigenfunction for the eigenvalue $\lambda_1^{\mathrm{L}}(a)$. Then φ_a is even, bounded, $\beta_+(\varphi_a) = \beta_-(\varphi_a)$, and*

$$\beta(\varphi_a)^2 = \frac{1}{2a}. \tag{4.3}$$

Proof. Dilation gives $\mathrm{d}\lambda_1^{\mathrm{L}}/\,\mathrm{d}a = -2/a$, the Hadamard formula gives $\mathrm{d}\lambda_1^{\mathrm{L}}/\,\mathrm{d}a = -4\beta(\varphi_a)^2$, and comparing the two expressions yields (4.3). The factor 4 instead of 2 appears because the symbol of L_Δ equals $2\ln|r|$, while the principal part of Q_c is $L_\Delta/2$.

The eigenvalue $\lambda_1^{\mathrm{L}}(a)$ is simple [6, Theorem 1.4(iii)], and φ_a can be chosen positive. The reflection $x \mapsto -x$ maps φ_a to a positive eigenfunction of the same eigenvalue, and by simplicity this function coincides with φ_a, that is, φ_a is even.

Dilation. For $u \in H((-1,1))$ the function $u^{(a)}$ lies in $H((-a,a))$, and $\widehat{u^{(a)}}(r) = a^{1/2}\hat{u}(ar)$. Since the symbol of L_Δ equals $2\ln|r|$ [6, Theorem 1.1(ii)], $E_{\mathrm{L}}(u^{(a)},u^{(a)}) = E_{\mathrm{L}}(u,u) - 2\ln a\cdot\|u\|^2$ (first for smooth u with compact support, then by density [6, Theorem 3.1]). Hence $\lambda_1^{\mathrm{L}}(a) = \lambda_1^{\mathrm{L}}(1) - 2\ln a$ (cf. [16, Lemma 2.5] for $L_\Delta/2$), $\varphi_a = \varphi_1^{(a)}$ and $\mathrm{d}\lambda_1^{\mathrm{L}}/\,\mathrm{d}a = -2/a$.

Boundedness and boundary expansion. The equation for φ_a has the form $\mathcal{B}_1(\varphi_a, v) = \langle(\lambda_1^{\mathrm{L}} + 2\gamma_{\mathrm{E}})\varphi_a + K_{\geq 1}\varphi_a, v\rangle$, where $\mathcal{B}_1$ is the form with kernel $\mathbb{1}_{\{|z|<1\}}/|z|$, and $(K_{\geq 1}u)(x) = \int_{|x-y|\geq 1} u(y)|x-y|^{-1}\,\mathrm{d}y$ for u supported in $(-a,a)$. The operator $K_{\geq 1}$ is positive, and its norms in L^2 and L^∞ do not exceed $2\ln^+(2a)$. The argument of Theorem 3.1, in which ϖ is replaced by the kernel $\mathbb{1}_{\{|z|<1\}}/|z|$, A_c by the operator $K_{\geq 1}$, $2\sigma_c$ by the constant $2\ln^+(2a)$, and where there is no pole term, shows that $\varphi_a \in L^\infty$. The boundary

coefficient $\beta(\varphi_a)$ exists by Theorem 3.8 with $f = \lambda_1^{\mathrm{L}}\varphi_a$, and by the dilation $\beta(\varphi_b)^2 = (a/b)\beta(\varphi_a)^2$ and the constants of the boundary estimate (3.4) for φ_b are uniform in b on any interval $[\alpha_0, \alpha_1]$, $\alpha_0 > 0$.

Hadamard's formula. Let $0 < \varepsilon < a$. The function $\varphi_a \mathbb{1}_{(-a+\varepsilon, a-\varepsilon)}$ lies in $H((-a+\varepsilon, a-\varepsilon))$ (as in the proof of Proposition 4.1). For disjoint supports $E_{\mathrm{L}}(u,v) = -\iint u(y)v(x)|x-y|^{-1}\,\mathrm{d}x\,\mathrm{d}y$: the local and far parts of the form lead to this expression with kernels $\mathbb{1}_{\{|z|<1\}}/|z|$ and $\mathbb{1}_{\{|z|\ge 1\}}/|z|$, and the term with γ_{E} equals zero. Repeating the same proof for the pair $(a-\varepsilon, a)$, we obtain

$$(\lambda_1^{\mathrm{L}}(a) - \lambda_1^{\mathrm{L}}(a-\varepsilon))\langle \varphi_{a-\varepsilon}, \varphi_a\rangle = -\int_{a-\varepsilon<|x|<a} \varphi_a(x)(|\cdot|^{-1} \star \varphi_{a-\varepsilon})(x)\,\mathrm{d}x.$$

For the kernel $1/|z|$, twice as large as the principal part of ϖ, the computation in the proof of Theorem 4.5(i) shows that the right-hand side equals $-4\varepsilon\beta(\varphi_a)\beta(\varphi_{a-\varepsilon}) + o(\varepsilon)$. Passing to the limit ($\varphi_{a-\varepsilon} \to \varphi_a$ in L^2 and $\beta(\varphi_{a-\varepsilon}) \to \beta(\varphi_a)$ by dilation), we arrive at $\mathrm{d}\lambda_1^{\mathrm{L}}/\,\mathrm{d}a = -4\beta(\varphi_a)^2$. Together with $\mathrm{d}\lambda_1^{\mathrm{L}}/\,\mathrm{d}a = -2/a$ this proves (4.3). □

4.5. **The Pohozaev identity.** For the fractional Laplacian this identity was proved by Ros-Oton and Serra [18], and for the logarithmic Laplacian it was stated in [1, Theorem 2.3] for bounded domains with boundary of class $C^{1,1}$ in $\mathbb{R}^N$, $N \ge 1$, under the assumption that at the boundary the limit $u^2\ln(\delta^{-2})$ exists, but the preprint was withdrawn by the authors. The difficulty is that near the boundary the solution is not smooth and at distance δ from the boundary it behaves like $\beta(\ln(1/\delta))^{-1/2}$, so that the boundary term is lost under approximation by smooth functions. We circumvent it by means of trial functions obtained by dilation, and only the boundary expansion of Theorem 3.8 is needed. The existence of the limit is not assumed in this argument but is part of its statement.

Let $\Omega = (y_-, y_+)$, $H(\Omega)$ be the space of [6] on Ω, $\lambda_1^{\mathrm{L}}(\Omega)$ be the smallest eigenvalue of L_Δ on Ω, and $d(x) = \operatorname{dist}(x, \mathbb{R}\setminus\Omega)$. We denote by $\beta_{y_+}(\varphi) = \lim_{\delta\to 0+}\varphi(y_+ - \delta)(\ln(1/\delta))^{1/2}$ and $\beta_{y_-}(\varphi) = \lim_{\delta\to 0+}\varphi(y_- + \delta)(\ln(1/\delta))^{1/2}$ the boundary coefficients of the function φ at the endpoints y_+ and y_-, if the limits exist. Theorem 3.8 applies to Ω without any changes: a shift takes Ω to $(-a, a)$, $a = (y_+ - y_-)/2$.

Lemma 4.7 (estimates for the solution of the resolvent problem)**.** *Let $\mu_0 < \lambda_1^{\mathrm{L}}(\Omega)$, $G \in L^\infty(\Omega)$ and $z \in H(\Omega)$ be a weak solution of the Dirichlet problem $(L_\Delta - \mu_0)z = G$. Then z is continuous, the boundary coefficients $\beta_{y_-}(z)$ and $\beta_{y_+}(z)$ exist, and*

$$|z(x)| \le C\|G\|_\infty \ell(d(x))^{1/2}, \quad |\beta_{y_-}(z)| + |\beta_{y_+}(z)| \le C\|G\|_\infty,$$

where C depends only on μ_0 and on the compact sets containing $y_+ - y_-$ and $\lambda_1^{\mathrm{L}}(\Omega) - \mu_0 > 0$.

Proof. Testing the equation with z itself, we find $(\lambda_1^{\mathrm{L}}(\Omega) - \mu_0)\|z\|^2 \le \langle G, z\rangle$, and hence $\|z\| \le \|G\|/(\lambda_1^{\mathrm{L}}(\Omega) - \mu_0)$. The formula of Lemma 3.5 is valid for all u, $\varphi \in H(\Omega)$: all its integrals converge absolutely. Applying it at small $r = \delta$ and repeating the argument of Theorem 3.1 with the cut-offs $(z-t)_+$ and $(-z-t)_+$ (there are no atoms and no pole term here), we obtain $\|z\|_\infty \le C(\|z\| + \|G\|_\infty)$. For bounded $L_\Delta z = G + \mu_0 z$, Theorem 3.8 yields the existence of β_{y_-}, β_{y_+} and the estimate (3.4) with an absolute constant. For $\delta_0 = \min(1, (y_+ - y_-)/2)/16$ it implies $|\beta_{y_+}(z)| \le C(\|z\|_\infty + \|G\|_\infty)$, and the same for β_{y_-}, while for $d(x) > \delta_0$ we have $|z(x)| \le \|z\|_\infty$ and $\ell(d(x)) \ge \ell(\delta_0)$. Continuity follows from [6, Theorem 1.11]. □

Theorem 4.8 (Pohozaev identity for the logarithmic Laplacian)**.** *Let $\mu_0 \in \mathbb{R}$, $f \in C(\overline{\Omega}) \cap W^{1,1}(\Omega)$ and $z \in H(\Omega)$ be the weak solution of the Dirichlet problem $(L_\Delta - \mu_0)z = f$. Then z is continuous, the boundary coefficients $\beta_{y_-}(z)$ and $\beta_{y_+}(z)$ exist, and*

$$\begin{aligned} 2(y_+ - y_-)\beta_{y_+}(z)^2 &= 2\|z\|^2 + \langle f, z\rangle + 2\langle z, (x-y_-)f'\rangle, \\ 2(y_+ - y_-)\beta_{y_-}(z)^2 &= 2\|z\|^2 + \langle f, z\rangle + 2\langle z, (x-y_+)f'\rangle. \end{aligned} \tag{4.4}$$

The left-hand sides of (4.4) correspond to the boundary term, which for the fractional Laplacian $(-\Delta)^s$ equals the square of the limit z/d^s at the boundary [18], and here is expressed in terms of the boundary coefficient. For $N = 1$ the statement [1, Theorem 2.3] with right-hand side $g(x) + \mu_0 u$ is the sum of the two equalities (4.4) with weights $y_+/(y_+ - y_-)$ and $-y_-/(y_+ - y_-)$, so that (4.4) is stronger.

Proof. Both sides of the first equality (4.4) express the flux through the layer near the right endpoint arising under dilation of the solution. The equation, tested on the dilated function, gives an exact identity at each $s > 1$. Divided by $s - 1$, it passes as $s \to 1+$ to (4.4), and for the limit on the left the boundary expansion suffices. Minimizers and problems on other intervals are not needed, therefore no condition on μ_0 arises. The second equality follows from the first after the reflection $x \mapsto y_- + y_+ - x$. It suffices to consider the right endpoint.

Boundedness and boundary expansion. The argument with the cutoff functions $(z-t)_+$ and $(-z-t)_+$ for small r from the proof of Lemma 4.7 does not use the inequality $\mu_0 < \lambda_1^{\mathrm{L}}(\Omega)$ and gives $\|z\|_\infty \le C(\|z\| + \|f\|_\infty)$. The continuity of z follows from [6, Theorem 1.11]. For bounded $L_\Delta z = f + \mu_0 z$ by Theorem 3.8 the boundary coefficients exist and $z(y_+ - \tau) = \beta \ell(\tau)^{1/2} + O(\ell(\tau))$, where $\beta = \beta_{y_+}(z)$. In particular, $z(y_+ - 0) = 0$.

Dilation identity. For $s > 0$ put $(\mathcal{R}_s\psi)(x) = s^{-1/2}\psi(y_- + (x - y_-)/s)$. This is a dilation centered at y_-, not at zero as $u^{(s)}$. The operator $\mathcal{R}_s$ maps $H((y_-, y_+))$ to $H((y_-, y_- + s(y_+ - y_-)))$, preserves the norm, and $\mathcal{R}_s^{-1} = \mathcal{R}_{1/s}$. Since the symbol $2\ln|r|$ changes under dilation by a constant, $E_{\mathrm{L}}(\mathcal{R}_s\psi, \mathcal{R}_s\chi) = E_{\mathrm{L}}(\psi, \chi) - 2\ln s \langle \psi, \chi\rangle$. Let $s > 1$ and $y_s = y_- + s(y_+ - y_-)$. Set $\varphi_1 = \mathcal{R}_{1/s} z$, $\varphi_2 = (\mathcal{R}_s z)\mathbb{1}_\Omega$ and $\varphi_3 = (\mathcal{R}_s z)\mathbb{1}_{(y_+, y_s)}$, and then $\mathcal{R}_s z = \varphi_2 + \varphi_3$. Here $\varphi_1 \in H((y_-, y_- + (y_+ - y_-)/s)) \subset H(\Omega)$. The inclusion $\varphi_2 \in H(\Omega)$ follows from boundedness: the change of the norm under truncation is estimated via $\int_\Omega \varphi_2^2 \ln(1/\operatorname{dist}(x, \partial\Omega))\,\mathrm{d}x$, as in the proof of Proposition 4.1. We test the equation on φ_2 and on φ_1 and subtract the resulting equalities. The terms with μ_0 cancel: $z = 0$ outside Ω, therefore $\langle z, \varphi_2\rangle = \langle z, \mathcal{R}_s z\rangle = \langle \mathcal{R}_{1/s} z, z\rangle = \langle z, \varphi_1\rangle$. By covariance $E_{\mathrm{L}}(z, \mathcal{R}_s z) = E_{\mathrm{L}}(\varphi_1, z) - 2\ln s \langle \varphi_1, z\rangle$, and on functions with disjoint supports $E_{\mathrm{L}}(\varphi, \psi) = -\iint \varphi(x)\psi(y)|x - y|^{-1}\,\mathrm{d}x\,\mathrm{d}y$. Hence

$$\int_{y_+}^{y_s} \varphi_3(y) P(y)\,\mathrm{d}y = \langle f, \varphi_2 - \varphi_1\rangle + 2\ln s \langle z, \varphi_1\rangle, \quad P(y) = \int_\Omega \frac{z(x)}{y - x}\,\mathrm{d}x.$$

Limit of the left-hand side. Set $\eta = y_s - y_+ = (s-1)(y_+ - y_-)$. From the boundary expansion for $y = y_+ + \sigma$

$$P(y) = 2\beta(\ln(1/\sigma))^{1/2} + o((\ln(1/\sigma))^{1/2}), \quad \sigma \to 0,$$

and $\varphi_3(y_s - \tau) = s^{-1/2} z(y_+ - \tau/s) = \beta \ell(\tau)^{1/2} + O(\ell(\tau) + (s-1))$ uniformly for $1 < s \le 2$, since $\ell(\tau/s)/\ell(\tau) \to 1$. Therefore as $s \to 1+$

$$\frac{1}{\eta}\int_{y_+}^{y_s} \varphi_3 P = \frac{2\beta^2}{\eta}\int_0^\eta \Big(\frac{\ln(1/\sigma)}{\ln(1/(\eta - \sigma))}\Big)^{1/2}\,\mathrm{d}\sigma + o(1),$$

and the last average tends to 1 (the substitution $\sigma = \eta t$ and the majorant $1 + (\ln(1/t))^{1/2}$). Thus, after division by $s - 1$ the left-hand side converges to $2(y_+ - y_-)\beta^2$.

Limit of the right-hand side. Here $2\ln s/(s-1) \to 2$ and $\langle z, \varphi_1\rangle \to \|z\|^2$ by the continuity of dilations in L^2. By a change of variable $\int_\Omega f \mathcal{R}_\sigma z = \sigma^{1/2} T(\sigma)$, where $T(\sigma) = \int_\Omega z(y)(\mathbb{1}_\Omega f)(y_- + \sigma(y - y_-))\,\mathrm{d}y$. Therefore $\langle f, \varphi_2 - \varphi_1\rangle = s^{1/2}T(s) - s^{-1/2}T(1/s)$, and the derivative of this quantity at $s = 1$ equals $T(1) + 2T'(1)$, with $T(1) = \langle f, z\rangle$. Substituting $f(x) = f(y_-) + \int_{y_-}^x f'$ and applying Fubini's theorem, we obtain

$$T(\sigma) = f(y_-) Z(p_\sigma(y_+)) + \int_\Omega f'(t)(Z(p_\sigma(y_+)) - Z(p_\sigma(t)))\,\mathrm{d}t, \quad Z(x) = \int_{y_-}^x z, \quad p_\sigma(x) = y_- + \frac{x - y_-}{\sigma}.$$

The antiderivative Z is Lipschitz, and $Z' = z$ on Ω, and also in y_+, since $z(y_+ - 0) = 0$. The difference quotients in σ of the quantities $Z(p_\sigma(t))$ are bounded, and by Lebesgue's theorem $T'(1) = \langle z, (x - y_-)f'\rangle$. The jump of $\mathbb{1}_\Omega f$ at y_+ makes no contribution. Thus, after division by $s - 1$ the right-hand side converges to $2\|z\|^2 + \langle f, z\rangle + 2\langle z, (x - y_-)f'\rangle$. Comparing the two limits gives the first equality (4.4). □

Corollary 4.9 (boundary coefficient of the eigenfunctions of the logarithmic Laplacian)**.** *Every real normalized eigenfunction φ of the operator L_Δ on $(-a, a)$ satisfies $\beta_+(\varphi)^2 = \beta_-(\varphi)^2 = 1/(2a)$.*

Proof. Let $L_\Delta \varphi = \lambda^{\mathrm{L}}\varphi$. Apply Theorem 4.8 to $z = \varphi$ with $\mu_0 = \lambda^{\mathrm{L}}$, $f = 0$, $y_- = -a$, $y_+ = a$. The right-hand side of the first equality (4.4) equals $2\|\varphi\|^2 = 2$, whence $4a\beta_+(\varphi)^2 = 2$. For the left endpoint we likewise obtain $4a\beta_-(\varphi)^2 = 2$. □

Corollary 4.10 (simplicity of the spectrum on an interval)**.** *All eigenvalues of the Dirichlet problem for L_Δ on $(-a, a)$ are simple.*

Proof. The real and imaginary parts of an eigenfunction correspond to the same eigenvalue, so it suffices to consider the real case. Let φ_1 and φ_2 be orthogonal normalized real eigenfunctions of the same eigenvalue and $\alpha_i = \beta_+(\varphi_i)$. By Corollary 4.9 $\alpha_i^2 = 1/(2a)$. The function $\psi = (\alpha_2\varphi_1 - \alpha_1\varphi_2)/(\alpha_1^2 + \alpha_2^2)^{1/2}$ is normalized and corresponds to the same eigenvalue. The boundary coefficient is linear, hence $\beta_+(\psi) = (\alpha_2\alpha_1 - \alpha_1\alpha_2)/(\alpha_1^2 + \alpha_2^2)^{1/2} = 0$, which contradicts this corollary applied to ψ. □

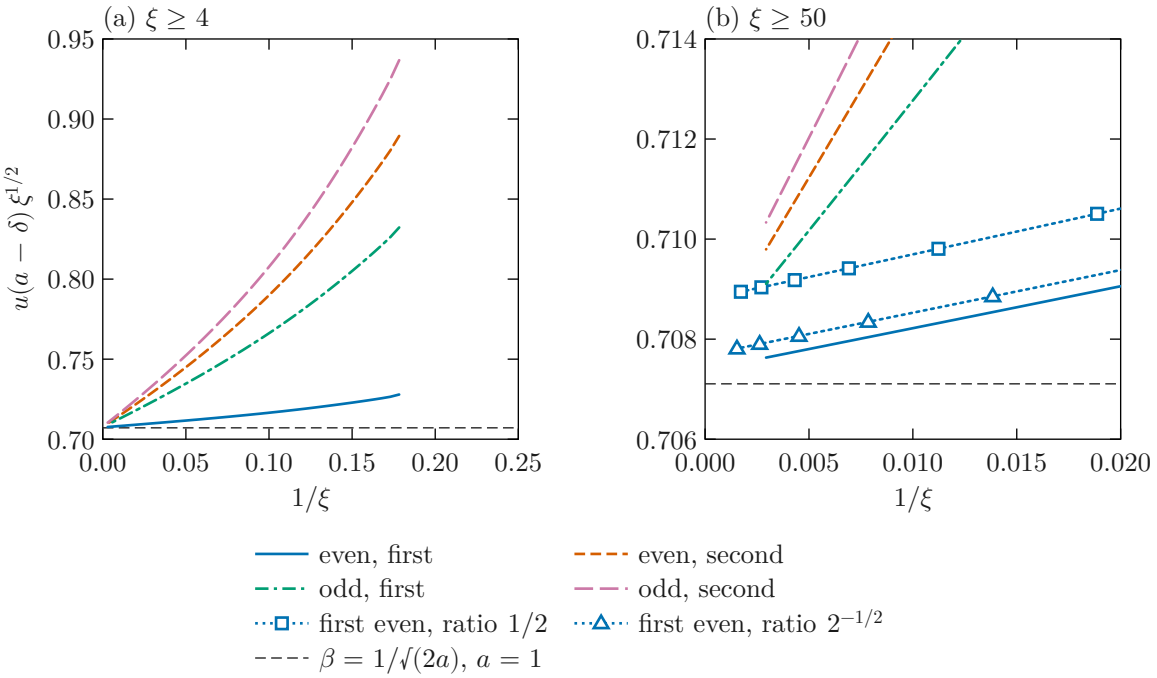


FIGURE 1. Boundary expansion of eigenfunctions of the logarithmic Laplacian on $(-1,1)$ (numerical, the method is described in Section A.4): the quantity $u(a-\delta)\xi^{1/2}$, $\xi = \ln(1/\delta)$, for the first two functions of each sector for (a) $\xi \geq 4$ and (b) $\xi \geq 50$, where the first even function is shown on three grids and the others on the finest one. The dashed horizontal line marks $\beta = 1/\sqrt{2a}$ (Theorem 4.6 and Corollary 4.9).

The boundary expansion is also visible in the eigenfunctions of L_Δ computed numerically (Figure 1). For the first even function the value $1/(2a) = 1/2$ is proved in Theorem 4.6, and for the other three in Corollary 4.9.

For the principal eigenfunction Corollary 4.9 repeats Theorem 4.6, proved above independently, from dilation and the Hadamard formula. The identity (4.4) is obtained only on an interval: in $\mathbb{R}^N$ the same argument requires a boundary expansion along the whole boundary with uniform constants.

5. Scale and boundary balance

The Hadamard formula turns every upper bound for the boundary coefficient, expressed in terms of the infimum itself, into a lower bound for λ. We first consider this transformation in general form, then study the scale Θ_c and introduce the boundary balance, that is, the ratio of the logarithmic decay rates of λ and Θ_c. All positivity arguments on the whole half-line rely on the following lemma.

5.1. **Continuation by continuity.**

Lemma 5.1 (continuity argument)**.** *Let $\lambda(5) > 0$, as proved in [5, Theorem 1.1], and let the function $\underline{\lambda} : [a_0, \infty) \to (0, \infty)$ be continuous and such that for every $b > a_0$ with $\lambda(e^{2a}) > 0$ at $a \in [a_0, b]$, one has $\lambda(e^{2b}) \geq \underline{\lambda}(b)$. Then $\lambda(c) > 0$ for all $c \geq 5$ and $\lambda(e^{2b}) \geq \underline{\lambda}(b)$ for $b > a_0$.*

Proof. Let a^* be the supremum of those $b \geq a_0$ for which $\lambda(e^{2a}) > 0$ for $a \in [a_0, b]$. From $\lambda(5) > 0$ [5, Theorem 1.1] and continuity of the function $a \mapsto \lambda(e^{2a})$ (Proposition 4.2) it follows that $a^* > a_0$. Suppose that $a^* < \infty$. For $a_0 < b < a^*$ we have $\lambda(e^{2b}) \geq \underline{\lambda}(b)$. By continuity of both sides $\lambda(e^{2a^*}) \geq \underline{\lambda}(a^*) > 0$. Then $\lambda > 0$ also on some interval to the right of a^*, which contradicts the definition of a^*. Hence $a^* = \infty$. □

Proposition 5.2 (sufficient boundary condition)**.** *Let $C \in L^1_{\mathrm{loc}}([a_0,\infty))$ and for almost all $a > a_0$ the inequality*

$$2\beta(u)^2 \le C(a)\lambda(e^{2a})$$

hold for every normalized minimizer u at $c = e^{2a}$. Then without any sign assumption on λ, for every $b > a_0$ one has $\lambda(e^{2b}) \ge \lambda(5)\exp(-\int_{a_0}^{b} C) > 0$, since $\lambda(5) > 0$ [5, Theorem 1.1].

Proof. Let $b > a_0$. Here $-\mathrm{d}\lambda/\mathrm{d}a = 2\beta(u_a)^2$ by Theorem 4.5(ii), where u_a is the limiting minimizer of Theorem 4.3. By assumption $\mathrm{d}\lambda/\mathrm{d}a + C\lambda \ge 0$ almost everywhere on $[a_0, b]$. The function $a \mapsto \lambda(e^{2a})$ is Lipschitz on $[a_0, b]$ (Proposition 4.2), and $a \mapsto \exp\int_{a_0}^{a} C$ is absolutely continuous. Their product is absolutely continuous, and its derivative $(\mathrm{d}\lambda/\mathrm{d}a + C\lambda)\exp\int_{a_0}^{a} C$ is almost everywhere nonnegative. Therefore it is nondecreasing, and

$$\lambda(e^{2b}) \ge \lambda(5)\exp\left(-\int_{a_0}^{b} C\right) > 0,$$

since $\lambda(5) > 0$ [5, Theorem 1.1]. □

Theorem 1.3 follows from the same argument for $C = 4\rho\kappa_\Theta$, if it is applied only on intervals where $\lambda > 0$, and then combined with Lemma 5.1 (Section 6).

5.2. **Scale.** The properties of the scale itself are collected in the following lemma.

Lemma 5.3 (exact expression and asymptotics of the scale)**.** *For all $c > 1$ one has $\Phi(a) > 0$ and $\kappa_\Theta > 0$. For $c \ge 5$, moreover, $0 < \Theta_c < 1$ and*

$$\Theta_c = 256\pi^8 c^5 e^{-4\pi c} R_\Theta(c)(1+\Delta(c))^4, \quad R_\Theta(c) = \left(\frac{1-3/(2\pi c)}{1-9/(4\pi c)}\right)^4, \tag{5.1}$$

where

$$\Delta(c) = \sum_{n\ge 2} n^4 \,\frac{1-3/(2\pi n^2 c)}{1-3/(2\pi c)}\, e^{-\pi(n^2-1)c},$$

and $|\Delta(c)| < 10^{-19}$. The function R_Θ decreases from $R_\Theta(5) < 1.2423$ to 1. Moreover, $R_\Theta(c) = 1 + 3/(\pi c) + O(c^{-2})$. For $c \to \infty$

$$\ln\frac{1}{\Theta_c} = 4\pi c - 5\ln c - 4\ln(4\pi^2) - \frac{3}{\pi c} + O(c^{-2}), \quad \kappa_\Theta = 2\pi c - \frac{5}{2} + \frac{3}{2\pi c} + O(c^{-2}). \tag{5.2}$$

Proof. By (2.2) for $x = a$, $e^{2a} = c$, $\Phi(a) = 4\pi^2 c^{9/4} e^{-\pi c}(1-3/(2\pi c))(1+\Delta(c))$. The series for Δ admits termwise differentiation, with $\Delta = O(e^{-3\pi c})$, $c\Delta' = O(ce^{-3\pi c})$. Since for $c \ge 5$ the factor $1 - 3/(2\pi c)$ is positive and $|\Delta| < 10^{-19}$, we have $\Phi(a) > 0$. Substituting $\Phi(a)$ into the definition of $\Theta_c = (2\pi\Phi(a)/\kappa)^4$, we obtain (5.1). The base of the power in $R_\Theta(c)$ equals $1 + (3/(4\pi c))/(1 - 9/(4\pi c))$ and decreases in c. Since $\mathrm{d}/\mathrm{d}a = 2c\,\mathrm{d}/\mathrm{d}c$,

$$\nu = -\frac{\mathrm{d}\ln\Phi(a)}{\mathrm{d}a} = 2\pi c - \frac{9}{2} - \frac{3/(\pi c)}{1-3/(2\pi c)} + O(c^2 e^{-3\pi c}), \quad \frac{4\pi c}{\kappa} = \frac{2}{1-9/(4\pi c)} = 2 + \frac{9}{2\pi c} + O(c^{-2}),$$

and from $\kappa_\Theta = \nu + 4\pi c/\kappa$ the second formula (5.2) follows. By the formula for ν for $c \ge 5$ one has $\nu \ge 2\pi c - 5 > 0$, therefore $\Phi(a)$ decreases, while κ increases, and $2\pi\Phi(a)/\kappa \le 2\pi\Phi(a_0)/\kappa(5) < 4.7\cdot 10^{-5}$, that is, $0 < \Theta_c < 1$. Next, $\ln(1/\Theta_c) = -4\ln(2\pi\Phi(a)) + 4\ln\kappa$ and $\ln\kappa = \ln(2\pi c) + \ln(1-9/(4\pi c))$, whence

$$\ln\frac{1}{\Theta_c} = 4\pi c - 5\ln c - 4\ln(4\pi^2) - 4\ln\left(1-\frac{3}{2\pi c}\right) + 4\ln\left(1-\frac{9}{4\pi c}\right) + O(e^{-3\pi c}),$$

and, expanding the logarithms, we obtain the first formula (5.2).

Positivity for all $c > 1$. For $y = e^{2a} = c$ the term of the series (2.2) with index n equals $t_n = 2\pi n^2 y^{5/4}(2\pi n^2 y - 3)e^{-\pi n^2 y}$. For $y \ge 1$ all t_n are positive, hence $\Phi(a) > 0$. Next, $\mathrm{d}t_n/\mathrm{d}a = 2y\,\mathrm{d}t_n/\mathrm{d}y$ and

$$\frac{\mathrm{d}t_n}{\mathrm{d}y} = 2\pi n^2 y^{1/4} e^{-\pi n^2 y}\big(-2\pi^2 n^4 y^2 + 7.5\pi n^2 y - 3.75\big).$$

Set $z = \pi n^2 y$. Then the bracket equals $-(2z^2 - 7.5z + 3.75)$. For $n \ge 2$ we have $z \ge 4\pi$, and the roots of this polynomial are less than 3.2, so that $\mathrm{d}t_n/\mathrm{d}a \le 0$. For $n = 1$ the inequality $\mathrm{d}t_1/\mathrm{d}a \le 2t_1$ is equivalent to $8z^2 - 22z + 3 \ge 0$, and the roots of this polynomial are less than $2.7 < \pi \le z$. The series admits termwise differentiation, hence $\Phi'(a) \le 2t_1 \le 2\Phi(a)$ and $\nu \ge -2$. Finally, $\kappa = 2\pi c - 9/2 > 0$ for $c > 1$ and $4\pi c/\kappa > 2$, so that $\kappa_\Theta = \nu + 4\pi c/\kappa > 0$ for all $c > 1$. □

The proof explains the constant $9/2$ in κ. The leading term of $\Phi(a)$ equals $4\pi^2 c^{9/4} e^{-\pi c}$, and $\kappa = -\,\mathrm{d}\ln(c^{9/4}e^{-\pi c})/\,\mathrm{d}a$ is the rate of its logarithmic decay, so that $\Theta_c^{1/4}$ is $2\pi\Phi(a)$ divided by it.

For the positivity of λ it suffices to compare κ_Θ with $2\pi c$: in the hypothesis of Theorem 1.3 the scale enters only through κ_Θ.

Lemma 5.4 (comparison of κ_Θ with $2\pi c$)**.** *For $c \geq 5$*

$$0.924 \leq \frac{\kappa_\Theta}{2\pi c} < 1. \tag{5.3}$$

Proof. By the proof of Lemma 5.3 $\kappa_\Theta = \nu + 4\pi c/\kappa$, where

$$\nu = 2\pi c - 9/2 - (3/(\pi c))(1 - 3/(2\pi c))^{-1} - 2c\Delta'/(1+\Delta),$$

and

$$1 - \frac{\kappa_\Theta}{2\pi c} = \frac{5\pi c - 81/4}{4\pi c(\pi c - 9/4)} + \frac{3/(2\pi^2 c^2)}{1 - 3/(2\pi c)} + \frac{\Delta'}{\pi(1+\Delta)}.$$

For $c \geq 5$ the first two summands are positive and decrease in c, and for $c = 5$ their sum is less than 0.07566. For the first one this is seen from the expression $(5\pi c - 81/4)/(4\pi^2 c^2 - 9\pi c)$: the numerator of its derivative equals $-20\pi^3 c^2 + 162\pi^2 c - 182.25\pi < 0$. Differentiating the series for Δ termwise, we obtain $|c\Delta'| \leq 170 c e^{-3\pi c} < 3 \cdot 10^{-18}$ (the term with $n = 2$ in $|\Delta'|$ is not greater than $167e^{-3\pi c}$, and the sum of the remaining ones is smaller than it by a factor of $e^{5\pi c}/14$). The last term does not exceed $55e^{-3\pi c}$ in absolute value and is negligibly small compared with the first one, which is not less than $0.34/c$. □

Let us compare the scale with the quantity $d_c^{\mathrm{K}} = (2^{14}/3)\sqrt{2}\pi^5 c^{9/2} e^{-4\pi c}$, that is, with the Fuchs asymptotics [11] for the quantity $1 - \chi_2$ of the band-limited signals problem in Connes's notation (χ_2^2 is the eigenvalue with index 4). With this asymptotics Connes [7, §6.4] compares the computed smallest eigenvalue of the form, and from (5.1) we find

$$\frac{\Theta_c}{d_c^{\mathrm{K}}} = \frac{3\pi^3}{64\sqrt{2}} \sqrt{c} R_\Theta(c)(1 + \Delta(c))^4,$$

so that the exponential factor of Θ_c and d_c^{K} is one and the same, while their ratio grows as $\sqrt{c}$. For the infimum itself the order $e^{-4\pi c}$ has been established only numerically and by rigorous estimates at individual c [5], and a comparison with these values is given in Section A.2.

Lemma 5.5 (the scale on the unit interval)**.** *For $c \geq 5$ and $0 \leq s \leq 1$ one has $\Theta_{c+s} \geq C_\Theta \Theta_c$, where κ_c is the quantity κ for the given c and*

$$C_\Theta = e^{-4\pi}(\kappa_5/\kappa_6)^4 (1 + 10^{-19})^{-4} > 1.50 \cdot 10^{-6}.$$

Proof. For $a = (\ln c)/2$ the series (2.2) is $\Phi(a) = \sum_{n \geq 1} t_n(c)$, where $t_n(c) = 2(2\pi^2 n^4 c^{9/4} - 3\pi n^2 c^{5/4}) e^{-\pi n^2 c}$, and all $t_n > 0$, since $2\pi n^2 c > 3$. The factor $T(c) = 2\pi^2 c^{9/4} - 3\pi c^{5/4}$ increases for $c \geq 1$: $T'(c) = c^{1/4}(9\pi^2 c/2 - 15\pi/4) > 0$, and $t_1(c+s) \geq e^{-\pi s} t_1(c)$. Next, $t_n/t_1 \leq n^4 (1 - 3/(2\pi c))^{-1} e^{-\pi(n^2-1)c}$. For $c \geq 5$ the sum of these ratios over $n \geq 2$ is less than 10^{-19}. As a result

$$\Phi(a_{c+s}) \geq t_1(c+s) \geq e^{-\pi s} t_1(c) \geq e^{-\pi s} \Phi(a_c)(1 + 10^{-19})^{-1}.$$

Finally, $\kappa_{c+s}/\kappa_c \leq 1 + 2\pi/\kappa_c \leq \kappa_6/\kappa_5$. It remains to raise the ratio $2\pi\Phi(a)/\kappa_c$ to the fourth power. □

5.3. **Boundary balance and its means.** The quantity

$$r_c = \frac{\beta(u_a)^2}{2\kappa_\Theta \lambda(c)},$$

where u_a is the limiting minimizer of Theorem 4.5(ii), compares the rates of decay of λ and of the scale. The minimizer u_a depends on the choice of the sequence ε_k in Theorem 4.3. However, at every point where λ is differentiable, by part (ii) of the Hadamard formula $\beta(u_a)^2 = -(1/2)\,\mathrm{d}\lambda/\,\mathrm{d}a$, and this value does not depend on the choice. By Corollary 8.2 the same value is attained by $\beta(u)^2$ for any normalized u from the eigenspace, hence r_c is defined unambiguously at all points where $\lambda \neq 0$, except on an at most countable set.

Proposition 5.6 (boundary balance and scale law)**.**

(a) *If $\lambda > 0$ on an interval, then almost everywhere on it*

$$\frac{\mathrm{d}\ln\vartheta}{\mathrm{d}a} = 4\kappa_\Theta(1 - r_c), \quad \vartheta = \lambda/\Theta_c,$$

where we call ϑ the level *of the infimum.*

(b) *The scale law* $\lambda(c) \ge \vartheta_* \Theta_c$ *for all* $c \ge 5$ *with constant* $\vartheta_* > 0$ *is equivalent to the existence of a number* M *for which*

$$\int_{a_0}^{a} 4\kappa_\Theta (r_c - 1)\,\mathrm{d}a' \le M \tag{5.4}$$

for every $a > a_0$ *such that* $\lambda(e^{2a'}) > 0$ *whenever* $a' \in [a_0, a]$. *Moreover, one can take* $\vartheta_* = \vartheta(5)e^{-M}$.

Proof. (a) On an interval where $\lambda > 0$, the function $\ln\lambda$ is absolutely continuous (Proposition 4.2). Almost everywhere $\mathrm{d}\ln\lambda/\,\mathrm{d}a = -2\beta(u_a)^2/\lambda = -4\kappa_\Theta r_c$ by Theorem 4.5(ii), and $\mathrm{d}\ln\Theta_c/\,\mathrm{d}a = -4\kappa_\Theta$ by definition.

(b) Suppose (5.4) holds and $\lambda(e^{2a}) > 0$ for $a \in [a_0, b]$. By (a) $\ln\vartheta(e^{2b}) = \ln\vartheta(5) - \int_{a_0}^{b} 4\kappa_\Theta(r_c - 1)\,\mathrm{d}a \ge \ln\vartheta(5) - M$, that is, $\lambda(e^{2b}) \ge \vartheta(5)e^{-M}\Theta_{e^{2b}}$. Applying Lemma 5.1 with $\underline{\lambda}(b) = \vartheta(5)e^{-M}\Theta_{e^{2b}}$, we find that the scale law holds for all $c \ge 5$ with $\vartheta_* = \vartheta(5)e^{-M}$. Conversely, under the scale law $\lambda > 0$ everywhere. By (a) the left-hand side of (5.4) equals $\ln\vartheta(5) - \ln\vartheta(c) \le \ln(\vartheta(5)/\vartheta_*)$. □

Part (b) is the scale law in integral form: the pointwise estimate $r_c \le 1 + d(a)$ implies (5.4) with $M = \int_{a_0}^{\infty} 4\kappa_\Theta d^+\,\mathrm{d}a$, provided this integral is finite. In other words, r_c may exceed 1 if the excess is integrable with respect to the measure $-\,\mathrm{d}\ln\Theta_c$.

What r_c depends on. By Proposition 5.6(a)

$$r_c - 1 = -\frac{1}{4\kappa_\Theta}\frac{\mathrm{d}\ln\vartheta}{\mathrm{d}a} = -\frac{c}{2\kappa_\Theta}\frac{\mathrm{d}\ln\vartheta}{\mathrm{d}c},$$

since $\mathrm{d}/\,\mathrm{d}a = 2c\,\mathrm{d}/\,\mathrm{d}c$. Here $2\kappa_\Theta/c = 4\pi - 5/c + O(c^{-2})$ by (5.2). The unit term in r_c accounts for the decay of the scale itself, and the deviation from it for the change of the level $\vartheta = \lambda/\Theta_c$. Even rigorous estimates of individual values of ϑ do not allow one to find r_c pointwise, since for that a lower bound for the derivative $\mathrm{d}\ln\vartheta/\,\mathrm{d}a$ is needed.

Corollary 5.7 (mean of the boundary balance)**.** *Let* $5 \le c_1 < c_2$, $a_i = (\ln c_i)/2$ *and* $\lambda > 0$ *on* $[c_1, c_2]$. *The mean of* r_c *with weight* $4\kappa_\Theta\,\mathrm{d}a = -\,\mathrm{d}\ln\Theta_c$ *equals*

$$\langle r\rangle_{[c_1,c_2]} \coloneqq \frac{\int_{a_1}^{a_2} 4\kappa_\Theta r_c\,\mathrm{d}a}{\int_{a_1}^{a_2} 4\kappa_\Theta\,\mathrm{d}a} = 1 - \frac{\ln(\vartheta(c_2)/\vartheta(c_1))}{\ln(\Theta_{c_1}/\Theta_{c_2})}. \tag{5.5}$$

(a) *If* $\vartheta_- \le \vartheta(c_i) \le \vartheta_+$ *for* $i = 1, 2$, *then* $|\langle r\rangle_{[c_1,c_2]} - 1| \le \ln(\vartheta_+/\vartheta_-)/\ln(\Theta_{c_1}/\Theta_{c_2})$.

(b) *If* $\lambda > 0$ *for all* c, *then* $\langle r\rangle_{[5,c]} \to 1$ *for* $c \to \infty$ *if and only if* $\ln\vartheta(c) = o(c)$.

(c) *By the rigorous bounds [5]* $\langle r\rangle_{[5,8]} \in [0.985, 1.017]$. *For integer* c_1 *and* c_2, *where* $5 \le c_1 \le 8$ *and* $c_1 < c_2 \le 31$ *or* $c_2 \in \{45, 47, 71\}$,

$$\langle r\rangle_{[c_1,c_2]} \le 1 + \frac{1.973}{\ln(\Theta_{c_1}/\Theta_{c_2})},$$

in particular, $\langle r\rangle_{[5,31]} \le 1.0023$, $\langle r\rangle_{[5,45]} \le 1.0020$, $\langle r\rangle_{[5,47]} \le 1.0017$, *and* $\langle r\rangle_{[5,71]} \le 1.0022$.

Proof. By Proposition 5.6(a) $\int_{a_1}^{a_2} 4\kappa_\Theta(1 - r_c)\,\mathrm{d}a = \ln\vartheta(c_2) - \ln\vartheta(c_1)$, since $\ln\vartheta$ is absolutely continuous on the interval where $\lambda > 0$, and $\int_{a_1}^{a_2} 4\kappa_\Theta\,\mathrm{d}a = \ln\Theta_{c_1} - \ln\Theta_{c_2} > 0$. Dividing the first equality by the second, we obtain (5.5). Part (a) follows from (5.5). In part (b) the denominator at $c_1 = 5$, $c_2 = c$ equals $4\pi c(1 + o(1))$ by (5.2), and the numerator is $\ln\vartheta(c) - \ln\vartheta(5)$.

(c) Denote by $\lambda_c^- \le \lambda(c)$ the rigorous lower bounds from [5, Table A.1] at $c = 5, \dots, 31$ and $c \in \{45, 47, 71\}$, rounded down, and by $\lambda_c^+ \ge \lambda(c)$ the rigorous upper bounds from Table A.3(a) of the same paper at $c = 5, \dots, 8$, rounded up. By [5, Theorem 1.1, Proposition 1.3] the bounds for $c = 45$ and 47 hold on the whole $L^2(I_c)$, and at $c = 71$ in the even sector, where λ is defined. By monotonicity $\lambda > 0$ on $[5, 71]$. Substituting the bounds $\lambda_c^\pm$ into (5.5), we obtain

$$\langle r\rangle_{[c_1,c_2]} \le 1 + \frac{\ln(\lambda_{c_1}^+\Theta_{c_2}/(\lambda_{c_2}^-\Theta_{c_1}))}{\ln(\Theta_{c_1}/\Theta_{c_2})},$$

and for $c_2 \le 8$ the lower bound

$$\langle r\rangle_{[c_1,c_2]} \ge 1 - \frac{\ln(\lambda_{c_2}^+\Theta_{c_1}/(\lambda_{c_1}^-\Theta_{c_2}))}{\ln(\Theta_{c_1}/\Theta_{c_2})}.$$

The largest value λ_c^+/Θ_c equals 3.0005 (at $c = 7$), the smallest for λ_c^-/Θ_c is 0.4172 (at $c = 71$), and $\ln(3.0005/0.4172) < 1.973$. For $[5, 8]$ we find $[0.9858, 1.0163]$, on $[5, 31]$: $1 + \ln(2.5006/1.2297)/317.78 < 1.0023$, on $[5, 45]$: $1 + \ln(2.5006/0.9723)/491.86 < 1.0020$, on $[5, 47]$: $1 + \ln(2.5006/1.0486)/516.78 < 1.0017$, on $[5, 71]$: $1 + \ln(2.5006/0.4172)/816.31 < 1.0022$. The values Θ_c are taken from the same Table A.1 with

ten digits. The error of their logarithms is less than 10^{-9} and is accounted for by rounding in the safe direction. □

5.4. **Lower bounds on a sparse sequence.** By formula (5.5) and Lemma 5.5, for bounding the mean boundary balance it suffices to have lower bounds on λ at individual points.

Corollary 5.8 (lower bounds on a sparse sequence)**.** *Let $5 \le c_1 < c_2 < \ldots \to \infty$, $D_k = c_{k+1} - c_k$ and $\lambda(c_k) \ge C c_k^{-A_0} \Theta_{c_k}$ for all k with constants $C > 0$ and $A_0 \ge 0$. Then $\lambda(c) > 0$ for all $c > 1$, and*

(a) *if $D_k = o(c_k)$, then $\limsup_{c\to\infty} \langle r\rangle_{[5,c]} \le 1$, whereas if $D_k \le \gamma c_k$ with constant $\gamma > 0$, then $\limsup_{c\to\infty} \langle r\rangle_{[5,c]} \le 1 + \gamma \ln(1/C_\Theta)/(4\pi) < 1 + 1.07\gamma$;*

(b) *for $c \in [c_k, c_{k+1}]$*

$$\lambda(c) \ge C C_\Theta^{\lceil D_k\rceil} (c + D_k)^{-A_0} \Theta_c.$$

Proof. For any $c > 1$ there is k with $c_k \ge c$. Then $\lambda(c) \ge \lambda(c_k) > 0$ by Proposition 2.1.

We begin with (b). Let $c \in [c_k, c_{k+1}]$ and $d = c_{k+1} - c \le D_k$. Split $[c, c_{k+1}]$ into $\lceil d\rceil$ intervals of length at most 1. Applying Lemma 5.5 to each of them, we obtain $\Theta_{c_{k+1}} \ge C_\Theta^{\lceil d\rceil} \Theta_c \ge C_\Theta^{\lceil D_k\rceil} \Theta_c$, since $C_\Theta < 1$. By Proposition 2.1 $\lambda(c) \ge \lambda(c_{k+1}) \ge C c_{k+1}^{-A_0} \Theta_{c_{k+1}}$, while $c_{k+1} \le c + D_k$.

(a) By (5.5) $\langle r\rangle_{[5,c]} = 1 + (\ln\vartheta(5) - \ln\vartheta(c))/\ln(\Theta_5/\Theta_c)$. The denominator here equals $4\pi c(1 + o(1))$ by (5.2). For $c \in [c_k, c_{k+1}]$, by (b) the numerator is at most $\ln(\vartheta(5)/C) + A_0 \ln(c + D_k) + \lceil D_k\rceil \ln(1/C_\Theta)$. If $D_k = o(c_k)$, then this is $o(c)$, since $c_k \le c$. If $D_k \le \gamma c_k \le \gamma c$, then the numerator is at most $\gamma c \ln(1/C_\Theta) + O(\ln c)$. It remains to observe that $\ln(1/C_\Theta) < 13.407$ and $13.407/(4\pi) < 1.067$. □

If $D_k \le D$ for all k, then the corollary can also be written in integral form: for $c \ge c_1$ it follows from (b) that $\ln\vartheta(c) \ge \ln C - \lceil D\rceil \ln(1/C_\Theta) - A_0 \ln(c + D)$, where $\ln(c + D) \le 2a + \ln(1 + D/5)$ for $c \ge 5$. Since $\lambda > 0$ on $[5, c]$, by Proposition 5.6(a) we obtain for the integral of the boundary balance the bound

$$\int_{a_0}^{a} 4\kappa_\Theta (r_c - 1)\,\mathrm{d}a' \le M_D + 2A_0 a, \quad M_D = \ln(\vartheta(5)/C) + \lceil D\rceil \ln(1/C_\Theta) + A_0 \ln(1 + D/5).$$

Together with the bounds [5, Table A.1], Lemma 5.5 makes it possible to bound λ rigorously also between integer points: for $5 \le c \le 31$ we have $\lambda(c) \ge \lambda(\lceil c\rceil) \ge 1.04\,\Theta_{\lceil c\rceil} \ge 1.04\, C_\Theta \Theta_c$.

6. Proof of the reduction to the boundary balance

We proceed to the proof of Theorem 1.3. Its hypothesis is given only on intervals where λ is already positive, and the proof consists of a lower bound on such an interval and of a continuity argument (Lemma 5.1), which extends it to all $c \ge 5$.

Proof of Theorem 1.3. Set $C(a) = 4\rho\kappa_\Theta$, where κ_Θ is taken at $c = e^{2a}$. The function C is continuous on $[a_0, \infty)$ by Lemma 5.3. Moreover, $\int_{a_0}^{b} C\,\mathrm{d}a = \rho(\ln\Theta_5 - \ln\Theta_{e^{2b}})$.

Let $b > a_0$ and $\lambda(e^{2a}) > 0$ for $a \in [a_0, b]$. The function $a \mapsto \lambda(e^{2a})$ is Lipschitz on $[a_0, b]$ (Proposition 4.2). By (1.1) almost everywhere on this interval, $\mathrm{d}\lambda/\mathrm{d}a + C\lambda \ge 0$. As in the proof of Proposition 5.2, the product $\lambda(e^{2a}) \exp\int_{a_0}^{a} C$ is absolutely continuous and nondecreasing, whence

$$\lambda(e^{2b}) \ge \lambda(5) \exp\left(-\int_{a_0}^{b} C\right) = \lambda(5) \left(\frac{\Theta_{e^{2b}}}{\Theta_5}\right)^{\rho} =: \underline{\lambda}(b).$$

For the continuous and positive function $\underline{\lambda}$, Lemma 5.1 gives $\lambda(c) > 0$ and $\lambda(e^{2b}) \ge \underline{\lambda}(b)$ for all $c \ge 5$ and $b > a_0$. □

The formulation of the hypothesis in terms of the boundary coefficient, given after Theorem 1.3, follows from Theorem 1.2: at a point of differentiability $-\mathrm{d}\lambda/\mathrm{d}a = 2\beta(u)^2$ for any normalized minimizer u, and the bound $\beta(u)^2 \le 2\rho\kappa_\Theta\lambda$ for almost all $a > a_0$ implies (1.1).

Lemma 5.4 allows one to write the hypothesis in the form $-\mathrm{d}\lambda/\mathrm{d}a \le 8\pi\rho' c\,\lambda(e^{2a})$: the original form with ρ implies the new one with $\rho' = \rho$, and the new one with ρ' implies the original one with $\rho = \rho'/0.924$. From the inequality with ρ' the same argument yields $\lambda(c) \ge \lambda(5) e^{-4\pi\rho'(c-5)}$. The exact form of Θ_c, including the constant $9/2$ and the factor c^5, is needed for the scale law ($\rho = 1$), but not for positivity.

7. Arithmetic form of the boundary balance

To see what cancellation the hypothesis of Theorem 1.3 requires, we show that it is equivalent to the following: the sum over primes, written in terms of the function $E(x) = \Psi(x) - x$, cancels at the minimizer the remaining part of the form with an absolute error of order $c \ln c \cdot \lambda$ (Section 7.2). This will require the dilation of the minimizer and the dilation identity, relying on the integrability of the derivative of the eigenfunctions (Section 3.4). Along the way a virial identity is derived for each eigenfunction of the form, on which Section 8 is based, although these results are not needed for the reduction to the boundary balance.

7.1. Dilation of the minimizer and the dilation identity.

Proposition 7.1 (dilation of the minimizer)**.** *Let $c > 1$, $a = (\ln c)/2$, u be a normalized minimizer at c, and $u^{(t)}(x) = t^{-1/2}u(x/t)$. For any $t > 0$ the function $u^{(t)}$ lies in $D(Q_{c^t})$ and $\|u^{(t)}\| = 1$. If $a \mapsto \lambda(e^{2a})$ is differentiable at the point a, $-\mathrm{d}\lambda/\,\mathrm{d}a = 2\beta(u)^2$ and $t \mapsto Q_{c^t}(u^{(t)})$ is differentiable at $t = 1$, then $\ln c \cdot \beta(u)^2 = -\,\mathrm{d}/\,\mathrm{d}t\, Q_{c^t}(u^{(t)})|_{t=1}$.*

Proof. The dilation is the same variation as in the proof of Bombieri's Theorem 5 [3, p. 199]. It maps $L^2(I_c)$ isometrically onto $L^2(I_{c^t})$, $I_{c^t} = [-ta, ta]$. Moreover $\widehat{u^{(t)}}(r) = t^{1/2}\hat{u}(tr)$. Hence $\int \ln(2 + |r|)|\widehat{u^{(t)}}(r)|^2\,\mathrm{d}r = \int \ln(2+|s|/t)|\hat{u}(s)|^2\,\mathrm{d}s < \infty$. By the description of the domain of the form in Section 2.2 $u^{(t)} \in D(Q_{c^t})$. In particular, $\lambda(e^{2ta}) \le Q_{c^t}(u^{(t)})$ for any $t > 0$, with equality at $t = 1$. The difference $Q_{c^t}(u^{(t)}) - \lambda(e^{2ta})$ is nonnegative and vanishes at $t = 1$. If both functions are differentiable at $t = 1$, this is a minimum point of the difference and its derivative vanishes there. It remains to note that $\mathrm{d}/\,\mathrm{d}t\,\lambda(e^{2ta})|_{t=1} = a\,\mathrm{d}\lambda/\,\mathrm{d}a = -2a\beta(u)^2 = -\ln c \cdot \beta(u)^2$. □

The hypotheses of the next proposition hold at every point of simplicity of λ (Theorem 4.5), and for any minimizer wherever λ is differentiable (Corollary 8.2).

Proposition 7.2 (dilation identity)**.** *Let u be a real even normalized minimizer at $c = e^{2a}$, $g = u \star \tilde{u}$, $P = P(u)$, and $m_1 = \int xu(x)\sinh(x/2)\,\mathrm{d}x$, and let λ be differentiable at the point a and $-\mathrm{d}\lambda/\,\mathrm{d}a = 2\beta(u)^2$. Then*

$$\ln c \cdot \beta(u)^2 = \frac{1}{2\pi}\int rh'_+(r)|\hat{u}(r)|^2\,\mathrm{d}r - 2\sum_{n<c}\frac{\Lambda(n)\ln n}{\sqrt{n}}g'(\ln n) - 2P(P + m_1). \tag{7.1}$$

Proof. The Archimedean and the pole terms are differentiable in t at $t = 1$. Indeed, $h'_+(r) = -(1/2)\operatorname{Im}\psi'(1/4 + ir/2)$ and $\psi'(z) = z^{-1} + O(z^{-2})$. Consequently, the function $|rh'_+(r)|$ is bounded. By Lebesgue's theorem at $t = 1$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2\pi}\int h_+(r)|\widehat{u^{(t)}}(r)|^2\,\mathrm{d}r = -\frac{1}{2\pi}\int rh'_+(r)|\hat{u}(r)|^2\,\mathrm{d}r,$$

and $\mathrm{d}/\,\mathrm{d}t\,P(u^{(t)}) = (P + m_1)/2$ (from $P(u^{(t)}) = t^{1/2}\int u(y)\cosh(ty/2)\,\mathrm{d}y$). The atom $n < c$ contributes the term $-2\Lambda(n)n^{-1/2}g(\ln n/t)$, differentiable at $t = 1$ by Theorem 3.11. At $c = p^m$ the atom $n = c$ with the term $-2\Lambda(c)c^{-1/2}g(2a/t)$ also enters the form Q_{c^t} at $t > 1$, its derivative at $t = 1 + 0$ being proportional to $g'(2a) = 0$, while at $t < 1$ it is absent. As a result $t \mapsto Q_{c^t}(u^{(t)})$ is differentiable at $t = 1$,

$$\frac{\mathrm{d}}{\mathrm{d}t}Q_{c^t}(u^{(t)})\Big|_{t=1} = -\frac{1}{2\pi}\int rh'_+(r)|\hat{u}(r)|^2\,\mathrm{d}r + 2\sum_{n<c}\frac{\Lambda(n)\ln n}{\sqrt{n}}g'(\ln n) + 2P(P + m_1),$$

and by Proposition 7.1 $-a\,\mathrm{d}\lambda/\,\mathrm{d}a = -\,\mathrm{d}/\,\mathrm{d}t\,Q_{c^t}(u^{(t)})|_{t=1}$, which together with $-\,\mathrm{d}\lambda/\,\mathrm{d}a = 2\beta(u)^2$ gives (7.1). □

This is a Pohozaev-type identity [18] for Q_c. Below it is derived from the equation for every eigenfunction (Corollary 7.5), for which it is convenient to write it in terms of the virials of the parts of the form.

Definition 7.3 (virial)**.** *The virial* of the symmetric part $\mathfrak{b}$ of the form on a function u supported in I_c is the derivative $V(\mathfrak{b}; u) = \mathrm{d}/\,\mathrm{d}t\,\langle \mathfrak{b}u^{(t)}, u^{(t)}\rangle|_{t=1}$, if it exists.

Lemma 7.4 (virials of the parts of the form)**.**

(a) *The dilation $u \mapsto u^{(t)}$ maps $L^2(I_c)$ unitarily onto $L^2(I_{c^t})$, and $\mathrm{d}/\,\mathrm{d}t\,u^{(t)}|_{t=1} = -\mathcal{X}u$, where $\mathcal{X} = x\,\mathrm{d}/\,\mathrm{d}x + 1/2$ and $\mathcal{X}^* = -\mathcal{X}$. For a smooth u compactly supported inside I_c, the virial (Definition 7.3) equals*

$$V(\mathfrak{b}; u) = \langle \mathfrak{b}u, u\rangle + 2\langle u, x(\mathfrak{b}u)'\rangle = \langle u, [\mathcal{X}, \mathfrak{b}]u\rangle. \tag{7.2}$$

(b) *The virial of the multiplier with symbol b equals* $-(2\pi)^{-1}\int rb'(r)|\hat{u}(r)|^2\,\mathrm{d}r$*, the virial of the pair of shifts* $S_{\tau_0}+S_{-\tau_0}$ *equals* $-2\tau_0 g'(\tau_0)$*, where* $g=u\star\tilde{u}$*, and the virial of the term* $2P(u)^2$ *equals* $2P(P+m_1)$*.*

(c) *For every* $u\in W^{1,1}(\mathbb{R})$ *supported in* I_c*, in particular for eigenfunctions of the form (Theorem 3.11), these virials exist and are expressed by formulas (a) and (b).*

Proof. (a) The unitarity of the dilation and the formula for $\mathcal{X}$ are verified by a change of variable, and $\mathcal{X}^*=-\mathcal{X}$ by integration by parts. The derivative equals $-\langle\mathfrak{b}\mathcal{X}u,u\rangle-\langle\mathfrak{b}u,\mathcal{X}u\rangle=\langle u,[\mathcal{X},\mathfrak{b}]u\rangle=2\langle u,\mathcal{X}\mathfrak{b}u\rangle$, and $2\langle u,\mathcal{X}\mathfrak{b}u\rangle=\langle\mathfrak{b}u,u\rangle+2\langle u,x(\mathfrak{b}u)'\rangle$.

(b) We have $\widehat{u^{(t)}}(r)=t^{1/2}\hat{u}(tr)$, the autocorrelation of $u^{(t)}$ equals $g(\cdot/t)$, and $\mathrm{d}/\,\mathrm{d}t\,P(u^{(t)})|_{t=1}=(P+m_1)/2$ (the proof of Proposition 7.2). It remains to differentiate at $t=1$.

(c) For the parts of the form Q_c from (b), each summand on the right-hand side of (7.2), when the support lies in a fixed interval, is bounded by $\|u\|_\infty\|u'\|_1$: the shifts do not change L^1-norms, the remainder of the Archimedean symbol is a sum of multiplication by a constant and the convolutions of Lemma 3.3, and the rank-one term is given by smooth functions. Therefore, if $\|u_n-u\|_\infty\to 0$ and $u_n'\to u'$ in L^1, the right-hand side of (7.2) for $u_n^{(t)}$ converges to the same expression for $u^{(t)}$ uniformly in t near 1. Approximating $u\in W^{1,1}(\mathbb{R})$ by smooth functions, we find that the mapping $t\mapsto\langle\mathfrak{b}u^{(t)},u^{(t)}\rangle$ is differentiable at $t=1$, its derivative equals the right-hand side of (7.2), and formulas (b) remain valid. □

The three expressions in (7.1) are the virials of the parts of the form, taken with the opposite sign. For the principal part of the symbol, $\ln|r|$, the virial equals $-\|u\|^2$, since $r(\ln|r|)'=1$. Write $Q_c(f)=(1/2)E_{\mathrm{L}}(f)+\mathfrak{b}_c(f)$, where E_{L} is the form of L_Δ (symbol $2\ln|r|$), and the *nonprincipal part* $\mathfrak{b}_c$ is composed of the remainder $h_+(r)-\ln|r|$ of the Archimedean symbol, the atoms, and the pole term. In this notation (7.1) takes the form

$$\ln c\cdot\beta(u)^2=1-V(\mathfrak{b}_c;u). \tag{7.3}$$

In other words, the virial of the nonprincipal part of the form on the minimizer differs from 1 by $\ln c\cdot\beta^2=2\kappa_\Theta r_c\lambda\ln c$.

The dilation identity is derived from the variational principle and the Hadamard formula and is therefore valid only for the minimizer at points where λ is differentiable. From the equation, via the Pohozaev identity (Theorem 4.8), the same equality is obtained for every eigenfunction.

Corollary 7.5 (virial identity for eigenfunctions)**.** *Let u be a real normalized eigenfunction of the form Q_c in any sector (the eigenvalue may be multiple), and $\mathfrak{b}_c$ be the nonprincipal part of the form Q_c. Then*

$$a(\beta_+(u)^2+\beta_-(u)^2)=1-V(\mathfrak{b}_c;u),\quad V(\mathfrak{b}_c;u)=\langle\mathfrak{b}_cu,u\rangle+2\langle u,x(\mathfrak{b}_cu)'\rangle,$$

and all terms are defined by Theorem 3.11. For an even function, the left-hand side equals $\ln c\cdot\beta(u)^2$*, and for the minimizer this equality is* (7.3)*.*

Proof. By Lemma 3.3 and its proof, the operator $\mathfrak{b}_c$ on functions supported in I_c is a sum of multiplication by a constant, convolutions with integrable kernels of bounded variation, the operator $-A_c$, and a finite-rank term with smooth functions, and u is a weak solution of the Dirichlet problem $L_\Delta u=2\lambda u-2\mathfrak{b}_cu$ on I_c. Denote the right-hand side $2\lambda u-2\mathfrak{b}_cu$ by f. The function u is continuous by Theorem 3.11, $u(\pm a)=0$ and $u\in W^{1,1}$, and hence $f\in C(\overline{I_c})\cap W^{1,1}(I_c)$. Apply Theorem 4.8 with $\mu_0=0$, $y_-=-a$, $y_+=a$ at both ends and add the equalities. Since $(x+a)+(x-a)=2x$,

$$4a(\beta_+^2+\beta_-^2)=4+2\langle f,u\rangle+4\langle u,xf'\rangle.$$

Since u is absolutely continuous and vanishes at the endpoints, $\langle u,u\rangle+2\langle u,xu'\rangle=0$. Hence $\langle f,u\rangle+2\langle u,xf'\rangle=-2V(\mathfrak{b}_c;u)$, which proves the statement. □

7.2. **Boundary balance and the Chebyshev function.** Let us now write the identity (7.1) in terms of the function $E(x)$.

Proposition 7.6 (boundary balance and the function $\Psi(x)-x$)**.** *Let u, g and a satisfy the hypotheses of Proposition 7.2 and*

$$\Psi(x)=\sum_{n\le x}\Lambda(n),\quad E(x)=\Psi(x)-x,\quad F(x)=x^{-1/2}\ln x\,g'(\ln x).$$

Then

$$\ln c\cdot\beta(u)^2=A(u)+R(u)+\mathcal{E}(u), \tag{7.4}$$

where

$$A(u)=\frac{1}{2\pi}\int rh'_+(r)|\hat u(r)|^2\,\mathrm{d}r,\quad R(u)=2\int_0^{\ln c}ye^{-y/2}g'(y)\,\mathrm{d}y,\quad \mathcal{E}(u)=-2\int_{[1,c)}F\,\mathrm{d}E.$$

The contribution of the principal part of the symbol to A equals exactly 1: $A(u)-1=(2\pi)^{-1}\int(rh'_+(r)-1)|\hat u(r)|^2\,\mathrm{d}r$. In particular, $r_c=(A+R+\mathcal{E})/(2\kappa_\Theta\lambda\ln c)$.

The main term of the sum over the atoms, with density $\mathrm{d}x$, cancels exactly against a part of the pole term (in the proof this is the quantity $2M^+$), and there remain the term R, independent of the weights $\Lambda(n)$, and the integral over $\mathrm{d}E$, where they enter only through the function $E(x)=\Psi(x)-x$. The terms R and $\mathcal{E}$ depend on u through g, and $\int_{[1,c)}F\,\mathrm{d}E=\sum_{n<c}\Lambda(n)F(n)-\int_1^cF(x)\,\mathrm{d}x$: there is no atom $n=c$ in Q_c, and the half-interval does not carry the mass of Ψ at the point $c=p^m$.

Proof. Let us write the identity of Proposition 7.2 in the form

$$\ln c\cdot\beta^2=A-2S-2P(P+m_1),\quad S=\sum_{n<c}\Lambda(n)n^{-1/2}\ln n\,g'(\ln n).$$

Pole term. Since $\cosh(x/2)\cosh(z/2)=(1/2)[\cosh((x+z)/2)+\cosh((x-z)/2)]$ and u is even, after the substitution $z\mapsto -z$ the first term coincides with the second. Consequently,

$$P^2=\iint u(x)u(z)\cosh((x-z)/2)\,\mathrm{d}x\,\mathrm{d}z=2\int_0^{\ln c}g(s)\cosh(s/2)\,\mathrm{d}s.$$

The autocorrelation of the dilation $u^{(t)}$ equals $g(s/t)$, and $P(u^{(t)})^2=2\int_0^{t\ln c}g(s/t)\cosh(s/2)\,\mathrm{d}s$. The function g' is continuous (Theorem 3.11) and $g(\ln c)=0$. Differentiating, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}P(u^{(t)})^2|_{t=1}=-2\int_0^{\ln c}sg'(s)\cosh(s/2)\,\mathrm{d}s.$$

On the other hand, $\mathrm{d}/\,\mathrm{d}t\,P(u^{(t)})|_{t=1}=(P+m_1)/2$ (the proof of Proposition 7.2). Equating, we find

$$2P(P+m_1)=-4\int_0^{\ln c}s\cosh(s/2)g'(s)\,\mathrm{d}s=-2M^+-2M^-,\quad M^\pm=\int_0^{\ln c}se^{\pm s/2}g'(s)\,\mathrm{d}s.$$

Atoms. The function F is continuous on $[1,c]$, $F(1)=0$, and $S=\int_{[1,c)}F\,\mathrm{d}\Psi=\int_1^cF(x)\,\mathrm{d}x+\int_{[1,c)}F\,\mathrm{d}E$, and the substitution $x=e^y$ shows that $\int_1^cF(x)\,\mathrm{d}x=M^+$.

Substituting both equalities, we obtain $\ln c\cdot\beta^2=A-2M^+-2\int_{[1,c)}F\,\mathrm{d}E+2M^++2M^-$, that is, (7.4) with $R=2M^-$. The assertion about the symbol follows from $r(\ln|r|)'=1$ and $(2\pi)^{-1}\int|\hat u|^2=1$. □

The proof relies only on $g(\ln c)=0$ and the continuity of g' on $[0,\ln c]$ (at $\ln c$, of the left derivative), so the equality $-2S-2P(P+m_1)=R+\mathcal{E}$ holds for any even function with support in I_c and such g', which allows one to verify it on Galerkin vectors (Section A.3).

Remark 7.7 (elementary estimates of the terms). For any normalized even u with support in I_c we have $|A-1|\le1.081$, since the values of $rh'_+(r)$ lie in $[0,2.081]$, and also $|R|\le2\int_0^{\ln c}ye^{-y/2}|g'|\,\mathrm{d}y$ and $|\mathcal{E}|\le2\sum_{n<c}\Lambda(n)n^{-1/2}\ln n|g'(\ln n)|+2\int_0^{\ln c}ye^{y/2}|g'|\,\mathrm{d}y$. The bound 2.081 is rigorous: it is obtained by partitioning the half-line into intervals and an explicit estimate of the tail of the series for h'_+, while the largest value, about 2.0807, is attained at $r\approx0.52$. Integration by parts, taking into account $F(1)=0$ and $F(c)=c^{-1/2}\ln c\,g'(\ln c)=0$ (Theorem 3.11), gives in addition $|\mathcal{E}|\le2\int_0^{\ln c}|E(e^y)|\,|\,\mathrm{d}(ye^{-y/2}g'(y))|$ with the exact function E. Such estimates lead only to $\beta^2=O(1)$, which already follows from Corollary 3.9, whereas the hypothesis requires $\beta^2=O(\kappa_\Theta\lambda)$. How much coarser they are is shown numerically in Section A.3.

By the identity (7.4) the hypothesis of Theorem 1.3 is equivalent to the inequality

$$\mathcal{E}(u_a) \le -(A+R)(u_a) + 2\rho\kappa_\Theta \lambda(e^{2a}) \ln c$$

almost everywhere on the intervals where $\lambda > 0$. In other words, the sum over primes, written in terms of $E(x)$, must cancel the terms $A + R$, which do not involve the primes, with absolute error $2\rho\kappa_\Theta \lambda \ln c$, that is, of order $c \ln c \cdot \lambda$.

8. Dependence of the infimum on the length of the interval

In this section we study the derivatives of λ with respect to $a = (\ln c)/2$ and the behavior of λ near powers of primes, and at the end we show what the proved properties do not imply.

8.1. **One-sided derivatives of the infimum.** The virial identity remains valid at a multiple eigenvalue as well, and from it both one-sided derivatives of λ are found at every point.

Theorem 8.1 (one-sided derivatives of the infimum)**.** *Let $\mathcal{N}_a$ be the space of real eigenfunctions for the eigenvalue $\lambda(e^{2a})$ in the even sector, and $\overline{b}(a)$ and $\underline{b}(a)$ be the largest and smallest values of $\beta(u)^2$ on its unit sphere. Then the function $y \mapsto \lambda(e^{2y})$ has at every point $a > 0$ both one-sided derivatives, and*

$$\lambda'_+(a) = -2\overline{b}(a), \quad \lambda'_-(a) = -2\underline{b}(a).$$

If $\dim \mathcal{N}_a \ge 2$, then $\underline{b}(a) = 0$, that is, $\lambda'_-(a) = 0$.

Proof. By dilating the eigenfunctions of the eigenvalue $\lambda(e^{2a})$ the difference quotients are estimated from one side, and the limits of minimizers at neighboring c show that these estimates are sharp.

Upper bound. Let $\varphi \in \mathcal{N}_a$ and $\|\varphi\| = 1$. By Theorem 3.11 and by the argument of Proposition 7.2, relying only on the continuity of g' and $g'(2a) = 0$, the function $t \mapsto Q_{c^t}(\varphi^{(t)})$ is differentiable at $t = 1$ (and at $c = p^m$), and by Lemma 7.4 and Corollary 7.5 its derivative equals $-1 + V(\mathfrak{b}_c; \varphi) = -\ln c \cdot \beta(\varphi)^2$. Since $\lambda(c^t) \le Q_{c^t}(\varphi^{(t)})$, for $t = 1 \pm \varepsilon/a$ we have $\lambda(e^{2(a\pm\varepsilon)}) \le \lambda(e^{2a}) \mp 2\varepsilon\beta(\varphi)^2 + o(\varepsilon)$. Choosing φ with $\beta(\varphi)^2 = \overline{b}(a)$ and with $\beta(\varphi)^2 = \underline{b}(a)$, we find that the limit superior of the right difference quotients does not exceed $-2\overline{b}$, while the limit inferior of the left ones is not less than $-2\underline{b}$.

Limits. Let $\varepsilon_k \to 0+$ and let v_k be a normalized minimizer at $e^{2(a+\varepsilon_k)}$. As in the proof of Lemma 4.4, some subsequence v_k converges in L^2 to a function $v \in \mathcal{N}_a$, $\|v\| = 1$. With the identity (4.1), valid for any two eigenfunctions (Proposition 4.1(a)), the computation of the proof of Theorem 4.5(i) with the inner function v and the outer ones v_k leads to the equality $(\lambda(e^{2(a+\varepsilon_k)}) - \lambda(e^{2a}))\langle v, v_k\rangle = -2\varepsilon_k\beta(v)\beta(v_k) + o(\varepsilon_k)$. Taking into account $\beta(v_k) \to \beta(v)$ (Theorem 3.10), the limit of the right difference quotients along this subsequence equals $-2\beta(v)^2 \ge -2\overline{b}$. Together with the upper bound this gives $-2\overline{b}$ for any subsequence, that is, $\lambda'_+ = -2\overline{b}$. On the left the same argument with minimizers at $e^{2(a-\varepsilon_k)}$ (the proof of Theorem 4.3) leads to the limit $-2\beta(u)^2 \le -2\underline{b}$, and the upper bound to the reverse inequality.

A multiple point. The linear functional $u \mapsto \beta(u)$ on $\mathcal{N}_a$ when $\dim \mathcal{N}_a \ge 2$ vanishes on some nonzero element of $\mathcal{N}_a$. Therefore $\underline{b}(a) = 0$. □

Corollary 8.2 (kink points)**.** *The function $a \mapsto \lambda(e^{2a})$ is differentiable everywhere except on an at most countable set of points a with $\overline{b}(a) > \underline{b}(a)$, and at every point of differentiability $\mathrm{d}\lambda/\,\mathrm{d}a = -2\beta(u)^2$ for every normalized minimizer u. Kinks occur only at points where $\lambda(c)$ is multiple, and there $\lambda'_- = 0 > \lambda'_+$. At every multiple point there exists a normalized eigenfunction $u \in \mathcal{N}_a$ with $\beta(u) = 0$, and for it $V(\mathfrak{b}_c; u) = 1$.*

Proof. If a is a kink, then $\lambda'_+(a) < \lambda'_-(a)$ and there exist rational numbers r and $\delta > 0$ for which $\lambda'_+(a) < r < \lambda'_-(a)$ and $\lambda(e^{2y}) < \lambda(e^{2a}) + r(y-a)$ for $0 < |y-a| < \delta$. Moreover, a is a point of strict maximum of the function $y \mapsto \lambda(e^{2y}) - ry$ on $(a-\delta, a+\delta)$, and the points with the same pair (r, δ) are separated from one another by at least δ. Hence on any bounded interval there are finitely many of them, and with countably many pairs the set of kinks is at most countable. By Theorem 8.1 the derivative exists if and only if $\overline{b}(a) = \underline{b}(a)$, that is, when β^2 is constant on the unit sphere of $\mathcal{N}_a$, and in this case $\mathrm{d}\lambda/\,\mathrm{d}a = -2\beta(u)^2$ for every minimizer. For $\dim \mathcal{N}_a = 1$ this is the case, while for a multiple λ we have $\underline{b} = 0$. The last assertion follows from Corollary 7.5 with $\beta(u) = 0$. □

Together with the Lipschitz property of λ on every closed interval, this yields Theorem 1.2. In particular, in the definition of the boundary flux (Section 4.2) and in Proposition 4.2 the words "almost everywhere" can be replaced by "everywhere except on an at most countable set", and the limiting minimizer is no longer needed. On the even sector the first variation of the form under a shift of the endpoints is a quadratic functional of rank one $-2\beta(u)^2$, which on an eigenspace of dimension $k \ge 2$ vanishes on a

hyperplane. Theorem 1.3 does not need this: for the Grönwall inequality an estimate almost everywhere suffices.

8.2. **Powers of primes.** Let us find how the infimum behaves to the right and to the left of a point $c_* = p^m$ at which a new atom enters the form. Set $a_* = (\ln c_*)/2$ and $w_* = \Lambda(c_*)c_*^{-1/2}$. Dilating the minimizer gives the upper bound for λ, in which the atoms $n < c_*$ contribute only a remainder $o(\delta/\ln(1/\delta))$, while the new atom contributes a correction of order $\delta/\ln(1/\delta)$ with an explicit constant. For this, two lemmas are needed: on the dilation remainder at points $\ln n$ and on the overlap of boundary layers.

For an eigenfunction u of the form Q_c set $g = u \star \tilde{u}$ and $H_u(\sigma) = \int u'(y)u'(y-\sigma)\,\mathrm{d}y$. The function H_u, the correlation of u' with itself, lies in $L^1(\mathbb{R})$ together with u' (Theorem 3.11). Since $g'(s) - g'(s') = \int (u(y-s) - u(y-s'))u'(y)\,\mathrm{d}y$ and $u(y-s) - u(y-s') = -\int_{s'}^{s} u'(y-\sigma)\,\mathrm{d}\sigma$,

$$g'(s) - g'(s') = -\int_{s'}^{s} H_u(\sigma)\,\mathrm{d}\sigma. \tag{8.1}$$

Lemma 8.3 (the dilation remainder at the shift points)**.** *Let u be an eigenfunction of the form Q_c (in any sector), $g = u \star \tilde{u}$ and $0 < s < 2a$. For $\tau \to 0$*

$$g(s+\tau) - g(s) - \tau g'(s) = o(|\tau|/\ln(1/|\tau|)). \tag{8.2}$$

Proof. For complex u, the function g is the sum of the autocorrelations of $\operatorname{Re} u$ and $\operatorname{Im} u$ (the paragraph after the proof of Theorem 3.11), therefore the real case is sufficient. The required estimate does not use the behavior of u' near the boundary: it is enough that $u' \in L^1$ and that the mass of the kernel $\mathcal{G}$ on a short interval near zero is logarithmically small by (3.6). First we bound the integral of $|u'|$ over short intervals away from the boundary, and then the integral of $|H_u|$ over the strip near the shift point.

Short intervals away from the boundary. For $d_0 > 0$ and $0 < h < 1$ denote by $W_{d_0}(h)$ the supremum of the integrals $\int_K |u'|$ over intervals $K \subset I_c$ of length h at a distance at least d_0 from the points $\pm a$, and by $W_0(r)$ the same quantity over all intervals of length r. From $u' \in L^1$ it follows that $W_0(r) \to 0$ for $r \to 0$. We will show that $W_{d_0}(h) = o(1/\ln(1/h))$.

We use the representation from the proof of Theorem 3.11:

$$u = u_1 + u_2, \quad u_1 = \mathcal{G} \star (f \mathbb{1}_{I_c}), \quad u_2 = \mathcal{G} \star \mathcal{U}_o.$$

The function u_2 is smooth on I_c, and the derivative of u_1 equals $\mathcal{G} \star \mathfrak{m}$ with a finite measure $\mathfrak{m} = (f\mathbb{1}_{I_c})'$. Inside I_c this measure equals $f'\,\mathrm{d}x$, and the rest of it is concentrated at the points $\pm a$. For $u \in W^{1,1}(\mathbb{R})$ we have $(A_c u)' = A_c u'$, and $|(k_1 \star u)'| = |k_1 \star u'| \le \|k_1\|_\infty \|u'\|_1$. For $C_f = \|k_1\|_\infty \|u'\|_1 + 2\|(\Pi u)'\|_{L^\infty(I_c)}$

$$|f'| \le C_\mu |u'| + 2A_c|u'| + C_f \quad (x \in I_c),$$

where, as in that proof, $C_\mu = |1 + 2\mu - \int k_1|$.

Let K be the interval from the definition of $W_{d_0}(h)$, $d = h^{1/2} \le d_0/2$, and K^d its d-neighborhood. For an even kernel $\mathcal{G}$ decreasing on $(0,\infty)$ and any y the integral $\int_K \mathcal{G}(x-y)\,\mathrm{d}x$ does not exceed $m_{\mathcal{G}}(h/2) = \int_{|x|<h/2} \mathcal{G}$, and for $\operatorname{dist}(y,K) \ge d$ it does not exceed $h\mathcal{G}(d)$. As a result,

$$\int_K |u_1'| \le m_{\mathcal{G}}(h/2)|\mathfrak{m}|(K^d) + h\mathcal{G}(d)|\mathfrak{m}|(\mathbb{R}).$$

The neighborhood K^d is at distance at least $d_0/2$ from $\pm a$, and therefore $|\mathfrak{m}|(K^d) \le (C_\mu + 4\sigma_c)W_0(h+2d) + C_f(h+2d)$: the operator $2A_c$ is a sum of shifts with weights of total mass $4\sigma_c$, while $\int_{K^d} |u'(x \pm \ln n)|\,\mathrm{d}x \le W_0(h+2d)$. From (3.6) it follows that $m_{\mathcal{G}}(h/2) \le C_{\mathcal{G}}/(1 + 2\ln(2/h)) + O(h^{1/2})$ and $h\mathcal{G}(h^{1/2}) = O(h^{1/2})$. Moreover, $\int_K |u_2'| \le C_{d_0} h$, where C_{d_0} is the supremum of $|u_2'|$ at points of I_c at distance at least d_0 from $\pm a$. Altogether,

$$W_{d_0}(h) \le \frac{C_{\mathcal{G}}}{1 + 2\ln(2/h)}[(C_\mu + 4\sigma_c)W_0(3h^{1/2}) + 3C_f h^{1/2}] + O(h^{1/2}) = o\left(\frac{1}{\ln(1/h)}\right).$$

The strip. Set $d_0 = \min(s, 2a-s)/4$ and $0 < h < d_0$. By Fubini's theorem

$$\int_{s-h}^{s} |H_u| \le \int |u'(y)| \int_{K_y} |u'(z)|\,\mathrm{d}z\,\mathrm{d}y, \quad K_y = [y-s, y-s+h].$$

For K_y outside I_c the inner integral equals zero, and for $K_y \subset I_c$ at a distance at least d_0 from $\pm a$ it does not exceed $W_{d_0}(h)$. If K_y touches $[a-d_0, a+d_0]$, then $y \ge a + s - d_0 - h > a$ and $u'(y) = 0$. It remains to consider the case where K_y touches $[-a-d_0, -a+d_0]$. In this case y lies in an interval

$J_y = [-a+s-d_0-h, -a+s+d_0]$ at a distance at least $2d_0$ from $\pm a$. Interchanging the order of integration, we obtain for this part the bound $\int |u'(z)| \int_{J_y \cap [z+s-h, z+s]} |u'(y)|\,\mathrm{d}y\,\mathrm{d}z \le \|u'\|_1 W_{d_0}(h)$. In total $\int_{s-h}^{s} |H_u| \le 2\|u'\|_1 W_{d_0}(h)$, and likewise for $[s, s+h]$.

Remainder. By (8.1) $g(s+\tau) - g(s) - \tau g'(s) = \int_0^\tau (g'(s+\sigma) - g'(s))\,\mathrm{d}\sigma$, and the absolute value of this quantity does not exceed $|\tau| \int_{s-|\tau|}^{s+|\tau|} |H_u| \le 4|\tau| \|u'\|_1 W_{d_0}(|\tau|)$, which proves (8.2). □

The condition $0 < s < 2a$ is essential: for $s = 2a$ the strip joins the boundary layers of the two ends, and this is the case of the new atom, treated in the next lemma.

Lemma 8.4 (overlap of the boundary layers)**.** *Let v be an even real-valued function supported in I_c, $\beta \in \mathbb{R}$, $C \ge 0$, and $0 < \delta_0 < 1$, with $|v(a-\sigma) - \beta(\ln(1/\sigma))^{-1/2}| \le C(\ln(1/\sigma))^{-1}$ for $0 < \sigma \le \delta_0$. For $0 < \eta \le \min(\delta_0, e^{-1})$ and $\xi_\eta = \ln(1/\eta)$*

$$
\begin{aligned}
(v \star \tilde{v})(2a-\eta) &= \int_0^\eta v(a-\sigma) v(a-\eta+\sigma)\,\mathrm{d}\sigma, \\
|(v \star \tilde{v})(2a-\eta) - \frac{\beta^2 \eta}{\xi_\eta}| &\le (\beta^2 + 2|\beta| C \xi_\eta^{1/2} + C^2) \frac{\eta}{\xi_\eta^2}.
\end{aligned} \tag{8.3}
$$

Proof. The integrand $v(x)v(x+2a-\eta)$ is nonzero only when $x \in [-a, -a+\eta]$. The substitution $x = -a+\sigma$ and the evenness of v give the first equality. Set $\xi(\sigma) = \ln(1/\sigma)$ and $J(\eta) = \int_0^\eta (\xi(\sigma)\xi(\eta-\sigma))^{-1/2}\,\mathrm{d}\sigma$. For $\sigma \le \eta$ we have $\xi(\sigma) \ge \xi_\eta$ and $J(\eta) \le \eta/\xi_\eta$. For $\sigma = \eta t$ we obtain $\xi(\sigma) = \xi_\eta(1+x)$, $x = \ln(1/t)/\xi_\eta$, and similarly for $\eta - \sigma$ with $y = \ln(1/(1-t))/\xi_\eta$. From $(1+x)^{-1/2} \ge 1 - x/2$ it follows that $(1+x)^{-1/2}(1+y)^{-1/2} \ge 1 - (x+y)/2$ for $x, y \ge 0$, while $\int_0^1 \ln(1/t)\,\mathrm{d}t = 1$. Therefore $J(\eta) \ge (\eta/\xi_\eta)(1 - 1/\xi_\eta)$. Now write $v(a-\sigma) = \beta\xi(\sigma)^{-1/2} + e(\sigma)$ with $|e(\sigma)| \le C\xi(\sigma)^{-1} \le C\xi_\eta^{-1}$. The cross terms contribute at most $2|\beta|C\eta\xi_\eta^{-3/2}$, and the product of the remainders at most $C^2\eta\xi_\eta^{-2}$, and (8.3) is proved. □

The hypothesis of Lemma 8.4 holds for every even eigenfunction of the form by Theorem 3.8, and for a family of such functions with common C and δ_0 by Corollary 3.9.

Theorem 8.5 (an estimate of the infimum by dilation)**.** *Let $a > 0$, $c = e^{2a}$, u be a real even normalized eigenfunction for the eigenvalue $\lambda(c)$, and $w_c = \Lambda(c)c^{-1/2}$, that is, $w_c = w_*$ for $c = p^m$ and $w_c = 0$ for the remaining c. As $\delta \to 0+$,*

$$
\begin{aligned}
\lambda(e^{2(a+\delta)}) &\le \lambda(c) - 2\beta(u)^2\delta - \frac{4w_c\beta(u)^2\delta}{\ln(1/\delta)} + o\left(\frac{\delta}{\ln(1/\delta)}\right), \\
\lambda(e^{2(a-\delta)}) &\le \lambda(c) + 2\beta(u)^2\delta + o\left(\frac{\delta}{\ln(1/\delta)}\right).
\end{aligned} \tag{8.4}
$$

In particular, if $c = p^m$, then

$$
\begin{aligned}
\lambda(e^{2(a_*+\delta)}) &\le \lambda(c_*) + \lambda'_+(a_*)\delta - \frac{2w_*|\lambda'_+(a_*)|\delta}{\ln(1/\delta)} + o\left(\frac{\delta}{\ln(1/\delta)}\right), \\
\lambda(e^{2(a_*-\delta)}) &\le \lambda(c_*) - \lambda'_-(a_*)\delta + o\left(\frac{\delta}{\ln(1/\delta)}\right).
\end{aligned} \tag{8.5}
$$

Proof. We bound $\lambda(c^t)$ from above by the value of the form at the dilated function u. Without the new atom the form gives the derivative $-2a\beta(u)^2$ and the remainder $o(|t-1|/\ln(1/|t-1|))$, while the new atom $n = c$ adds the overlap of the boundary layers (Lemma 8.4).

The form on the dilated function. Let $t > 0$ be close to 1, $c' = c^t$ and $u^{(t)}(x) = t^{-1/2}u(x/t)$. The function $u^{(t)}$ is normalized and lies in $D(Q_{c'})$ (Proposition 7.1), and $\lambda(c') \le Q_{c'}(u^{(t)})$. The autocorrelation of $u^{(t)}$ equals $g(\cdot/t)$, where $g = u \star \tilde{u}$. For $t < 1$ close to 1, the atoms $n < c'$ enter the form $Q_{c'}$. The atoms $c' \le n < c$ can be added to them, since for them $g(\ln n/t) = 0$. For $t > 1$, the atom $n = c$ enters as well, provided $c = p^m$. Consequently,

$$
\begin{aligned}
Q_{c'}(u^{(t)}) &= \mathcal{F}(t) - 2w_c g(2a/t)\mathbb{1}_{t>1}, \\
\mathcal{F}(t) &= \frac{1}{2\pi}\int h_+(r/t)|\hat{u}(r)|^2\,\mathrm{d}r - 2\sum_{n<c} \frac{\Lambda(n)}{\sqrt{n}} g\left(\frac{\ln n}{t}\right) + 2P(u^{(t)})^2.
\end{aligned}
$$

By the proof of Theorem 8.1 the function $t \mapsto Q_{c^t}(u^{(t)})$ is differentiable at $t = 1$ with derivative $-\ln c \cdot \beta(u)^2 = -2a\beta(u)^2$, and the contribution of the atom $n = c$ to it for $t = 1+0$ is proportional to $g'(2a) = 0$ (Theorem 3.11), and $\mathcal{F}'(1) = -2a\beta(u)^2$.

The remainder without the new atom. Let us bound $\mathcal{F}(t) - \mathcal{F}(1) - (t-1)\mathcal{F}'(1)$. Since

$$h'_+(r) = -(1/2)\operatorname{Im}\psi'(1/4 + ir/2), \quad h''_+(r) = -(1/4)\operatorname{Re}\psi''(1/4 + ir/2),$$

and $\psi^{(k)}(z) = O(|z|^{-k})$ for $|z| \to \infty$, the functions $rh'_+(r)$ and $r^2h''_+(r)$ are bounded. Consequently, the Archimedean term is twice differentiable in t with a uniformly bounded second derivative. The pole term is smooth: $P(u^{(t)}) = t^{1/2}\int u(y)\cosh(ty/2)\,\mathrm{d}y$. For an atom $n < c$ with $0 < \ln n < 2a$, Lemma 8.3 with $s = \ln n$ and $\tau = (\ln n)(1/t - 1)$ estimates its contribution to the remainder as $o(|t-1|/\ln(1/|t-1|))$. Collecting the terms, we find $\mathcal{F}(t) = \lambda(c) - 2a\beta(u)^2(t-1) + o(|t-1|/\ln(1/|t-1|))$.

The new atom. Take $t = 1 \pm \delta/a$. Then $c' = e^{2(a\pm\delta)}$ and $2a(t-1) = \pm 2\delta$. With the minus sign this is precisely the second inequality (8.4). With the plus sign the new atom term remains. Set $\eta = 2a - 2a/t = 2a\delta/(a+\delta)$. Applying Lemma 8.4 to u with the constants of Theorem 3.8, we find

$$g(2a/t) = g(2a-\eta) = \frac{\beta(u)^2\eta}{\ln(1/\eta)} + O(\eta(\ln(1/\eta))^{-3/2}) = \frac{2\beta(u)^2\delta}{\ln(1/\delta)} + O(\delta(\ln(1/\delta))^{-3/2})$$

(here $\eta = 2\delta + O(\delta^2)$ and $\ln(1/\eta) = \ln(1/\delta) + O(1)$), and the first inequality (8.4) follows. If $c = p^m$, choose $u \in \mathcal{N}_{a_*}$ with $\beta(u)^2 = \overline{b}(a_*)$ in the first inequality and with $\beta(u)^2 = \underline{b}(a_*)$ in the second. It remains to recall that $\lambda'_+(a_*) = -2\overline{b}(a_*)$ and $\lambda'_-(a_*) = -2\underline{b}(a_*)$ by Theorem 8.1, which is precisely (8.5). □

For $c \neq p^m$ the first inequality (8.4) is the usual dilation estimate, and for $c = p^m$ the term $-4w_c\beta(u)^2\delta/\ln(1/\delta)$ appears in it from the atom $n = c$, which joins the layers at the two ends (simplicity of the eigenvalue $\lambda(c_*)$ in Theorem 8.5 is not assumed). The contribution of this atom to the form at the dilated minimizer equals $-2w_*g(2a-\eta)$, and by Lemma 8.4 the overlap of the layers of width η equals $\beta^2\eta/\ln(1/\eta)$ up to $O(\eta(\ln(1/\eta))^{-3/2})$. On the left, for $c_- < c < c_*$, where c_- is the largest power of a prime smaller than c_*, all shifts $\ln n$ of the form are at distance at least $\ln(c/c_-)$ from $2a$: the layers at the ends are not joined by them.

Corollary 8.6 (the modulus of continuity of the derivative near a power of a prime)**.** *Let $c_* = p^m$ and $\overline{b} = \overline{b}(a_*) > 0$. Denote by λ' the derivative of the function $y \mapsto \lambda(e^{2y})$ wherever it exists, that is, everywhere outside an at most countable set (Corollary 8.2). For $\delta \to 0+$*

(a)
$$\frac{1}{\delta}\int_{a_*}^{a_*+\delta} (\lambda'_+(a_*) - \lambda'(y))\,\mathrm{d}y \geq \frac{4w_*\overline{b} + o(1)}{\ln(1/\delta)},$$
and hence the supremum of $\lambda'_+(a_*) - \lambda'(y)$ over the points of differentiability $y \in (a_*, a_*+\delta)$ is no less than $(4w_*\overline{b} + o(1))/\ln(1/\delta)$. In particular, if $\omega(\delta)\ln(1/\delta) \to 0$, then ω does not serve as a modulus of continuity of λ' to the right of a_*, and $\lambda \notin C^{1,\alpha}([a_*, a_*+\delta_0])$ for any $\alpha > 0$ and $\delta_0 > 0$;

(b) *from the left*
$$\frac{1}{\delta}\int_{a_*-\delta}^{a_*} (\lambda'(y) - \lambda'_-(a_*))\,\mathrm{d}y \geq -o\left(\frac{1}{\ln(1/\delta)}\right);$$

(c) *for minimizers u_y the supremum of $\beta(u_y)^2$ over the same points y is no less than $\overline{b}(1 + (2w_* + o(1))/\ln(1/\delta))$, and for $\lambda(c_*) > 0$ the same holds for r_c with the right-hand side $r_*(1 + (2w_* + o(1))/\ln(1/\delta))$, where $r_* = \overline{b}/(2\kappa_\Theta\lambda(c_*))$ and κ_Θ is taken at $c = c_*$.*

Proof. For a function $y \mapsto \lambda(e^{2y})$, Lipschitz by Proposition 4.2, we have $\lambda(e^{2(a_*+\delta)}) - \lambda(c_*) = \int_{a_*}^{a_*+\delta}\lambda'$. Item (a) is the first inequality (8.5) divided by δ, and (b) is the second one. At points of differentiability $\lambda' = -2\beta(u_y)^2$ for any minimizer (Corollary 8.2). Moreover, $\lambda'_+(a_*) = -2\overline{b}$, which proves the first claim of (c). Next, $0 < \lambda(e^{2y}) \leq \lambda(c_*)$ for small δ by continuity and monotonicity, while κ_Θ depends smoothly on c (the series for Δ in Lemma 5.3 is differentiated termwise) and is Lipschitz near c_*. Hence, $r_c \geq \beta(u_y)^2/(2\kappa_\Theta(c_*)\lambda(c_*)) \cdot (1 + O(\delta))$. □

The exact r_c has no jump at the point $c_* = p^m$ if $\lambda(c_*)$ is a simple eigenvalue (Theorem 4.5(i)), but to the right of c_* the function r_c has a logarithmic cusp (Corollary 8.6), while on the left a correction of the same sign is impossible.

8.3. **Contribution of the new atom.** Let us compare λ to the right of c_* with the infimum of the form from which the atom $n = c_*$ is excluded. Let $c_+ > c_*$ be the next power of a prime. For $c \in [c_*, c_+)$ set

$$Q_c^\circ(f) = Q_c(f) + 2w_*(f \star \tilde f)(2a_*),$$

that is, Q_c° is the form Q_c without the atom $n = c_*$, and denote by $\lambda^\circ(c)$ its infimum in the even sector. On functions supported in I_{c_*} we have $(f \star \tilde f)(2a_*) = 0$, and therefore $\lambda^\circ(c_*) = \lambda(c_*)$. The proofs of Theorems 3.1, 3.10, Corollary 3.9 and Lemma 4.4 carry over without changes to the form Q_c° of the same kind as Q_c, with atoms $n < c_*$.

Theorem 8.7 (contribution of the new atom)**.**

(a) *For any $c \in (c_*, c_+)$*

$$-2w_*(u \star \tilde u)(2a_*) \le \lambda(c) - \lambda^\circ(c) \le -2w_*(v \star \tilde v)(2a_*), \tag{8.6}$$

where u and v are the normalized minimizers of the forms Q_c and Q_c° in the even sector.

(b) *If $a_* \in \mathcal{P}$ and $\beta_* = \beta(u_{a_*})$, then for $\delta \to 0+$*

$$\lambda(e^{2(a_*+\delta)}) - \lambda^\circ(e^{2(a_*+\delta)}) = -(4w_*\beta_*^2 + o(1))\frac{\delta}{\ln(1/\delta)}.$$

Proof. (a) We have

$$\lambda(c) \le Q_c(v) = Q_c^\circ(v) - 2w_*(v \star \tilde v)(2a_*) = \lambda^\circ(c) - 2w_*(v \star \tilde v)(2a_*),$$
$$\lambda^\circ(c) \le Q_c^\circ(u) = \lambda(c) + 2w_*(u \star \tilde u)(2a_*).$$

(b) Let $a = a_* + \delta$, $u = u_a$ with sign $\langle u_a, u_{a_*}\rangle \ge 0$ and $v = v_a$ a minimizer of $Q^\circ_{e^{2a}}$ with the same sign condition. From Lemma 4.4 and Theorem 3.10 it follows that $\beta(u_a) \to \beta_*$ as $\delta \to 0+$. For v_a the argument is the same. By compactness of the embedding (the proof of Theorem 4.3) every sequence $a_k \downarrow a_*$ has a subsequence along which v_{a_k} converges in L^2 to a function v with $\|v\| = 1$ and support in I_{c_*}. The energy is lower semicontinuous, and the remaining terms of the form Q° are continuous, hence $Q_{c_*}(v) \le \liminf \lambda^\circ(e^{2a_k}) \le \lambda(c_*)$ (the function λ° is nonincreasing). By simplicity $v = u_{a_*}$, and $\beta(v_{a_k}) \to \beta_*$ by the theorem on continuity of the boundary coefficient, applied to Q°. By Corollary 3.9 (for Q_c and for Q_c° with $\overline\lambda = \lambda(c_*) + 1$) the condition of Lemma 8.4 holds for u_a and v_a with common C and δ_0. For $\eta = 2\delta$, when $(f \star \tilde f)(2a_*) = (f \star \tilde f)(2a - \eta)$, by (8.3) both bounds in (8.6) equal

$$-4w_*(\beta_*^2 + o(1))\frac{\delta}{\ln(1/(2\delta))} = -(4w_*\beta_*^2 + o(1))\frac{\delta}{\ln(1/\delta)},$$

where it is taken into account that $\ln(1/(2\delta)) = \ln(1/\delta) - \ln 2$. □

8.4. **Convergence rate of the boundary coefficient.** We find how fast the boundary coefficient of the minimizer of a form with a fixed set of atoms changes as the length of the interval varies (Theorem 8.12). It follows from this estimate that at points of the set $\mathcal{P}$ to the left of c_* and between powers of primes the modulus of continuity of the derivative of λ is $O((\ln(1/h))^{-5/4})$ (Corollary 8.13), and Section 8.5 gives the exact asymptotics of the derivative to the right of c_*. Forms with a fixed set of atoms are defined as follows.

Definition 8.8 (form with a prescribed set of atoms)**.** Let $\mathcal{S}$ be a finite set of natural numbers $n \ge 2$. Denote by $Q_a^{\mathcal{S}}$ the form on $L^2(I_a)$, $I_a = [-a, a]$, which arises from (2.1) when the summation is taken only over $n \in \mathcal{S}$, and by $A^{\mathcal{S}}$, $\lambda^{\mathcal{S}}(a)$, and $\lambda_2^{\mathcal{S}}(a)$ its atom operator and the first two eigenvalues of the even sector. We say that $\mathcal{S}$ *is separated from the edges* on a compact interval J with margin $\varrho > 0$ if $4\varrho \le \ln n \le 2a - 4\varrho$ for all $n \in \mathcal{S}$ and $a \in J$.

For $\mathcal{S} = \{n : n < e^{2a}\}$ the form $Q_a^{\mathcal{S}}$ is Q_c, and for $\mathcal{S} = \{n : n < c_*\}$ and $a \ge a_*$ it is the form Q_c°. All statements of Sections 3, 4 and Theorem 3.11 carry over to $Q_a^{\mathcal{S}}$ with the same proofs, provided $\ln n < 2a$ for $n \in \mathcal{S}$. Instead of $\{n : n < c_*\}$ one can take the set of powers of primes $n < c_*$: for the other n the weight $\Lambda(n)$ is zero. It is separated from the edges on $J = [a_* - \varrho, a_* + \varrho]$, $0 < \varrho < a_*$, for $8\varrho \le \ln(c_*/c_-)$ and $4\varrho \le \ln 2$, while for $c_* = 2$ this set is empty and the condition holds for any such ϱ.

Lemma 8.9 (uniform estimates of eigenfunctions)**.** *Let $\mathcal{S}$ be as in Definition 8.8, $J \subset (0, \infty)$ be a compact interval, $\varrho > 0$ and $\overline\lambda \in \mathbb{R}$. There exist constants C and $h_0 > 0$ such that every normalized real-valued eigenfunction u of the form $Q_a^{\mathcal{S}}$, $a \in J$, with eigenvalue at most $\overline\lambda$ satisfies the conditions*

(a) $\|u'\|_{L^1(\mathbb{R})} \le C$;

(b) $|u(x+h) - u(x)| \le C/\ln(1/h)$ *for $0 < h \le h_0$ and $a - |x| \ge \varrho$;*

(c) $|u(x+h)-u(x)| \le C(\ln(1/h))^{-1/2}$ *for* $0<h\le h_0$ *and all* $x\in\mathbb{R}$.

Proof. Atoms with $\ln n \ge 2a$ do not act on functions supported in I_a: the shift by $\pm\ln n$ takes I_a into an interval intersecting I_a in at most one point. Hence one may assume $\ln n < 2a$ for $n\in\mathcal{S}$, and the transfer of the statements to $Q_a^{\mathcal{S}}$ after Definition 8.8 applies.

(a) All the constants in the proof of Theorem 3.11 are expressed in terms of $\|u\|_\infty$, $|\mu|$, $\|f\|_\infty$, K, $\sigma=\sum_{n\in\mathcal{S}}\Lambda(n)n^{-1/2}$, a, and the values of the kernel $\mathcal{G}$ at the point r_0. The first four quantities are uniformly bounded by Theorem 3.1 and Corollary 3.9, while σ and a lie in compact sets. Hence the constant C_1 of that proof is bounded, r_0 can be chosen uniformly, with $2m_{\mathcal{G}}(r_0)C_1\le 1/2$, and the same holds for the remaining constants C_2, C_3, $C_4(r_0)$. The constant C_3 equals $\int_{I_a}|u_2'|+4(\|u\|_\infty+\|u_1\|_\infty)$. Here $\|u_1\|_\infty\le\|f\|_\infty$, since $\int\mathcal{G}=1$, and the integral is bounded by the estimate $|\mathcal{U}_o(\pm(a+\varepsilon))|\le C(1+\ln^+(1/\varepsilon))^{1/2}e^{-\varepsilon}$ with C depending only on K and $\|u\|_\infty$. The estimate $\mathcal{T}_h(u)\le 2C_4(r_0)h$, valid for all $h>0$, with which that proof concludes, together with $\|u'\|_1=\sup_{h>0}\mathcal{T}_h(u)/h$ gives (a).

(b) By the proof of Theorem 3.11, $u=\mathcal{G}\star\mathcal{U}$, where $\mathcal{U}=f$ on I_a, while $\|f\|_{L^\infty(I_a)}$ and $\|\mathcal{U}\|_2$ are uniformly bounded. For an even function $\mathcal{G}$ decreasing on $(0,\infty)$ we have $\int|\mathcal{G}(z+h)-\mathcal{G}(z)|\,\mathrm{d}z=2m_{\mathcal{G}}(h/2)$, since $\mathcal{G}(z)>\mathcal{G}(z+h)$ only when $z>-h/2$. If $a-|x|\ge\varrho$, then the points y with $|x-y|<\varrho/2$ lie in I_a, and their contribution to $u(x+h)-u(x)$ is at most $2m_{\mathcal{G}}(h/2)\|f\|_{L^\infty(I_a)}$. For the remaining y we have $|\mathcal{G}(x+h-y)-\mathcal{G}(x-y)|\le\mathcal{G}(\varrho/4)$ and $\int_{|z|\ge\varrho/2}|\mathcal{G}(z+h)-\mathcal{G}(z)|\,\mathrm{d}z\le 2h\mathcal{G}(\varrho/4)$ when $h\le\varrho/4$. By the Cauchy–Schwarz inequality their contribution does not exceed $(2h)^{1/2}\mathcal{G}(\varrho/4)\|\mathcal{U}\|_2$. Here $m_{\mathcal{G}}(h/2)\le C_{\mathcal{G}}/(1+2\ln(2/h))+O(h^{1/2})$ by (3.6).

(c) Let r be the distance from x to the nearest of the endpoints $\pm a$. If $r\ge h^{1/3}+h$, then the argument of (b) with ϱ replaced by $h^{1/3}$ leads to the estimate $|u(x+h)-u(x)|\le 2m_{\mathcal{G}}(h/2)\|f\|_\infty+Ch^{1/6}$, since $\mathcal{G}(r/4)\le C/r$. If, however, $r<h^{1/3}+h$, then both points x, $x+h$ lie outside I_a or at a distance less than $2h^{1/3}$ from the edge. The modulus of the value of u at each of them, by Corollary 3.9, is at most $K(\ln(1/(2h^{1/3})))^{-1/2}$. □

The next lemma refines Theorem 3.1 for solutions of the equation with a right-hand side: the kernel ϖ near zero is truncated at a distance ε, and the denominator of the estimate grows like $\ln(1/\varepsilon)$.

Lemma 8.10 (a logarithmic gain in L^∞)**.** *Let* $\mu\in\mathbb{R}$, $\mathfrak{g}\in L^\infty(I_a)$, *and let a real function* $d\in D(Q_a^{\mathcal{S}})$ *satisfy the equation*

$$\mathcal{B}(d,\varphi)=\langle\mu d+A^{\mathcal{S}}d-\Pi d+\mathfrak{g},\varphi\rangle,\quad \varphi\in D(Q_a^{\mathcal{S}}).$$

Set $\sigma=\sum_{n\in\mathcal{S}}\Lambda(n)n^{-1/2}$ *and suppose that* $m_\varepsilon>\mu+2\sigma$ *in the notation of Theorem 3.1. Then*

$$\|d\|_\infty\le\frac{(\|\varpi_\varepsilon\|_2+C_\Pi)\|d\|_2+\|\mathfrak{g}\|_\infty}{m_\varepsilon-\mu-2\sigma}. \tag{8.7}$$

Moreover $m_\varepsilon=\ln(1/\varepsilon)+O(1)$ *and* $\|\varpi_\varepsilon\|_2=O(\varepsilon^{-1/2})$ *as* $\varepsilon\to 0$.

Proof. We repeat the proof of Theorem 3.1 with $q=(d-t)_+$. Since $d\le t\mathbb{1}_{I_a}+q$, we have $\langle A^{\mathcal{S}}d,q\rangle\le 2\sigma t\int q+2\sigma\|q\|^2$, then $\langle\varpi_\varepsilon\star d,q\rangle\le\|\varpi_\varepsilon\|_2\|d\|_2\int q$, $-\langle\Pi d,q\rangle\le C_\Pi\|d\|_2\int q$ and $\langle\mathfrak{g},q\rangle\le\|\mathfrak{g}\|_\infty\int q$. Hence $q=0$ with t equal to the right-hand side of (8.7), and likewise for $-d$. The asymptotics follow from $\varpi(z)=1/(2|z|)+O(1)$ near zero and the exponential decay of ϖ. □

Lemma 8.11 (the boundary coefficient as a functional of the right-hand side)**.** *Let* z *be the weak solution of the Dirichlet problem* $L_\Delta z=f$ *in* $I=(-a,a)$, *where* $z\in H(I)\cap L^\infty(\mathbb{R})$ *and* $f\in L^\infty(I)$, φ_δ *are the bump functions of Lemma 3.6, and* $\xi_1\ge\ln(4/\min(1,a))$. *Then*

$$\beta_+(z)=\frac14\int_{\xi_1}^\infty\langle f,\varphi_{e^{-\xi}}\rangle\xi^{-3/2}\,\mathrm{d}\xi+R,\quad |R|\le C_\chi\xi_1^{1/2}\|z\|_\infty, \tag{8.8}$$

where C_χ *depends only on* χ *and on an upper bound for* a.

Proof. In the proof of Lemma 3.6(a) all terms $e(\delta)$ except $\langle f,\varphi_\delta\rangle$ are estimated through $\|z\|_\infty$, that is,

$$e(\delta)=\langle f,\varphi_\delta\rangle+e_0(\delta),\quad |e_0(\delta)|\le C_\chi\|z\|_\infty.$$

In part (b) of the same proof, $U_\zeta=\zeta\star U$ with $\zeta(\sigma)=\chi(e^\sigma)e^\sigma$ and $\tilde e=2\xi U_\zeta'-U_\zeta$ are introduced, and $\tilde e(\xi)-e(e^{-\xi})=U(\xi-\ln 4)-U_\zeta(\xi)$ is in absolute value at most $3\ln 2\cdot\|z\|_\infty$. Moreover, $(\xi^{-1/2}U_\zeta)'=\tilde e/(2\xi^{3/2})$. Integrating from ξ_1 to ∞, we obtain $B=\xi_1^{-1/2}U_\zeta(\xi_1)+\int_{\xi_1}^\infty\tilde e(2\xi^{3/2})^{-1}\,\mathrm{d}\xi$, where $|U_\zeta(\xi_1)|\le\|z\|_\infty(\xi_1+|\ln 2a|+3)$. The last estimate follows from the inequalities $|U(\xi)|\le\|z\|_\infty(\xi+|\ln 2a|)$ and $|U_\zeta-U|\le\ln 2\cdot\|z\|_\infty$. It remains to substitute $\tilde e(\xi)=\langle f,\varphi_{e^{-\xi}}\rangle+O(\|z\|_\infty)$: the remainder contributes $O(\xi_1^{-1/2}\|z\|_\infty)$ to the integral, and $\beta_+=B/2$ (Theorem 3.8), and (8.8) is proved. □

Lemma 8.11 shows that the boundary coefficient of a solution is determined by the values of the right-hand side in an arbitrarily narrow boundary layer, provided the function itself is small in L^∞, while the contribution of a strip of width $e^{-\xi_1}$ enters with weight $\xi_1^{-1/2}$.

Theorem 8.12 (the rate of convergence of the boundary coefficient)**.** *Let $\mathcal{S}$ be separated from the edges on $J=[a_*-\varrho,a_*+\varrho]$ with margin ϱ, $0<\varrho<a_*$, let $\lambda^{\mathcal{S}}(a_*)$ be a simple eigenvalue, u_* the normalized minimizer of $Q^{\mathcal{S}}_{a_*}$, and for $a\in J$ let the function v_a be the corresponding minimizer of $Q^{\mathcal{S}}_a$ with sign $\langle v_a^{(a_*/a)},u_*\rangle\ge 0$. For $\delta=|a-a_*|\to 0$*

$$|\beta(v_a)-\beta(u_*)|=O((\ln(1/\delta))^{-5/4}). \tag{8.9}$$

Proof. We dilate the minimizer at a onto the interval I_{a_*} and compare it with u_*. Their difference is small in L^2 by the minimax principle, Lemma 8.10 converts this into an estimate in L^∞ with a logarithmic gain, and the boundary coefficient of the difference is then estimated by Lemma 8.11.

Set $t=a/a_*$, $\xi_\delta=\ln(1/\delta)$ and $\tilde v=v_a^{(1/t)}$, that is, $\tilde v(y)=t^{1/2}v_a(ty)$. This is a normalized even function supported in I_{a_*}.

The equation. Dilation is unitary, takes $D(Q^{\mathcal{S}}_{a_*})$ to $D(Q^{\mathcal{S}}_a)$ and changes the energy by the multiplier M_t with symbol $h_+(r)-h_+(r/t)$, that is, $\mathcal{B}(f^{(t)},\varphi^{(t)})=\mathcal{B}(f,\varphi)-\langle M_tf,\varphi\rangle$. Substituting into the equation for v_a the test functions $\varphi^{(t)}$, we obtain for all $\varphi\in D(Q^{\mathcal{S}}_{a_*})$ the equality $\mathcal{B}(\tilde v,\varphi)=\langle\mu_a\tilde v+A^{(t)}\tilde v-\Pi^{(t)}\tilde v+M_t\tilde v,\varphi\rangle$. Here $\mu_a=\lambda^{\mathcal{S}}(a)-h_+(0)$, $A^{(t)}$ is the operator of the atoms $n\in\mathcal{S}$ with shifts $(\ln n)/t$, $\Pi^{(t)}\varphi=2\langle\varphi,c_t\rangle c_t$ with $c_t(y)=t^{1/2}\cosh(ty/2)$. Let $s=\langle\tilde v,u_*\rangle\ge 0$ and $d=\tilde v-su_*$. For this difference $\mathcal{B}(d,\varphi)=\langle\mu_*d+A^{\mathcal{S}}d-\Pi d+\mathfrak{g},\varphi\rangle$, $\mu_*=\lambda^{\mathcal{S}}(a_*)-h_+(0)$, where

$$\mathfrak{g}=(\mu_a-\mu_*)\tilde v+(A^{(t)}-A^{\mathcal{S}})\tilde v-(\Pi^{(t)}-\Pi)\tilde v+M_t\tilde v.$$

Estimates of $\mathfrak{g}$. By Proposition 4.2, transferred to $Q^{\mathcal{S}}_a$ (see after Definition 8.8), $|\mu_a-\mu_*|=O(\delta)$. Next,

$$h_+(r)-h_+(r/t)=\int_{1/t}^1(r\tau)h_+'(r\tau)\frac{\mathrm{d}\tau}{\tau}=\ln t+m_t(r),$$

where $(1+r^2)|rh_+'(r)-1|$ is a bounded function: $\psi'(z)=z^{-1}+z^{-2}/2+O(|z|^{-3})$. Consequently, $|m_t(r)|\le C|\ln t|(1+r^2)^{-1}$ for $t\in[1/2,2]$, whence by the inversion formula and the Cauchy–Schwarz inequality $\|m_t(D)\tilde v\|_\infty\le(2\pi)^{-1}\|m_t\|_2\|\widehat{\tilde v}\|_2=O(\delta)$ and $\|M_t\tilde v\|_\infty=O(\delta)$. The term with Π equals $O(\delta)$ in L^∞. For the atoms

$$|(A^{(t)}-A^{\mathcal{S}})\tilde v(x)|\le\sum_{n\in\mathcal{S}}\Lambda(n)n^{-1/2}\sum_{\pm}|\tilde v(x\pm(\ln n)/t)-\tilde v(x\pm\ln n)|$$

with $|(\ln n)/t-\ln n|=O(\delta)$. Then $\|\mathfrak{g}\|_\infty=O(\xi_\delta^{-1/2})$ by Lemma 8.9(c). If $a_*-|x|<\varrho$, then among the points $x\pm\ln n$, $x\pm(\ln n)/t$ those lying in I_{a_*} are at distance at least 2ϱ from $\pm a_*$. Therefore $|\mathfrak{g}(x)|=O(\xi_\delta^{-1})$ in boundary strips of width ϱ by item (b) of the same lemma.

Estimate in L^2. Since $Q^{\mathcal{S}}_{a_*}(u_*,d)=\lambda^{\mathcal{S}}(a_*)\langle u_*,d\rangle=0$ and $s^2+\|d\|^2=1$, $(\lambda_2^{\mathcal{S}}(a_*)-\lambda^{\mathcal{S}}(a_*))\|d\|^2\le Q^{\mathcal{S}}_{a_*}(\tilde v)-\lambda^{\mathcal{S}}(a_*)$ by the minimax principle. The form on $\tilde v$ differs from $Q^{\mathcal{S}}_a(v_a)=\lambda^{\mathcal{S}}(a)$ by $O(\delta)$: the Archimedean term by $(2\pi)^{-1}\int|h_+(r)-h_+(r/t)||\widehat{\tilde v}(r)|^2\,\mathrm{d}r\le C_h|\ln t|$ with $C_h=\sup|rh_+'(r)|<\infty$ (proof of Proposition 7.2), the pole term by $O(\delta)$, the atoms by $2\sum\Lambda(n)n^{-1/2}|g(\ln n)-g(t\ln n)|\le C\delta\|v_a\|_\infty\|v_a'\|_1$ by Lemma 8.9(a), where $g=v_a\star\tilde v_a$. Together with $|\lambda^{\mathcal{S}}(a)-\lambda^{\mathcal{S}}(a_*)|=O(\delta)$ this gives $\|d\|_2=O(\delta^{1/2})$.

Estimate in L^∞. By Lemma 8.10 with $\varepsilon=\delta^{1/2}$ we have

$$\|d\|_\infty\le\frac{O(\delta^{-1/4}\delta^{1/2})+O(\xi_\delta^{-1/2})}{\xi_\delta/2-O(1)}=O(\xi_\delta^{-3/2}).$$

The boundary coefficient. The difference d, as in Lemma 3.3, is a weak solution of the Dirichlet problem $L_\Delta d=f_d$ in I_{a_*} with $f_d=2(\mu_*d+A^{\mathcal{S}}d-\Pi d+\mathfrak{g})+\mathcal{V}d$. In boundary strips of width ϱ we have $|f_d|\le C\|d\|_\infty+2|\mathfrak{g}|=O(\xi_\delta^{-1})$. By Lemma 8.11 with $\xi_1=\xi_\delta^{1/2}$ (for small δ the supports of $\varphi_{e^{-\xi}}$, $\xi\ge\xi_1$, lie in the boundary strip)

$$|\beta(d)|\le\frac14O(\xi_\delta^{-1})\cdot2\xi_1^{-1/2}+O(\xi_1^{1/2}\xi_\delta^{-3/2})=O(\xi_\delta^{-5/4}).$$

Here the strip of width $e^{-\xi_1}$, where $|f_d|=O(\xi_\delta^{-1})$, contributes to $\beta(d)$ a quantity of order $\xi_1^{-1/2}\xi_\delta^{-1}$, and the remainder of the same lemma equals $O(\xi_1^{1/2}\xi_\delta^{-3/2})$. The choice $\xi_1=\xi_\delta^{1/2}$ balances the two contributions and gives the exponent $-5/4$.

Return to v_a. The boundary coefficient is linear, and

$$\beta(\tilde v) = s\beta(u_*) + \beta(d), \quad s = (1-\|d\|^2)^{1/2} = 1 - O(\delta).$$

Moreover, from $v_a(a-\sigma) = t^{-1/2}\tilde v(a_* - \sigma/t)$ it follows that $\beta(v_a) = t^{-1/2}\beta(\tilde v)$, and (8.9) is proved. □

Corollary 8.13 (modulus of continuity of the derivative for a fixed set of atoms)**.** *Suppose that $\mathcal S$ is separated from the edges on a compact interval J with a margin ϱ, and $\mathcal K \subset J$ is a compact set at every point a of which $\lambda^{\mathcal S}(a)$ is a simple eigenvalue. There exists $C_{\mathcal K}$ such that $|(\lambda^{\mathcal S})'(a) - (\lambda^{\mathcal S})'(b)| \le C_{\mathcal K}(\ln(1/|a-b|))^{-5/4}$ for $a \in \mathcal K$ and $|a-b|$ sufficiently small. In particular, the function $y \mapsto \lambda(e^{2y})$ has such a modulus of continuity of the derivative on any compact set $\mathcal K \subset \mathcal P$ containing no points $(\ln p^m)/2$, and at every point $a_* \in \mathcal P$ with $c_* = p^m$ from the left:*

$$\lambda'(a_* - \delta) = \lambda'(a_*) + O((\ln(1/\delta))^{-5/4}), \quad \delta \to 0+.$$

Proof. The set of simplicity is open: the proof of Lemma 4.4 carries over to $Q^{\mathcal S}$ and gives the continuity of $\lambda_2^{\mathcal S} - \lambda^{\mathcal S}$. Near $\mathcal K$ by Theorem 4.5(i) $(\lambda^{\mathcal S})' = -2\beta(v)^2$. The estimates in the proof of Theorem 8.12 depend on a_* only through the constants of Lemma 8.9 and the gap $\lambda_2^{\mathcal S} - \lambda^{\mathcal S}$, which is continuous and is bounded away from zero on $\mathcal K$. If $c_* = p^m$, then for $a \le a_*$ we have $\lambda(e^{2a}) = \lambda^{\mathcal S}(a)$ (the atoms with $\ln n \ge 2a$ do not act on functions supported in I_a), where $\mathcal S$ consists of powers of primes $n < c_*$ and is separated from the edges on $[a_* - \varrho, a_* + \varrho]$ for small ϱ. □

8.5. **Derivative to the right of a prime power.** To the right of a prime power the set of atoms of Q_c is not separated from the edges: the shift $2a_*$ differs from $2a$ by 2δ. The same argument, in which this atom is taken into account separately, leads to the pointwise asymptotics.

Theorem 8.14 (derivative of the infimum to the right of a prime power)**.** *Let $c_* = p^m$, $a_* \in \mathcal P$ and $\beta_* = \beta(u_{a_*})$. For $\delta \to 0+$*

$$\begin{aligned} \beta(u_{a_*+\delta}) &= \beta_*\left(1 + \frac{w_*}{\ln(1/\delta)}\right) + O((\ln(1/\delta))^{-5/4}), \\ \lambda'(a_*+\delta) &= \lambda'(a_*)\left(1 + \frac{2w_*}{\ln(1/\delta)}\right) + O((\ln(1/\delta))^{-5/4}), \end{aligned} \tag{8.10}$$

where the sign of $u_{a_*+\delta}$ is chosen as in Theorem 8.12, and

$$\frac{\lambda(e^{2(a_*+\delta)}) - \lambda(c_*)}{\delta} = -2\beta_*^2 - \frac{4w_*\beta_*^2}{\ln(1/\delta)} + O((\ln(1/\delta))^{-5/4}). \tag{8.11}$$

Proof. The new atom transports the values of the minimizer at the opposite end into a layer of width η at the endpoint. From Lemma 8.11 it is seen that such an addition to the right-hand side changes the boundary coefficient by $w_*\beta/\ln(1/\eta)$, while a term of the same amplitude at a distance of order η from the endpoint changes it only by $O(\beta/\ln^2(1/\eta))$. The boundary expansion of the derivative u' will not be needed for this.

Adding a new atom. Let $a = a_* + \delta$, $\mathcal S$ be the set of prime powers smaller than c_*, $v = u_a$, and t, $\tilde v$, s, d be as in the proof of Theorem 8.12 with $u_* = u_{a_*}$. The estimates of Lemma 8.9 are also valid for v. The atom $n = c_*$, entering the equation for v, adds to $\mathfrak g$ the term $\mathfrak N(x) = w_*(\tilde v(x + 2a_*/t) + \tilde v(x - 2a_*/t))$. This function is concentrated in layers of width $\eta = 2a_* - 2a_*/t = 2a_*\delta/a$ at the endpoints: for $0 \le \sigma < \eta$ one has $\mathfrak N(a_* - \sigma) = w_*\tilde v(a_* - (\eta - \sigma))$, and for $\sigma \ge \eta$ (and $\sigma \le a_*$) $\mathfrak N(a_* - \sigma) = 0$. Moreover $\|\mathfrak N\|_\infty = O(\xi_\delta^{-1/2})$, $\xi_\delta = \ln(1/\delta)$, by Corollary 3.9. The remaining terms of $\mathfrak g$ and the estimates in L^2 and L^∞ from the same proof remain unchanged: $Q_{c_*}(\tilde v) = \lambda(e^{2a}) + 2w_*(v \star \tilde v)(2a_*) + O(\delta)$, and $(v \star \tilde v)(2a_*) = O(\delta/\xi_\delta)$ by Lemma 8.4. Altogether $\|d\|_2 = O(\delta^{1/2})$ and $\|d\|_\infty = O(\xi_\delta^{-3/2})$.

The boundary coefficient of the difference. By Lemma 8.11 with $\xi_1 = \xi_\delta^{1/2}$, as in Theorem 8.12, $\beta(d) = (1/4)\int_{\xi_1}^\infty \langle 2\mathfrak N, \varphi_{e^{-\xi}}\rangle \xi^{-3/2}\,\mathrm d\xi + O(\xi_\delta^{-5/4})$. Let $\xi_\eta = \ln(1/\eta)$ and $M = \xi_\delta^{1/2}$. The support of $\varphi_{e^{-\xi}}$ lies in $a_* - \sigma$ with $\sigma \in [e^{-\xi}/2, 2e^{-\xi}]$, and for $\xi < \xi_\eta - \ln 2$ the integrand vanishes, on the interval $[\xi_\eta - \ln 2, \xi_\eta + M]$ it does not exceed $2\|\mathfrak N\|_\infty \xi^{-3/2}$ and contributes $O(M\xi_\delta^{-2})$, while for $\xi > \xi_\eta + M$ we

have $\eta-\sigma\in[\eta(1-2e^{-M}),\eta]$ and, by Corollary 3.9, $\langle 2\mathfrak{N},\varphi_{e^{-\xi}}\rangle = 2w_*\beta(\tilde v)\xi_\eta^{-1/2}+O(\xi_\delta^{-1})$. Consequently,

$$\beta(d)=\frac14\cdot 2(\xi_\eta+M)^{-1/2}\cdot 2w_*\beta(\tilde v)\xi_\eta^{-1/2}+O(\xi_\delta^{-5/4})=\frac{w_*\beta(\tilde v)}{\xi_\delta}+O(\xi_\delta^{-5/4}),$$

since $\xi_\eta=\xi_\delta+O(1)$. Substituting this into $\beta(\tilde v)=s\beta_*+\beta(d)$, we obtain

$$\beta(\tilde v)\left(1-\frac{w_*}{\xi_\delta}\right)=\beta_*+O(\xi_\delta^{-5/4}),$$

and $\beta(u_a)=t^{-1/2}\beta(\tilde v)$ with $t^{-1/2}=1+O(\delta)$. Division by $1-w_*/\xi_\delta$, that is, multiplication by $1+w_*/\xi_\delta+O(\xi_\delta^{-2})$, gives the first formula (8.10). The second follows from $\lambda'=-2\beta(u_a)^2$ (Theorem 4.5(i), $\mathcal{P}$ open). Integrating it over $(a_*,a_*+\delta)$ and using $\int_0^\delta \ln(1/y)^{-1}\,\mathrm{d}y=(\delta/\xi_\delta)(1+O(\xi_\delta^{-1}))$, we obtain (8.11). □

Thus for a simple eigenvalue $\lambda(c_*)$ the first inequality (8.5) turns into an equality, and from Theorem 8.7 and (8.11) for the form Q° it follows that $(\lambda^\circ(e^{2(a_*+\delta)})-\lambda(c_*))/\delta=-2\beta_*^2+o(1/\ln(1/\delta))$. Corollary 8.13, applied to $\lambda^{\mathcal{S}}$ with the set $\mathcal{S}$ of prime powers smaller than c_*, refines the remainder to $O((\ln(1/\delta))^{-5/4})$.

Remark 8.15 (the sign of the overlap for the form Q°). The inequality (8.6) is not asymptotic. If for some $c\in(c_*,c_+)$ it turned out that $\lambda^\circ(c)<0<\lambda(c)$, then $(v\star\tilde v)(2a_*)<0$ for the minimizer v of the form Q_c°: its values in the boundary layers of width $2(a-a_*)$ at the two ends are of opposite sign on average, and v is no longer close to u_{a_*} in the sense of Lemma 8.4. This concerns Q° to the right of c_* and says nothing about λ to the left, where $\lambda=\lambda^\circ$. A numerical example for $c_*=7$ is given in Section A.2.

8.6. **What the proved properties do not imply.** We show that the properties of the function $a\mapsto\lambda(e^{2a})$ established above do not settle the questions of the boundedness of r_c and of the smoothness of λ between prime powers. For this we construct a model function with the same properties and a boundary balance tending to infinity. Items (a)–(d) of the following proposition cover all the assertions of the paper about the function $a\mapsto\lambda(e^{2a})$:

- monotonicity and the Lipschitz property (Propositions 2.1 and 4.2);
- one-sided derivatives and kinks (Theorem 8.1 and Corollary 8.2), since a continuously differentiable function has no kinks;
- the behavior at prime powers (Theorems 8.5 and 8.14, Corollary 8.6), where for $\lambda^\dagger$ one has to take $\overline{b}=\underline{b}=-(\lambda^\dagger)'(a_*)/2$;
- the modulus of continuity of the derivative (Corollary 8.13);
- the rigorous values [5], from which Corollaries 5.7 and 5.8 were derived.

The proposition shows that these assertions imply neither the boundedness of r_c, nor that λ belongs to the class $C^{1,\alpha}$ between prime powers, nor that it does not.

Proposition 8.16 (the proved properties are insufficient)**.** *There exists a function $\lambda^\dagger$ mapping $[a_0,\infty)$ to $(0,\infty)$, with the following properties.*

(a) *The function $\lambda^\dagger$ is strictly decreasing and continuously differentiable.*

(b) *At every point $a_*=(\ln c_*)/2$, $c_*=p^m\ge 5$, for $\delta\to 0+$*

$$(\lambda^\dagger)'(a_*+\delta)=(\lambda^\dagger)'(a_*)\left(1+\frac{2w_*}{\ln(1/\delta)}\right)+O(\delta),$$
$$(\lambda^\dagger)'(a_*-\delta)=(\lambda^\dagger)'(a_*)+O(\delta),$$

where $w_*=\Lambda(c_*)c_*^{-1/2}$ and $(\lambda^\dagger)'(a_*)<0$. The second equality refers to $c_*>5$: the point a_0 is the left endpoint of the domain of $\lambda^\dagger$.

(c) *On any compact subset of $[a_0,\infty)$ avoiding the points $(\ln p^m)/2$ the function $\lambda^\dagger$ is infinitely differentiable.*

(d) *For $c=e^{2a}\in\{5,\dots,8\}$ one has $\lambda_c^-\le\lambda^\dagger(a)\le\lambda_c^+$, and for $c\in\{9,\dots,31\}\cup\{45,47,71\}$ the lower bound $\lambda^\dagger(a)\ge\lambda_c^-$ holds, where λ_c^- and λ_c^+ are the rigorous bounds [5] from Corollary 5.7.*

(e) *The quantity $-(\lambda^\dagger)'/(4\kappa_\Theta\lambda^\dagger)$ tends to infinity as $a\to\infty$.*

(f) *For any compact set $\mathcal{K}\subset[a_0,\infty)$ with interior points, lying strictly inside the interval between adjacent values $(\ln p^m)/2$, the function $\lambda^\dagger$ can be chosen so that it retains properties (a), (b), (d), (e) and does not belong to $C^{1,\alpha}(\mathcal{K})$ for any $\alpha>0$, although its derivative has on $\mathcal{K}$ the modulus of continuity $O((\ln(1/h))^{-5/4})$.*

Proof. We construct $\lambda^\dagger$ in two steps: first a smooth function λ_0, consistent with the rigorous bounds [5], then we add to it logarithmic cusps to the right of the points a_* and verify properties (a)–(f).

The smooth part. Set $\theta_k = (\lambda_k^- + \lambda_k^+)/(2\Theta_k)$ for $k = 5, \dots, 8$, $\theta_k = 2\lambda_k^-/\Theta_k$ for $k = 9, \dots, 31$ and $k \in \{45, 47, 71\}$ and $\theta_k = 2$ for the remaining $k = 32, \dots, 70$. The ratios λ_k^+/Θ_k and λ_k^-/Θ_k lie in $[0.41, 3.01]$ (Corollary 5.7 and [5, Tables A.1, A.3]), and for $k \le 31$ the second is not less than 1.04. Hence $\theta_k \in [0.82, 6.02]$ and $|\ln\theta_{k+1} - \ln\theta_k| \le \ln(6.02/0.82) < 2$. Let $\sigma_0 \in C^\infty(\mathbb{R})$ be nondecreasing, $\sigma_0 = 0$ on $(-\infty, 0]$, $\sigma_0 = 1$ on $[1, \infty)$ and $0 \le \sigma_0' \le 2$. Set $\ln\theta(c) = \ln\theta_k + (\ln\theta_{k+1} - \ln\theta_k)\sigma_0(c-k)$ for $c \in [k, k+1]$, $5 \le k \le 70$, further $\ln\theta(c) = \ln\theta_{71} - (\ln\theta_{71} + c^2)\sigma_0(c-71)$ for $c \ge 71$ (for $c \ge 72$ this is $-c^2$) and $\lambda_0(a) = \Theta_c\theta(c)$, $c = e^{2a}$. The function λ_0 is infinitely differentiable and positive. Since $\mathrm{d}/\,\mathrm{d}a = 2c\,\mathrm{d}/\,\mathrm{d}c$, on $[5, 71]$ we have $|\,\mathrm{d}\ln\theta/\,\mathrm{d}a| \le 2c \cdot 2 \cdot 2 = 8c$, and $4\kappa_\Theta \ge 4 \cdot 0.924 \cdot 2\pi c > 23c$ by (5.3). For $c \ge 71$ the factor θ does not increase (here $\ln\theta_{71} + c^2 > 0$). Therefore $(\ln\lambda_0)' = -4\kappa_\Theta + (\ln\theta)' \le -2\kappa_\Theta$ everywhere ($8c < 23c/2 < 2\kappa_\Theta$), and for $c \ge 72$ one has $(\ln\lambda_0)' = -4\kappa_\Theta - 4c^2$.

Logarithmic cusps. Let $G_1(\delta) = \int_0^\delta \mathrm{d}y/\ln(1/y)$ for $0 < \delta < e^{-1}$ and $G_1(\delta) = 0$ for $\delta \le 0$. The function G_1 is continuously differentiable, $G_1'(\delta) = 1/\ln(1/\delta) \to 0$ for $\delta \to 0+$ and $G_1(\delta) \le \delta/\ln(1/\delta)$ ($1/\ln(1/y)$ is increasing). Let $\omega_0 \in C^\infty(\mathbb{R})$, $\omega_0 = 1$ on $(-\infty, 1]$, $\omega_0 = 0$ on $[2, \infty)$, and $-2 \le \omega_0' \le 0$. For each prime power $c_* \ge 5$ set $k_* = 2w_*\lambda_0'(a_*) < 0$, take $\varepsilon_* \in (0, e^{-24}/2)$ and

$$\lambda^\dagger(a) = \lambda_0(a) + \sum_{c_*} k_*\omega_0((a - a_*)/\varepsilon_*)G_1(a - a_*).$$

The numbers ε_* are chosen so small that the intervals $[a_*, a_* + 2\varepsilon_*]$ are pairwise disjoint and contain no points $(\ln k)/2$ with integer k except, possibly, a_*, on them $|\lambda_0'| \ge |\lambda_0'(a_*)|/2$ and $2|k_*|\varepsilon_* \le \lambda_0/2$.

Verification. The terms with cusps are nonzero only on the pairwise disjoint intervals $(a_*, a_* + 2\varepsilon_*)$. Away from the points a_* they are infinitely differentiable, and at a_* they have a continuous zero derivative: $G_1(0) = G_1'(0+) = 0$. This gives (a) in the smoothness part and (c). For $0 < \delta < \varepsilon_*$ we have

$$(\lambda^\dagger)'(a_* + \delta) = \lambda_0'(a_* + \delta) + \frac{k_*}{\ln(1/\delta)} = \lambda_0'(a_*) + O(\delta) + \frac{2w_*\lambda_0'(a_*)}{\ln(1/\delta)},$$

while on the left $(\lambda^\dagger)' = \lambda_0'$. Part (b) is satisfied: $(\lambda^\dagger)'(a_*) = \lambda_0'(a_*) \le -2\kappa_\Theta\lambda_0 < 0$. Next, $(\lambda^\dagger)' = \lambda_0' + \sum_{c_*} k_*[\omega_0 G_1' + \varepsilon_*^{-1}\omega_0' G_1]$, where $k_*\omega_0 G_1' \le 0$, and on $[a_* + \varepsilon_*, a_* + 2\varepsilon_*]$

$$|k_*\varepsilon_*^{-1}\omega_0' G_1| \le \frac{2|k_*|}{\varepsilon_*} \cdot \frac{2\varepsilon_*}{\ln(1/(2\varepsilon_*))} = \frac{8w_*|\lambda_0'(a_*)|}{\ln(1/(2\varepsilon_*))} \le \frac{6|\lambda_0'(a_*)|}{24} \le \frac{|\lambda_0'|}{2},$$

since $w_* = \Lambda(c_*)c_*^{-1/2} \le \max_{x \ge 1} \ln x/\sqrt{x} = 2/e < 3/4$. Hence $(\lambda^\dagger)' \le \lambda_0'/2 < 0$, and $\lambda^\dagger$ decreases. The terms with cusps are nonpositive and in absolute value do not exceed $2|k_*|\varepsilon_*$. Consequently, $\lambda^\dagger \ge \lambda_0/2 > 0$. At the points $(\ln k)/2$, $k = 5, \dots, 71$, these summands vanish, therefore $\lambda^\dagger = \theta_k\Theta_k$, and (d) holds. Finally, $\lambda^\dagger \le \lambda_0$ and $-(\lambda^\dagger)' \ge -\lambda_0'/2$, and therefore for $c \ge 72$

$$-\frac{(\lambda^\dagger)'}{4\kappa_\Theta\lambda^\dagger} \ge -\frac{\lambda_0'}{8\kappa_\Theta\lambda_0} = \frac{1}{2}\left(1 + \frac{c^2}{\kappa_\Theta}\right) \to \infty,$$

which proves (e).

Modulus of continuity (item (f)). Let $\mathcal{K}$ lie in the interval between neighboring values $(\ln p^m)/2$, let b be an interior point of $\mathcal{K}$, and $G_{5/4}(\delta) = \int_0^\delta (\ln(1/|y|))^{-5/4}\,\mathrm{d}y$ for $|\delta| < e^{-3}$. The function $\varphi_{5/4}(t) = (\ln(1/t))^{-5/4}$ is increasing and concave on $(0, e^{-3})$:

$$\varphi_{5/4}''(t) = \frac{5}{4}t^{-2}\left(\ln\frac{1}{t}\right)^{-13/4}\left(\frac{9}{4} - \ln\frac{1}{t}\right) < 0.$$

Hence it is subadditive, and

$$|G_{5/4}'(x) - G_{5/4}'(y)| = |\varphi_{5/4}(|x|) - \varphi_{5/4}(|y|)| \le \varphi_{5/4}(|x - y|).$$

Add to $\lambda^\dagger$ the term $-\eta\chi(a - b)G_{5/4}(a - b)$, where χ is a smooth cut-off equal to 1 near zero, with support in $(-e^{-3}, e^{-3})$ lying together with b inside the interval, and $\eta > 0$ so small that monotonicity, positivity, and (d) are preserved. Properties (b) and (e) are unchanged, the derivative has modulus $O(\varphi_{5/4}(h)) = O((\ln(1/h))^{-5/4})$ on $\mathcal{K}$, and $|(\lambda^\dagger)'(b + h) - (\lambda^\dagger)'(b)| = \eta\varphi_{5/4}(h) + O(h)$, which is not $O(h^\alpha)$ for any $\alpha > 0$. □

9. Conclusion

The paper studied how the infimum $\lambda(c)$ of the truncated Weil form decays as the interval lengthens. Every bounded weak solution of the Dirichlet problem for the logarithmic Laplacian on an interval with bounded right-hand side has a boundary coefficient with an explicit convergence rate to it, and for the eigenfunctions of L_Δ itself on $(-a,a)$ one has $\beta^2 = 1/(2a)$ (Theorem 1.1). The function λ is locally Lipschitz in a, has one-sided derivatives everywhere (Theorem 8.1), is differentiable outside an at most countable set of values a at which $\lambda(c)$ is a multiple eigenvalue, and at points of differentiability $\mathrm{d}\lambda/\,\mathrm{d}a = -2\beta^2$ with the boundary coefficient β of any minimizer. The atoms of the explicit formula do not enter this equality, and if $\lambda(c)$ is a simple eigenvalue, the derivative of λ exists also at $c = p^m$ (Theorem 1.2).

To the right of a power of a prime the derivative of λ has a logarithmic cusp: when the right derivative is negative the function λ does not belong there to the class $C^{1,\alpha}$ for any $\alpha > 0$, and for a simple eigenvalue λ the exact asymptotics of the derivative at this point is known (Sections 8.2–8.5). Boundedness of the boundary balance on intervals where $\lambda > 0$ implies a lower bound for λ in the form of a power of Θ_c and the positivity of λ for all c (Theorem 1.3), while from the proved properties of λ this boundedness does not follow (Proposition 8.16). Along the way one obtains a Pohozaev identity for the Dirichlet problem $(L_\Delta - \mu_0)z = f$ on an interval for every μ_0, the simplicity of the spectrum of L_Δ on an interval, a virial identity for the eigenfunctions of the form, and the membership of these functions in the class $W^{1,1}$, and for the scale, an exact formula and an estimate of its decrease on an interval of length 1.

Limitations. Only the means of the boundary balance over intervals inside $[5, 71]$ are known rigorously, in particular $\langle r\rangle_{[5,71]} \le 1.0022$ (Corollary 5.7). The pointwise values r_c are found numerically, by the Galerkin method, and the Rayleigh–Ritz results, that is, estimates of λ from above, give only means of r_c over unit steps. Near prime powers the Galerkin values r_c at the computed N do not converge in either of the two bases (Appendix A).

Open questions. We note three of them, and the first is the hypothesis of Theorem 1.3, that is, the estimate $r_c \le \rho$ for all c.

The second question is the simplicity of the eigenvalue $\lambda(c)$, in particular for $c = p^m$: Theorems 4.5(i), 8.7(b), 8.12, 8.14 and Corollary 8.13 rely on it. Theorem 1.3 does not depend on it: it needs the Hadamard formula only almost everywhere, and there the formula holds without the simplicity condition. Proving simplicity requires a rigorous lower bound for the second eigenvalue of the even sector, while the paper contains only Galerkin upper estimates, and in [5] lower estimates for a single λ. The Galerkin values of the first two eigenvalues at $c = 7$ differ strongly (Section A.2), but this does not prove the simplicity of the eigenvalue.

The third question, a computational one, concerns rigorous pointwise values of r_c at prime powers, for which one needs proved two-sided estimates of λ with relative accuracy about 1% at noninteger points near p^m (Section A.2).

Data availability

The data from which the numerical statements of the paper and the figures were obtained are deposited in an open Zenodo record doi:10.5281/zenodo.23186383. This DOI refers to the whole record and resolves to its latest version. The data of the paper correspond to version 1.1 of 31 October 2026, and the first version of the record is dated 6 October 2026. The record consists of the following parts:

- the Galerkin values λ_N, β_N^2 and r_c for $c = 6$, 10, 12, 14.5 ($N = 64$, 128, 192), the coefficients of the minimizers and verification runs (comparison with the difference of eigenvalues, control distortion, recomputation, elementary estimates and computations in double precision);
- the summands of the boundary balance decomposition for $c = 6, 6.5, \dots, 20$ ($N = 128$ and 192);
- data on the grid $c \in [6, 14]$ with step 0.1 ($N = 128$ and 192) and a comparison of the two bases for $c = 6.81, \dots, 6.99$ (N from 128 to 384);
- one-sided values at powers of primes (N from 64 to 384), values at powers of primes in the Dirichlet basis (N from 128 to 384) and convergence of the Galerkin values at $c = 7$, 8, 9 and at $c = 6$, 6.5 for N up to 1024 (in the cosine basis up to 768), including the form without the atom $n = 7$;
- Rayleigh–Ritz results for $N = 512$ for $c = 40$, 41, 50, 51, 60, 61, the scale Θ_c and κ_Θ for integer c from 5 to 1000, the rigorous estimates of [5] used in Corollary 5.7, and the figure data, including those for $N = 256$.

The contents of the package and the verification procedure are described in its README.md file, and the SHA-256 checksums of all files are listed in SHA256SUMS. The availability of the source code

TABLE A.1. Stability of the Galerkin values of r_c in N and in the basis.

quantity	points and N	value
r_c away from prime powers	$c = 6, 6.5, \dots, 20$, excluding $c = p^m$; $N = 192$	$[0.913, 1.086]$
same, shift for $N = 128 \to 192$	same points, cosine basis	≤ 0.040
r_c away from prime powers	grid of step 0.1 on $[6, 14]$ without $c = p^m$, 76 nodes; $N = 128$	$[0.85, 1.12]$
same, shift for $N = 128 \to 192$	$[6, 8] \cup [12, 14]$, 39 nodes	≤ 0.042
basis discrepancy	$[6, 6.85]$; $N = 64$	≤ 0.033
same	$c = 6.99$; $N = 64$	0.29
Dirichlet basis, scatter in N	$c = 6.99$; $N = 192, \dots, 1024$	≤ 0.017
basis discrepancy	$c \le 6.95$; $N = 384$	≤ 0.012
same	$c = 6.97$; $N = 384$	0.03
same	$c = 6.99$; $N = 384$	0.06

The shift is the largest change of r_c over the points, and the basis discrepancy is the maximum difference of the values in the cosine basis and in the Dirichlet basis. The endpoints of the ranges are rounded outwards, bounds of the form $\le x$ are rounded up, and the remaining values are rounded to two decimal places.

used to compute these results is described there as well. The value at $N = 768$ for $c = 40$ is taken from [5, Table A.3(b)].

Appendix A. Numerical data on the boundary balance

This appendix collects numerical data concerning the hypothesis of Theorem 1.3, and they do not enter the proofs. What is known rigorously about it and what is not provided by the proved properties of λ is said in the introduction and in Section 8.6. The method is described in Section A.1, the Galerkin values r_c and the means from the Rayleigh–Ritz results are given in Section A.2, the terms of the identity (7.4) at the Galerkin minimizers in Section A.3, and the computation for Figure 1 in Section A.4.

A.1. **The Galerkin method.** To compute the boundary balance in the even sector one takes the basis of normalized functions $\cos(\pi j x/a)$, $j = 0, \dots, N$, on I_c [5, §2.2], which we call the *cosine basis*. For comparison, a basis of functions $\cos(\pi(j + 1/2)x/a)$ equal to zero at the endpoints is also used (*Dirichlet basis*). The matrix of the form is computed to precision 600 bits. Its elements are expressed in terms of the autocorrelations of the basis functions, which are known in closed form: the atoms and the pole term enter exactly, while the integral with the Archimedean density 2ϖ is computed by composite Gauss–Legendre quadrature on $N + 8$ equal subintervals of 32 nodes each. We call the smallest eigenvalue λ_N (an upper bound for λ) and eigenvector u_N Galerkin. Under the dilation taking the basis of the interval I_c to the basis of I_{c^t}, the derivative of the form at fixed coefficients of the vector equals $-\ln c \cdot \beta_N^2$ by the envelope theorem, where $\beta_N^2 = -(1/2)\,\mathrm{d}\lambda_N/\,\mathrm{d}a$. This derivative is computed by Richardson extrapolation with step 10^{-25}, and the Galerkin value of the boundary balance is the ratio $r_N = \beta_N^2/(2\kappa_\Theta \lambda_N)$. Note that β_N^2 is merely notation for $-(1/2)\,\mathrm{d}\lambda_N/\,\mathrm{d}a$, not the boundary coefficient of the vector u_N, which in the cosine basis has $u_N(\pm a) \ne 0$ and no boundary expansion of Theorem 3.8.

Set $\Upsilon = 2\kappa_\Theta \lambda_N \ln c$, that is, the value of $\ln c \cdot \beta_N^2$ at $r_c = 1$. The derivative given by the envelope theorem was checked against the difference of the numbers $\lambda_N(c^t)$ themselves: for $N = 64$, 128, 192 and $c = 6$, 10, 12, 14.5 the discrepancy does not exceed $4 \cdot 10^{-36}\Upsilon$ (the data for each pair c, N are given in the open record, see "Data availability"). Recomputation of r_c at $N = 128$ and 192 agrees with the first one in all printed digits.

A.2. **Galerkin values and means.** *The scale and the Fuchs benchmark.* We begin with how close the infimum is to the scale. By the formula for $\Theta_c/d_c^{\mathrm{K}}$ from Section 5.2 the scale is larger than the Fuchs benchmark by a factor of 2.85, 5.90, 7.04, and 8.78 at $c = 5$, 31, 45, and 71. As seen from Figure 2, the rigorous lower bounds of [5] range from 0.41 to 1.61 of the scale Θ_c and exceed d_c^{K}, while the Rayleigh–Ritz values at $c \le 61$ range from 1.77 to 3.69 (at all $c \ge 40$, the results at $N = 512$). The lower bounds λ_c^- here are taken at $c = 5, \dots, 31$ and $c \in \{45, 47, 71\}$, the upper ones λ_c^+ at $c = 5, \dots, 8$ [5, Tables A.1, A.3], and the Rayleigh–Ritz values are the same ones from which the means of r_c are found below.

The boundary balance. The rigorous data concerning the hypothesis reduce to the means of Corollary 5.7. Away from the prime powers and their neighborhoods of width about 0.01, the Galerkin values r_c lie in $[0.85, 1.12]$ and are stable in N and in the basis (Table A.1 and Figure 3), while at the prime powers themselves and near them they do not converge for the computed N in either basis. These numbers serve as evidence, not as a bound.

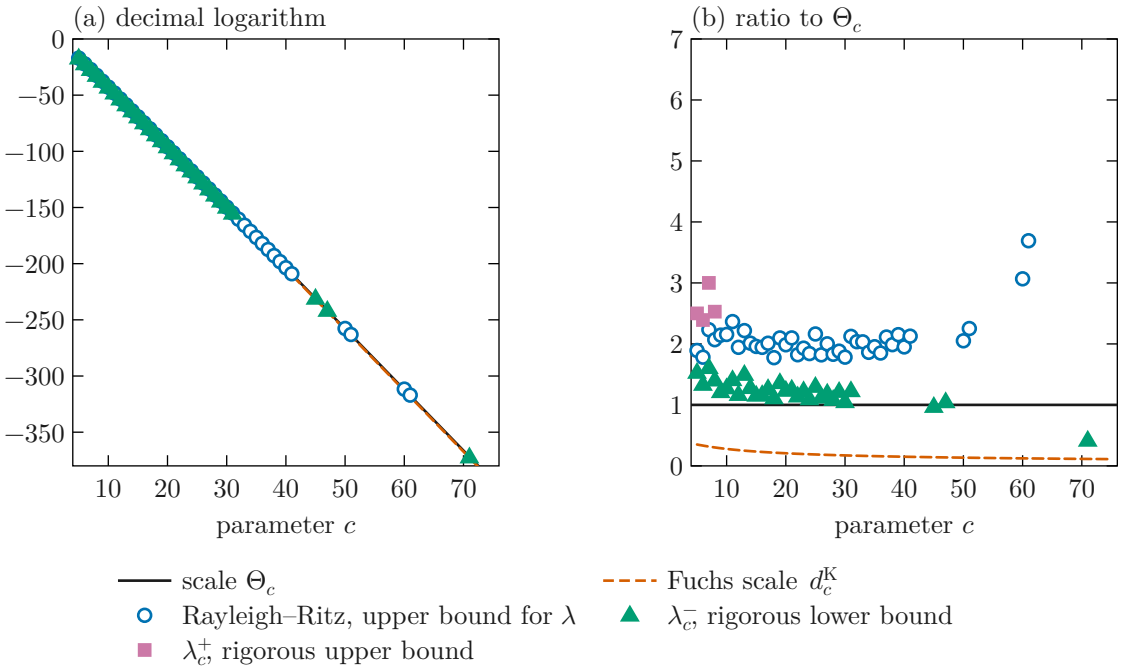


FIGURE 2. The scale Θ_c, the Fuchs benchmark d_c^{K}, the rigorous bounds for λ from [5], and the Rayleigh–Ritz values, numerical upper bounds for λ, not converged in N at $c \geq 22$: (a) decimal logarithms, (b) the same quantities divided by Θ_c. Between such points by the monotonicity $\lambda_{c'}^- \leq \lambda(c) \leq \lambda_{c''}^+$, where $c' > c$ and $c'' \leq \min(c, 8)$ are the values nearest to c at which they are known. At $c = 71$ the lower bound is computed with slope 0.6 instead of the measured 0.277 [5, proof of Proposition 1.3], so it is about half as large as that given by the rule [5, Remark 6.2].

TABLE A.2. The Galerkin boundary balance r_N near $c = 7$.

N	cos., $c = 6.99$	cos., $c = 7 + 0$	Dir., $c = 7$
128	0.965	1.24	1.05
192	1.047	1.35	1.09
256	1.099	1.35	1.08
384	1.141	1.28	1.05
512	1.154		1.01
768			0.96
1024			0.93

Cos. is the cosine basis, Dir. is the Dirichlet basis. In the cosine basis the values at $c = 7 + 0$ were computed only for $N \leq 384$, and those at $c = 6.99$ only for $N \leq 512$.

On the grid for $N = 128$ (all 81 nodes) the values oscillate: adjacent local maxima are separated by gaps of 0.2–0.5 in c, and $\mathrm{d}r_c/\,\mathrm{d}c$ changes sign 48 times. The oscillations do not contradict Theorem 1.3: its hypothesis allows any constant ρ and does not require monotonicity of r_c.

A computation in both bases at $N = 128$, 192, 256, 384 at the points $c = 6.81, 6.83, \ldots, 6.99$ shows that the discrepancy of the bases at $c = 7$ on the left is the error of the cosine basis. It converges slowly there: at $c = 6.99$ the values grow with N (Table A.2) and approach 1.199, the result of the Dirichlet basis at $N = 384$.

Prime powers. The Galerkin values r_N do not converge here: λ_N are upper bounds for λ, while r_N is expressed in terms of their derivative. The λ_N themselves converge numerically: at $c = 7$ in the Dirichlet basis λ_N decrease from $6.886 \cdot 10^{-28}$ at $N = 128$ to $6.601 \cdot 10^{-28}$ at $N = 1024$, and in the cosine basis at $N = 768$ they are already $6.503 \cdot 10^{-28}$. Convergence of the functions does not imply convergence of the derivatives, which fails where the derivative of the exact λ changes sharply.

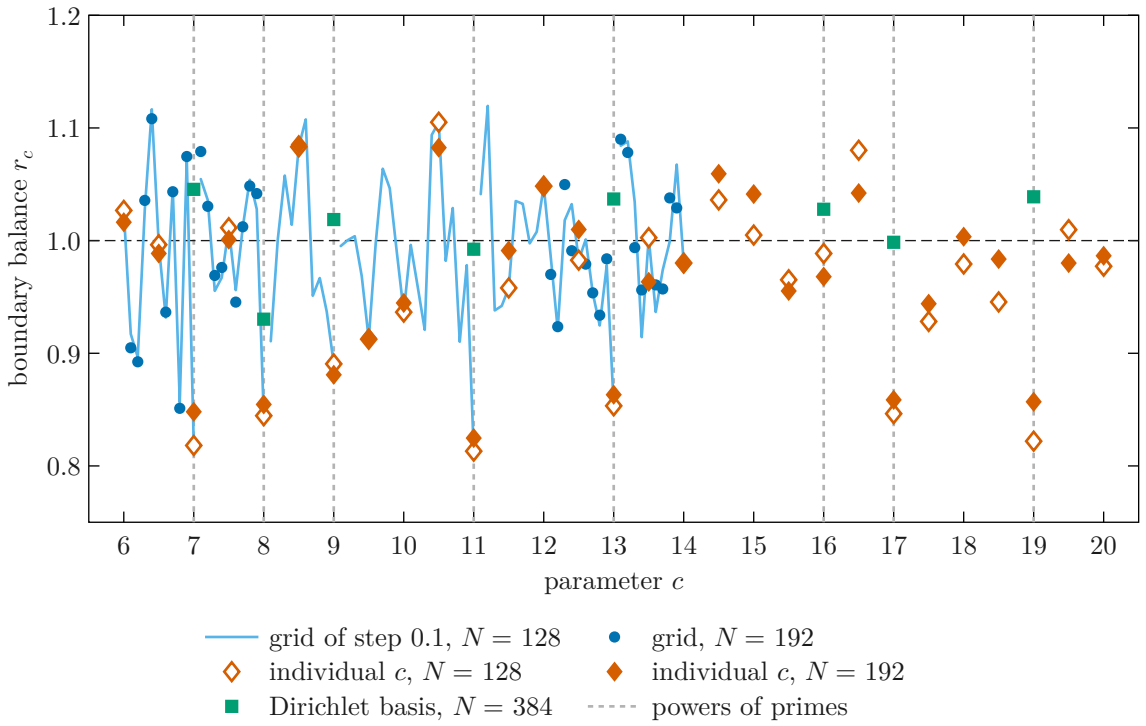


FIGURE 3. The Galerkin boundary balance r_c: the line and the circles show the grid of step 0.1 at $N = 128$ and 192, the diamonds show the points $c = 6, 6.5, \dots, 20$, the squares show the values in the Dirichlet basis at the prime powers at $N = 384$, which in N did not converge. Vertical lines mark the prime powers at which the left limits $c = p^m - 0$ of the cosine-basis values are shown.

TABLE A.3. Dip of the Galerkin r_N to the right of the prime power c_* in the Dirichlet basis.

c_*	N	minimum of r_N	minimum point
7	256	0.966	7.007
	384	0.948	7.007
	512	0.930	7.004
8	256	0.889	8.010
9	256	0.934	9.007

The minimum is taken over the nine shifts $\delta = 0.0005$, 0.001, 0.002, 0.004, 0.007, 0.01, 0.02, 0.03, 0.05 in $c = c_* + \delta$, and the values of r_N are rounded to three decimal places.

In the cosine basis the Galerkin value r_c at the points $c = p^m$ has an upward jump: to the right of $c = 7$ it is greater than one (Table A.2), and at $c = 7 - 0$ it lies in $[0.81, 0.86]$ at $N = 128, \dots, 384$. The jump of β_N^2 at $c = 7$ coincides with $2\Lambda(7)7^{-1/2}u_N(a)^2$ with relative accuracy better than $2 \cdot 10^{-8}$ for all four N (and at $N = 64$): it is generated by the nonzero boundary value $u_N(a)$.

In the Dirichlet basis, where $u_N(a) = 0$ and there is no jump, the values at the very point $c = 7$ in N likewise do not converge. At fixed N the function r_N near $c = 7$ shows a smooth dip of width about 0.02, which as N grows deepens and shifts toward $c = 7$ (Table A.3). The dip is also visible at $c = 8$ and $c = 9$ (at $N = 256$), but not at $c = 6$, which is not a power of a prime. On $[5.99, 6.05]$ the values on the whole decrease: for $N = 256$ and 384 the rise between neighboring grid points is at most 0.007.

In the Dirichlet basis the approximation error resembles a small shift of the argument: on $[6.975, 6.9985]$ the values r_N at $N = 512$ are close to r_N at $N = 256$ shifted to the left by 0.002. The root-mean-square residual equals 0.008 or 0.013 (depending on which of the two grids is interpolated) versus 0.032 without the shift. Such an error is small where the slope of r_c is small: at $c = 6.5$ the values in the Dirichlet basis at $N = 512$, 768, 1024 and in the cosine basis at $N = 512$, 768 differ by no more than 0.004.

TABLE A.4. Means of the boundary balance and the level according to the Rayleigh–Ritz values at large c (for $c \geq 40$, $N = 512$).

c	$\langle r\rangle_{[c-1,c]} - 1$	$c(\langle r\rangle_{[5,c]} - 1)$	$\vartheta(c)$
41	-0.007	-0.011	2.13
51	-0.008	-0.016	2.25
61	-0.015	-0.059	3.69

The means were found by the formula (5.5), the values are rounded to three decimal places, and the level ϑ to two.

TABLE A.5. Stability of the Rayleigh–Ritz values.

test	result
last step in N at $c \geq 22$ [5, Table A.3]	decrease by 4.9–17.4%
$c = 40$, transition from $N = 512$ to $N = 768$	decrease by 12%
$c = 40$ and 61: 20 additional digits	the digits do not change
$c = 40$ and 61: doubled number of quadrature nodes	the digits do not change
the bound 0.0191 at $N = 512$ or 768 for $c = 40$ [5, Table A.3(b)]	does not change

Decrease is the relative reduction of the upper bound for λ, "the digits do not change" means the coincidence of all printed digits of the Rayleigh–Ritz value.

The dip at $c = p^m$ is consistent with the results of Section 8.2: if $\lambda(c_*)$ is a simple eigenvalue, the exact r_c has no jump at this point (Theorem 4.5(i)), but to the right of c_* the derivative of λ varies logarithmically fast (Theorem 8.14), while to the left its modulus of continuity is $O((\ln(1/\delta))^{-5/4})$ (Corollary 8.13).

Heuristically: if the basis permits distances of about h near the boundary, the apparent jump has order $2w_*/\ln(1/h)$. For $c = 7$, where $2w_* = 1.47$, this is 21% for $h = 10^{-3}$ and 4% for $h = 10^{-15}$. The correction is the contribution of the new atom, which connects the boundary layers of the two ends (Lemma 8.4). Relative to the form without it, the infimum decreases by exactly $(4w_*\beta_*^2 + o(1))\delta/\ln(1/\delta)$ (Theorem 8.7).

If the atom $n = 7$ is removed and the form is continued beyond $c = 7$, then its Galerkin matrix at $c = 7.07$ is no longer positive definite at $N = 256$ and 384, although $\lambda(7.07) \geq \lambda(8) > 0$ [5, Theorem 1.1]. By Remark 8.15 this means that the minimizer of the form Q° at $c = 7.07$ in the boundary layers is far from the original one at $c = 7$.

A basis with resolution h near the boundary smooths the logarithmic cusp into a dip of width of order h: the value at the point itself approaches the limit only as the dip narrows. Near the powers of primes the Galerkin values r_c do not serve as evidence about the exact r_c. The cusp does not interfere with the hypothesis of Theorem 1.3, but the pointwise estimate of r_c via the derivative at prime powers is not applicable.

Finding r_c from the boundary profile u_N is not possible in practice. The boundary coefficient of the normalized minimizer, whose square is comparable to λ by the Hadamard formula, is about $4 \cdot 10^{-12}$ ($\beta_N^2 = 1.6 \cdot 10^{-23}$ at $c = 6.5$ and $N = 512$–1024), which is many orders of magnitude smaller than the error of u_N near the edge. The derivative, however, expresses β^2 in terms of the energy, where the relative accuracy of λ_N is preserved.

At $c = 7$ the Galerkin $\lambda_N = 7.18 \cdot 10^{-28}$ and the second eigenvalue $1.74 \cdot 10^{-21}$ ($N = 192$) differ by more than a factor of 10^6. However, these are upper bounds, which do not prove the simplicity of the eigenvalue.

Means over Rayleigh–Ritz values. For these we took the upper bounds for λ (with $6c + 40$ digits) from Table A.3 of [5] at $c \leq 39$ and the results computed by us at $N = 512$ for $c = 40$, 41, 50, 51, 60, 61. In Figure 4 one can see that the means r_c over steps $[c, c+1]$ fluctuate around one: on all 38 such steps the deviation is at most 0.0191. As c grows it does not decrease, and for the mean over $[5, c]$ the deviation from one, multiplied by c, grows in absolute value together with the level ϑ (Table A.4). The Rayleigh–Ritz values at $c \geq 22$ have not converged: for fixed N the upper estimate is the coarser the larger c (Table A.5), and the increase of ϑ at $c = 60$, 61 may be an artifact of the approximation.

What rigorous values at powers of primes require. Two-sided bounds for λ are needed at noninteger points: by (5.5) on the interval $[c_*, c_* + h]$ their relative errors ε at the endpoints lead to an uncertainty of the mean r_c of about $2\varepsilon/\ln(\Theta_{c_*}/\Theta_{c_*+h})$. For $c_* = 7$ and $h = 0.01$ the denominator equals 0.119, and the excess over r_* by Theorem 8.14 is of order $2w_*/\ln(1/\delta) \approx 0.20$ for $\delta = \ln(7.01/7)/2$, so to distinguish these quantities one needs accuracy $\varepsilon \approx 1\%$. The method of [5, Theorem 5.10] is applicable for any

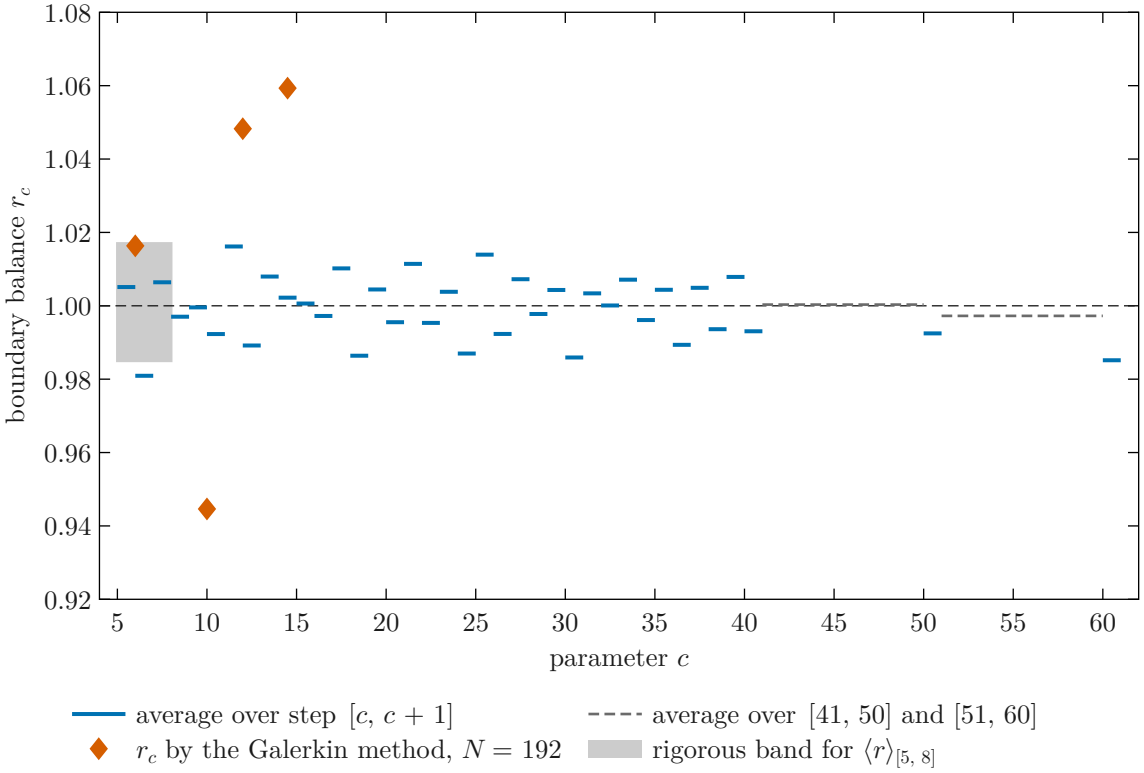


FIGURE 4. Boundary balance from Rayleigh–Ritz values, numerical upper bounds for λ, which at $c \geq 22$ have not converged in N: segments show means (5.5) over unit steps, dashes show means over $[41, 50]$ and $[51, 60]$, diamonds show r_c by the Galerkin method at $N = 192$. The rectangle is the rigorous strip $\langle r\rangle_{[5,8]} \in [0.985, 1.017]$ (Corollary 5.7), while the means over the Rayleigh–Ritz values bound the exact means neither from above nor from below.

TABLE A.6. The terms of identity (7.4) at the Galerkin minimizers of the form Q_c, $N = 192$.

c	Υ	A	R	$\mathcal{E}$	$A+R$	r_c
6	$9.47 \cdot 10^{-21}$	1.16727	−0.57411	−0.59317	0.59317	1.0163
10	$4.42 \cdot 10^{-41}$	1.17225	−0.58880	−0.58345	0.58345	0.9446
12	$1.65 \cdot 10^{-51}$	1.17344	−0.59230	−0.58114	0.58114	1.0483
14.5	$1.26 \cdot 10^{-64}$	1.17445	−0.59525	−0.57920	0.57920	1.0593

Here $\Upsilon = 2\kappa_\Theta \lambda_N \ln c$ is the value of $\ln c \cdot \beta^2$ at $r_c = 1$, and $r_c = (A + R + \mathcal{E})/\Upsilon$. The columns $A + R$ and $-\mathcal{E}$ coincide in all printed digits, since their difference $r_c\Upsilon$ is small. The factor $2\kappa_\Theta \ln c$ in Υ equals 126 for $c = 6$ and 474 for $c = 14.5$.

$c > 1$, but there the bounds are computed only at integer points: for $3 \leq c \leq 31$ the lower bounds are no less than $1.04\,\Theta_c$, the Galerkin values for $5 \leq c \leq 31$ do not exceed $2.37\,\Theta_c$, and the upper bounds reach $3.0005\,\Theta_c$. With such bounds the uncertainty of the mean over the same interval would be about $\ln(3.0005/1.04)/0.119 \approx 8.9$.

A.3. **The terms at the minimizers.** Consider how the terms of identity (7.4) behave at the Galerkin minimizers. For fixed coefficients of the vector, the derivative of each part of the form with respect to dilation is found by the same Richardson extrapolation. The quantities $M^\pm$ from the proof of Proposition 7.6 are the derivatives with respect to dilation of the forms $-\int_0^{\ln c} e^{\pm s/2} g(s)\,\mathrm{d}s$, so that $R = 2M^-$ and $\mathcal{E} = -2S + 2M^+$. The unit-size terms cancel down to $r_c\Upsilon$ (Table A.6), while the elementary estimates of Remark 7.7 are many orders of magnitude coarser than what the hypothesis requires.

At $c = 6, 6.5, \ldots, 20$ ($N = 192$) the terms change slowly and monotonically: A grows from 1.167 to 1.176, R decreases from −0.574 to −0.599, and $\mathcal{E}$ increases from −0.593 to −0.577. Inside $\mathcal{E}$ the atoms contribute $-2S$ from 0.307 to 0.388, and the main term $2M^+$ from −0.900 to −0.965. The sum

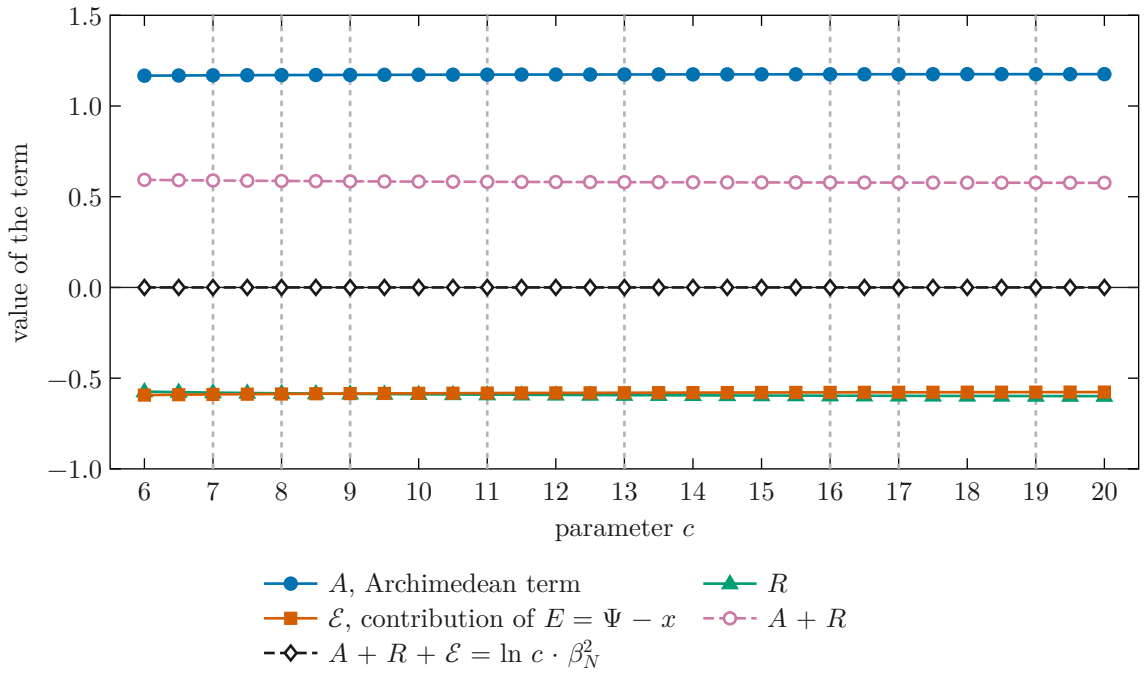


FIGURE 5. The terms of identity (7.4) at the Galerkin minimizers for $c = 6, 6.5, \dots, 20$, $N = 192$. The sum $A + R + \mathcal{E} = r_c \Upsilon$ is indistinguishable from zero in the figure (Υ from 10^{-93} to 10^{-20}), and the vertical lines mark the powers of primes.

$A + R + \mathcal{E}$ equals $r_c \Upsilon$, that is, it is smaller than the terms by a factor of 10^{20}–10^{93}, and in Figure 5 it is indistinguishable from zero. Figure 6 shows how the parts of size one cancel down to a quantity of order $\lambda_N \ln(1/\lambda_N)$.

Note that the order of this cancellation is prescribed by the identity (7.4) and the smallness of λ: for $r_c \approx 1$ it amounts to $\log_{10}(1/\Upsilon)$ decimal places and by itself says nothing about the weights $\Lambda(n)$.

In the same notation, the virials of the remainder of the Archimedean symbol, of the atoms, and of the pole term from (7.3) for $c = 6$ and $N = 192$ equal -0.1673, -0.3072, and $+1.4744$, and their sum differs from 1 by $9.6 \cdot 10^{-21}$.

Checks. The agreement of the sum $A + R + \mathcal{E}$ with the full derivative follows from the identity and therefore does not test it. Two checks are independent of it: the comparison with the difference of the numbers $\lambda_N(c^t)$ themselves (Section A.1) and the test perturbation of the density $e^{s/2}$ by 10^{-12}, which is detected. The data record includes both these checks and the others: the pole term as a rank-one term and as a measure, the recomputation without the splitting into A, R, $\mathcal{E}$, the computations in double precision, and the dependence on N.

Elementary bounds. At the Galerkin minimizers for $c = 6$, 10, 12, 14.5 ($N = 192$), Remark 7.7 gives $|R| \le 0.596$ and the two bounds $|\mathcal{E}| \le 1.331$ and $|\mathcal{E}| \le 3.60$. All three depend weakly on c, and $\ln c \cdot \beta^2 = A + R + \mathcal{E}$ does not exceed 4.01. The second bound for $|\mathcal{E}|$ is $3.8 \cdot 10^{20}$ times larger than Υ for $c = 6$, and $2.9 \cdot 10^{64}$ times for $c = 14.5$.

A.4. **Eigenfunctions of the logarithmic Laplacian.** For Figure 1 the eigenfunctions of L_Δ on $(-1, 1)$ were computed by a piecewise-constant Galerkin method. The cells are refined towards the boundary in a geometric progression with ratio $2^{-1/4}$, and all four functions were also found with ratios $1/2$ and $2^{-1/2}$. The figure shows the boundary expansion for sign-changing solutions as well: three of the four functions change sign (Theorem 3.8 does not require $u \ge 0$). The shift of order 10^{-3} at $1/\xi \to 0$ in Figure 1(b) is a grid error, which decreases as the ratio approaches one. To estimate the limit (3.5), the quantity $(1/2)\xi^{-1/2} \int_{e^{-\xi}}^1 u(a - t)\, \mathrm{d}t/t$ at $\xi \ge 20$ was approximated by a polynomial of degree two in $\xi^{-1/2}$ on grids with ratios $2^{-1/2}$ and $2^{-1/4}$. The square of the constant term was then extrapolated to ratio 1 under the assumption that the grid error is linear in the ratio's logarithm. For all four functions the squares of the limits lie between 0.49997 and 0.50008.

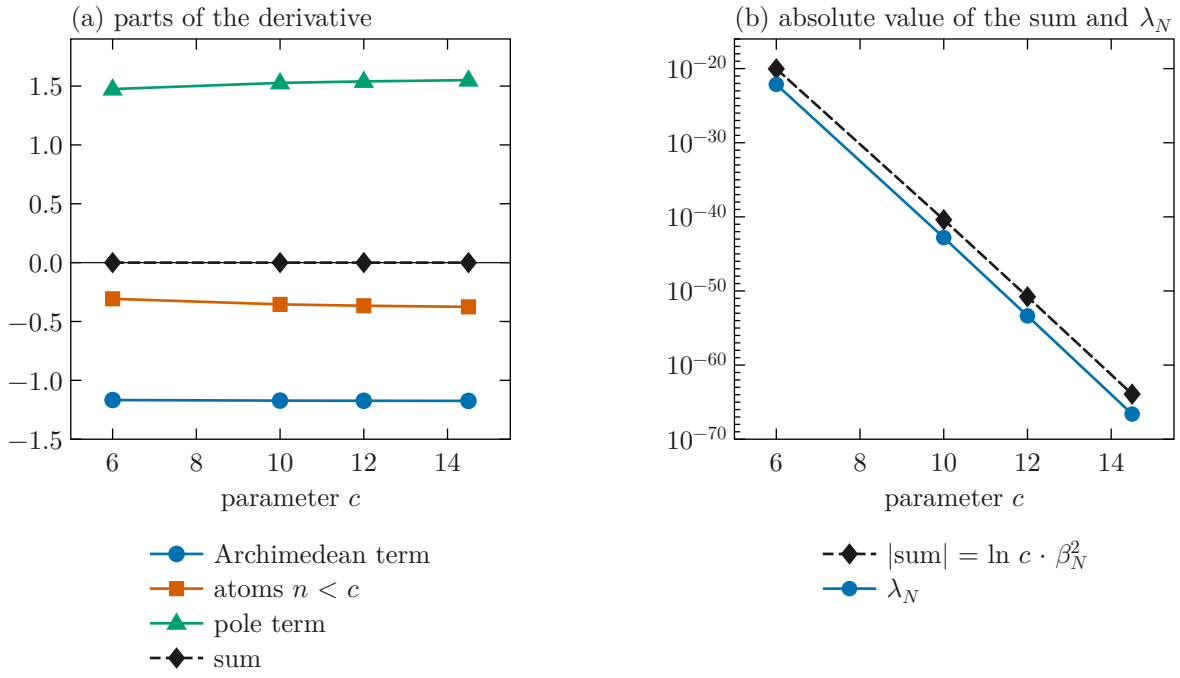


FIGURE 6. The parts of the derivative $\mathrm{d}/\,\mathrm{d}t\, Q_{c^t}(u_N^{(t)})|_{t=1} = -\ln c \cdot \beta_N^2$ on the Galerkin space, $N = 256$: (a) the Archimedean term, the atoms $n < c$, the pole term, and their sum for $c = 6$, 10, 12, 14.5, (b) the modulus of the sum and λ_N on a logarithmic scale. Contributions of size one cancel by a factor of 10^{20}–10^{64}, down to a magnitude of order $\lambda_N \ln(1/\lambda_N)$.

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